MATH READING NOTES: *Homogenization and Asymptotic Analysis for Highly Oscillating Periodic Structures*

Chuanwan (Christine) Wu (cw4459@nyu.edu)

Based on the book *Asymptotic Analysis for Periodic Structures*
by A. Bensoussan, J.-L.Lions, and G.Papanicolaou

July 31, 2025

Note: *The reading was done for a side-project during the author's master's at Courant Institute.*

Disclaimer: *These notes are informal, and have not been peer-reviewed, which may contain errors. Feedback is welcome (contact the author via email).* ♟️🐰

CONTENTS

1 Overview **3**
1.1 Introduction to the Problem . . . 3

2 Elliptic Operators **5**
2.1 Setting of the problem (I). . . . 5
2.2 Setting of the Problem (II): Boundary Conditions . . . 7
2.3 A One-Dimensional Example . . . 9
2.4 Asymptotic Expansions . . . 10
2.4.1 Orientation . . . 10
2.4.2 The Method of Multiple Scales . . . 10
2.4.3 Properties of the Homogenized Operator . . . 17
2.4.4 Justification of the Asymptotic Expansion for Dirichlet Problems . . . 18
2.4.5 Higher-Order Corrections in the Expansion . . . 20
2.5 Energy Proof of the Homogenization . . . 23
2.5.1 Orientation: Statement of the Main Result . . . 23
2.5.2 Proof of the Convergence Theorem . . . 24
2.5.3 Alternative Justification via Adjoint Expansion . . . 29
2.5.4 Comparison Theorems for Homogenized Coefficients . . . 33

2.6 Correctors . . . 40
2.6.1 Orientation . . . 40
2.6.2 Structure of the First Corrector . . . 40
2.6.3 Detailed Proof of Theorem 5.1 . . . 42
2.6.4 First-Order System and Asymptotic Expansion . . . 48
2.6.5 Correctors and Error Estimates for the Dirichlet Problem . . . 53

3 Evolution Operators **59**
3.1 Evolution Operators of Hyperbolic, Petrowsky, or Schrödinger Type . . . 59
3.1.1 Orientation . . . 59
3.1.2 Linear Operators with Coefficients Regular in Time . . . 59
3.1.3 Linear Operators with Coefficients Irregular in Time . . . 67
3.1.4 Asymptotic Expansions (I) . . . 74
3.1.5 Asymptotic Expansions (II): Rigorous Justification for a First-Order Term . . . 81

1 Overview

1.1 Introduction to the Problem

Homogenization is a mathematical framework for deriving macroscopic, effective descriptions of materials or processes with fine-scale microscopic structures. This is formalized by studying a family of partial differential operators A^ε, parameterized by a small scale parameter $\varepsilon > 0$.

1. **Properties of A^ε:** The operator family can be characterized by:
    1. **Time Dependence**: Stationary (time-independent) or evolutionary.
    2. **Nature of the Problem**: Describing a steady state or a transient evolution.
    3. **Linearity**: Linear or nonlinear.
    4. **Coefficients**: Feature rapidly oscillating functions (often periodic or almost periodic) with a characteristic period proportional to ε.
    5. **Scale Parameter** ε: Represents the ratio of the microscopic scale to the macroscopic scale ($\varepsilon \ll 1$).
2. **General Setup:** For a fixed $\varepsilon > 0$, we consider a boundary value problem in a domain Ω:

$$\begin{cases} A^\varepsilon u_\varepsilon = f & \text{in } \Omega, \\ \text{Boundary Conditions} & \text{on } \partial\Omega. \end{cases} \tag{1.1}$$

    Here, u_ε is the microscopic solution and f is a source term. The problem is well-posed for each ε, but our interest lies in the **limit** as $\varepsilon \to 0$.

    The Core Goal of Homogenization. The primary objective is to find a *homogenized*, macroscopic solution u_0 such that:

$$u_\varepsilon \to u_0 \quad \text{(in an appropriate sense as } \varepsilon \to 0\text{)},$$

    where u_0 is the solution to a (simpler) *homogenized equation* $\mathcal{A}u_0 = f$.

3. **The Homogenized Equation and Effective Coefficients:** Under suitable conditions, the limit u_0 satisfies:

$$\mathcal{A}u_0 = f \quad \text{in } \Omega, \tag{1.2}$$

    where $\mathcal{A}$ is the **homogenized operator**. Its coefficients are constant **effective coefficients** that describe the overall, averaged macroscopic properties of the medium. This equation is typically much simpler to solve than the original, oscillatory problem.

4. **Cell Problems:** The effective coefficients of $\mathcal{A}$ are not simple averages; they are computed by solving **cell problems** (also called *local problems*) on a single periodic cell Y. These problems capture the interaction of the microstructure within itself.

5. **Key Homogenization Methods:** There are several analytical approaches to derive $\mathcal{A}$:

    1. **Multiple-Scale Asymptotic Expansions**
        - 1.1 Introduce a *fast variable* $y = x/\varepsilon$.
        - 1.2 Assume an ansatz: $u_\varepsilon(x) = u_0(x, y) + \varepsilon u_1(x, y) + \varepsilon^2 u_2(x, y) + \cdots$.
        - 1.3 Substitute into the PDE and solve order-by-order in ε.
    2. **Energy (Variational) Methods**
        - 2.1 Use weak formulations and test functions.
        - 2.2 Prove convergence $u_\varepsilon \rightharpoonup u_0$ in Sobolev spaces via compactness arguments and energy estimates.
    3. **Probabilistic Methods**
        - 3.1 Interpret the process stochastically (e.g., diffusion in random media).
        - 3.2 Use laws of large numbers, central limit theorems, and ergodicity.
    4. **Bloch Wave Expansion**
        - 4.1 Used for periodic, spectral problems (e.g., in photonic crystals).
        - 4.2 Decompose the solution using the eigenfunctions of the periodic operator.

Note: A significant challenge lies not only in proving the existence of the homogenized operator $\mathcal{A}$ but also in **explicitly computing its coefficients**. While analytical formulas exist, numerically solving the required cell problems remains non-trivial due to their own computational complexity and potential ill-conditioning.

2 ELLIPTIC OPERATORS

2.1 Setting of the problem (I).

Summary of Key Concepts

Symbol	Name	Description
$\mathcal{O}$	Macroscopic Domain	Bounded open set in $\mathbb{R}^n$ where the PDE is posed.
Y	Unit Cell	$\prod_{j=0}^{n}]0, y_j^\circ[$; represents the periodic microstructure.
ε	Scale Parameter	Small parameter governing the microstructure size.
$a_{ij}(y), a_0(y)$	Coefficients	**Y-periodic** functions defining material properties.
A^ε	Micro-Scale Operator	$-\partial_i(a_{ij}(x/\varepsilon)\partial_j) + a_0(x/\varepsilon)$. Uniformly elliptic.
u_ε	Micro-Scale Solution	Solution to $A^\varepsilon u_\varepsilon = f$.
$\mathcal{A}$	**Homogenized Operator**	Constant-coefficient effective operator.
u	Macro-Scale Solution	Solution to $\mathcal{A}u = f$; $u_\varepsilon \rightharpoonup u$.

Core Assumptions:

1. **Periodicity:** Coefficients are Y-periodic.
2. **Uniform Ellipticity:** $a_{ij}(y)\xi_i\xi_j \geq \alpha|\xi|^2$ for all y, ξ, with $\alpha > 0$ **independent of** ε.
3. **Boundedness:** $a_{ij}, a_0 \in L^\infty(\mathbb{R}^n)$.

Let $\mathcal{O}$ be a bounded open set of $\mathbb{R}^n$. We consider boundary value problems in $\mathcal{O}$ associated to the operator A^ε, which is uniformly elliptic as $\varepsilon \to 0$, and whose coefficients are rapidly oscillating (with "period" ε). Mathematically, this is formalized as:

1. **The Unit Cell and Periodicity:** We define the unit cell:

$$Y = \prod_{j=0}^{n}]0, y_j^\circ[\subset \mathbb{R}^n. \tag{2.1}$$

[1] A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is called **Y-periodic** if it is periodic with period y_j° in each coordinate direction y_j.

2. **Coefficient Functions:** We consider functions a_{ij} (for $i, j = 1, \ldots, n$) and a_0 satisfying:

$$\begin{cases} a_{ij}(y) \in \mathbb{R}, & a_{ij} \text{ is Y-periodic}, \quad a_{ij} \in L^\infty(\mathbb{R}^n), \\ a_{ij}(y)\xi_i\xi_j \geq \alpha\xi_i\xi_i, & \alpha > 0, \quad \text{a.e. } y \in \mathbb{R}^n. \end{cases} \tag{2.2}$$

[1] Note on notation: The indexing $j = 0$ to n (vs. the conventional $j = 1$ to n) may indicate an extra dimension or follow the book's specific convention. The brackets $]a, b[$ denote open intervals, equivalent to (a, b) in Anglo-American notation.

(Note: we do not assume symmetry, i.e., $a_{ij} \neq a_{ji}$ in general). We also consider a_0 such that:

$$\begin{cases} a_0 \in L^\infty(\mathbb{R}^n), & a_0 \text{ is Y-periodic,} \\ a_0(y) \geq \gamma > 0 \quad (\text{though often } a_0 \equiv 0 \text{ in applications}). \end{cases} \tag{2.3}$$

3. **The Operator:** The family of operators is defined as:

$$A^\varepsilon = -\frac{\partial}{\partial x_i}\left(a_{ij}\left(\frac{x}{\varepsilon}\right)\frac{\partial}{\partial x_j}\right) + a_0\left(\frac{x}{\varepsilon}\right). \tag{2.4}$$

Here $\varepsilon > 0$ is a small parameter.

Remark 2.1. The operator (2.4) models simple composite materials with a microscopic period of size ε.

Remark 2.2. The family A^ε is uniformly elliptic, with an ellipticity constant α independent of ε.

4. **Problem Statement:** We consider the boundary value problem:

$$\begin{cases} A^\varepsilon u_\varepsilon = f & \text{in } \mathcal{O}, \\ u_\varepsilon = 0 & \text{on } \Gamma = \partial\mathcal{O}. \end{cases} \tag{2.5}$$

We study the asymptotic behavior of u_ε as $\varepsilon \to 0$.

5. **Main Result:**

Homogenization Theorem. Under the given conditions, there exists a second-order elliptic operator $\mathcal{A}$ (the homogenized operator) with constant, explicitly computable coefficients such that:

$$u_\varepsilon \rightharpoonup u \quad (\text{weakly in, e.g., } H_0^1(\mathcal{O})),$$

where u is the solution to the effective homogenized problem:

$$\begin{cases} \mathcal{A}u = f & \text{in } \mathcal{O}, \\ u = 0 & \text{on } \Gamma. \end{cases} \tag{2.6}$$

The operator $\mathcal{A}$ is constructed via the solution of **cell problems** on Y, and its coefficients represent the effective material properties.

In the next section, we will discuss different boundary conditions.

2.2 Setting of the Problem (II): Boundary Conditions

The variational (weak) formulation is essential for proving well-posedness and, later, convergence results. It provides a robust framework for handling the oscillatory coefficients and various boundary conditions.

Sobolev Spaces Framework. The analysis takes place within Sobolev spaces, which are function spaces suitable for solving PDEs.

1. **Sobolev Spaces:** We denote by $H^1(\mathcal{O})$ the Sobolev space of functions with square-integrable first derivatives:

$$H^1(\mathcal{O}) = \left\{ v \in L^2(\mathcal{O}) \;\middle|\; \frac{\partial v}{\partial x_i} \in L^2(\mathcal{O}),\ i = 1, \dots, n \right\}. \tag{2.7}$$

   equipped with the norm:

$$\|v\|^2_{H^1(\mathcal{O})} = \|v\|^2_{L^2(\mathcal{O})} + \|\nabla v\|^2_{L^2(\mathcal{O})}. \tag{2.8}$$

   We define the space for Dirichlet boundary conditions:

$$H^1_0(\mathcal{O}) = \text{closure of } \mathcal{D}(\mathcal{O}) = C^\infty_0(\mathcal{O}) \text{ in } H^1(\mathcal{O}). \tag{2.9}$$

   It is characterized by the trace property: $H^1_0(\mathcal{O}) = \{v \in H^1(\mathcal{O}) \mid v = 0 \text{ on } \Gamma\}$.
   Poincaré's inequality [2](which holds on bounded domains) implies the norms $\|v\|_{H^1(\mathcal{O})}$ and $\|\nabla v\|_{L^2(\mathcal{O})}$ are equivalent on $H^1_0(\mathcal{O})$.

2. **The General Space V:** The boundary conditions are encoded by choosing the space:

$$V = \text{a closed subspace of } H^1(\mathcal{O}) \text{ such that } H^1_0(\mathcal{O}) \subset V \subset H^1(\mathcal{O}). \tag{2.10}$$

   This defines the set of *admissible functions* that satisfy the homogeneous part of the BCs. A typical example is $V = \{v \in H^1(\mathcal{O}) \mid v|_{\Gamma_0} = 0\}$ for some $\Gamma_0 \subset \Gamma$.

3. **Bilinear Form and Coercivity:** For $u, v \in H^1(\mathcal{O})$, define the bilinear form associated with A^ε:

$$a^\varepsilon(u, v) = \int_{\mathcal{O}} a_{ij}\left(\frac{x}{\varepsilon}\right) \frac{\partial u}{\partial x_j} \frac{\partial v}{\partial x_i}\, dx + \int_{\mathcal{O}} a_0\left(\frac{x}{\varepsilon}\right) uv\, dx \tag{2.11}$$

$$= \int_{\mathcal{O}} a^\varepsilon_{ij} \partial_j u\, \partial_i v\, dx + \int_{\mathcal{O}} a^\varepsilon_0 uv\, dx. \tag{2.12}$$

[2]Poincaré's inequality states that for a bounded domain $\mathcal{O}$ and $v \in H^1_0(\mathcal{O})$, there exists a constant $C = C(\mathcal{O}) > 0$ such that $\|v\|_{L^2(\mathcal{O})} \leq C\|\nabla v\|_{L^2(\mathcal{O})}$. This implies the equivalence of the norms $\|v\|_{H^1(\mathcal{O})}$ and $\|\nabla v\|_{L^2(\mathcal{O})}$ on $H^1_0(\mathcal{O})$.

The assumptions (2.2) and (2.3) imply this form is **coercive**:

$$a^\varepsilon(v,v) \geq \min(\alpha,\gamma)\,\|v\|^2_{H^1(\mathcal{O})} \quad \forall v \in H^1(\mathcal{O}). \tag{2.13}$$

Crucially, the constant $\min(\alpha,\gamma) > 0$ **is independent of** ε, ensuring uniform well-posedness.

4. **The Abstract Variational Problem:** For a given $f \in L^2(\mathcal{O})$ (so $\langle f, v\rangle = \int_{\mathcal{O}} fv\,dx$ defines a functional in V'), the solution u_ε is defined by:

$$u_\varepsilon \in V, \quad a^\varepsilon(u_\varepsilon, v) = \langle f, v\rangle \quad \forall v \in V. \tag{2.14}$$

The **Lax-Milgram lemma** guarantees existence and uniqueness of a solution $u_\varepsilon \in V$ for each $\varepsilon > 0$.

5. **The Boundary Value Problem (BVP).** [3]The function u_ε is the unique solution to the variational problem:

$$u_\varepsilon \in V, \quad a^\varepsilon(u_\varepsilon, v) = \int_{\mathcal{O}} fv dx \quad \forall v \in V. \tag{2.15}$$

This is the precise weak formulation of the original boundary value problem (2.5), with the choice of space V encoding the specific boundary conditions.

The choice of V determines the **boundary conditions**. Here are key examples:

Space V	**Corresponding Boundary Value Problem**	**Type**
$H^1_0(\mathcal{O})$	$\begin{cases} A^\varepsilon u_\varepsilon = f & \text{in } \mathcal{O}, \\ u_\varepsilon = 0 & \text{on } \Gamma. \end{cases}$	Dirichlet
$H^1(\mathcal{O})$	$\begin{cases} A^\varepsilon u_\varepsilon = f & \text{in } \mathcal{O}, \\ \frac{\partial u_\varepsilon}{\partial \nu_{A^\varepsilon}} = 0 & \text{on } \Gamma. \end{cases}$ $\frac{\partial v}{\partial \nu_{A^\varepsilon}} = a^\varepsilon_{ij}\frac{\partial v}{\partial x_j}\nu_i$	Neumann
$\{v \in H^1(\mathcal{O}) \mid v\vert_{\Gamma_0} = 0\}$	$\begin{cases} A^\varepsilon u_\varepsilon = f & \text{in } \mathcal{O}, \\ u_\varepsilon = 0 \text{ on } \Gamma_0, \\ \frac{\partial u_\varepsilon}{\partial \nu_{A^\varepsilon}} = 0 \text{ on } \Gamma_1. \end{cases}$	Mixed
$\{v \in H^1(\mathcal{O}) \mid v\vert_{\Gamma} = \text{const}\}$	$\begin{cases} A^\varepsilon u_\varepsilon = f & \text{in } \mathcal{O}, \\ u_\varepsilon = c_\varepsilon \text{ on } \Gamma, \\ \int_\Gamma \frac{\partial u_\varepsilon}{\partial \nu_{A^\varepsilon}}\,d\Gamma = 0. \end{cases}$	Const. Dirichlet

[3]The Lax-Milgram lemma states that for a bounded, coercive bilinear form $a(\cdot,\cdot)$ on a Hilbert space V, and any bounded linear functional $f \in V'$, there exists a unique $u \in V$ such that $a(u,v) = \langle f, v\rangle$ for all $v \in V$. This provides the theoretical foundation for the well-posedness of elliptic PDEs in variational form. See **[?]** for the reference.

2.3 A One-Dimensional Example

This example illustrates homogenization's core concepts in the simplest setting. Consider:

$$-\frac{d}{dx}\left[a^\varepsilon(x)\frac{du_\varepsilon}{dx}\right] = f \quad \text{in } (x_0, x_1), \quad u_\varepsilon(x_0) = u_\varepsilon(x_1) = 0, \tag{2.16}$$

where $a^\varepsilon(x) = a(x/\varepsilon)$, $a(y)$ is Y-periodic on $Y = (0, L)$, and $a(y) \geq \alpha > 0$.

Key Steps:

1: Uniform Estimates

The weak formulation yields a uniform energy bound:

$$\|u_\varepsilon\|_{H^1} \leq C\|f\|_{L^2}, \tag{2.17}$$

implying $u_\varepsilon \rightharpoonup u$ weakly in H_0^1.

2: The Averaging Pitfall

While $a^\varepsilon \rightharpoonup \mathcal{M}(a)$ weakly-*, the naive homogenized equation

$$-\frac{d}{dx}\left(\mathcal{M}(a)\frac{du}{dx}\right) = f \quad \text{(incorrect)} \tag{2.18}$$

fails. The effective property is *not* the arithmetic mean.

3: Flux Reformulation and Convergence

Define the flux $\sigma^\varepsilon := a^\varepsilon \frac{du_\varepsilon}{dx}$, giving:

$$-\frac{d\sigma^\varepsilon}{dx} = f. \tag{2.19}$$

3.1 **Flux boundedness:** $\{\sigma^\varepsilon\}$ bounded in H^1

3.2 **Strong convergence:** $\sigma^\varepsilon \to \sigma$ strongly in L^2

3.3 **Weak convergence:** $1/a^\varepsilon \rightharpoonup \mathcal{M}(1/a)$ weakly-*

3.4 **Limit passage:** From $\frac{du_\varepsilon}{dx} = \frac{\sigma^\varepsilon}{a^\varepsilon}$, we obtain:

$$\frac{du}{dx} = \mathcal{M}\left(\frac{1}{a}\right)\sigma. \tag{2.20}$$

3.5 **Homogenized system:** Combining with $-\frac{d\sigma}{dx} = f$ gives:

$$-\frac{d}{dx}\left(\frac{1}{\mathcal{M}(1/a)}\frac{du}{dx}\right) = f. \tag{2.21}$$

Homogenized Result. The limit u satisfies:

$$-\frac{d}{dx}\left(a^*\frac{du}{dx}\right) = f, \quad u(x_0) = u(x_1) = 0, \tag{2.22}$$

where the effective coefficient is the harmonic mean:

$$a^* = \frac{1}{\mathcal{M}(1/a)} = \left(\frac{1}{L}\int_0^L \frac{dy}{a(y)}\right)^{-1}. \tag{2.23}$$

Remark 2.5. Harmonic-Arithmetic Inequality:

$$a^* \leq \mathcal{M}(a), \tag{2.24}$$

with equality only if $a(y)$ is constant. This shows macroscopic resistance is dominated by the weakest microscopic link.

Remark 2.6. Beyond Periodicity:
For non-periodic coefficients, the limit may depend on the subsequence $\varepsilon' \to 0$, yielding different homogenized operators. Periodicity ensures uniqueness.

2.4 Asymptotic Expansions

2.4.1 Orientation

This section presents the multiple-scale asymptotic expansion method, a powerful formal procedure for deriving homogenized equations and effective coefficients. While the formulas obtained are correct, their rigorous justification often requires additional functional analytic tools.

2.4.2 The Method of Multiple Scales

The core idea is to assume the solution depends on two variables:

i. The **slow (macroscopic) variable**: x

ii. The **fast (microscopic) variable**: $y = x/\varepsilon$

We seek a solution $u_\varepsilon(x)$ in the form of an asymptotic expansion:

$$u_\varepsilon(x) = u_0(x, x/\varepsilon) + \varepsilon u_1(x, x/\varepsilon) + \varepsilon^2 u_2(x, x/\varepsilon) + \cdots, \tag{2.25}$$

where each function $u_j(x, y)$ is Y-periodic in y.

Remark. It is technically complicated to handle boundary conditions within this ansatz, which is a source of serious difficulties in rigorously justifying the method. Nevertheless, the method yields the correct homogenized operator.

1. Transformation of Derivatives and Operator Decomposition.

1.1 Chain Rule. For a function $\phi(x, y)$ with $y = x/\varepsilon$, the derivative operator becomes:

$$\frac{\partial}{\partial x_j}\phi(x, x/\varepsilon) = \left(\frac{\partial \phi}{\partial x_j} + \frac{1}{\varepsilon}\frac{\partial \phi}{\partial y_j}\right)\Bigg|_{y=x/\varepsilon}. \tag{2.26}$$

1.2 Operator Expansion. This leads to a decomposition of the differential operator A^ε into powers of ε:

$$A^\varepsilon = \varepsilon^{-2}A_1 + \varepsilon^{-1}A_2 + \varepsilon^0 A_3, \tag{2.27}$$

where

$$A_1 = -\frac{\partial}{\partial y_i}\left(a_{ij}(y)\frac{\partial}{\partial y_j}\right), \tag{2.28}$$

$$A_2 = -\frac{\partial}{\partial y_i}\left(a_{ij}(y)\frac{\partial}{\partial x_j}\right) - \frac{\partial}{\partial x_i}\left(a_{ij}(y)\frac{\partial}{\partial y_j}\right), \tag{2.29}$$

$$A_3 = -\frac{\partial}{\partial x_i}\left(a_{ij}(y)\frac{\partial}{\partial x_j}\right) + a_0(y). \tag{2.30}$$

2. Hierarchy of Equations.

2.1 Substitution. We substitute the asymptotic expansion (2.25), $u_\varepsilon = u_0(x, y) + \varepsilon u_1(x, y) + \varepsilon^2 u_2(x, y) + \cdots$ (where $y = x/\varepsilon$), and the decomposed operator $A^\varepsilon = \varepsilon^{-2}A_1 + \varepsilon^{-1}A_2 + \varepsilon^0 A_3$ into the original equation $A^\varepsilon u_\varepsilon = f$:

$$\left(\varepsilon^{-2}A_1 + \varepsilon^{-1}A_2 + A_3\right)\left(u_0 + \varepsilon u_1 + \varepsilon^2 u_2 + \varepsilon^3 u_3 + \cdots\right) = f(x). \tag{2.31}$$

2.2 Full Distribution (The "Cross-Multiplication"). We now perform the multiplication on the left-hand side of (2.31). This involves multiplying each term in the first parenthesis by each term in the second parenthesis:

$$\begin{aligned}
&\left(\varepsilon^{-2}A_1\right)(u_0) + \left(\varepsilon^{-2}A_1\right)(\varepsilon u_1) + \left(\varepsilon^{-2}A_1\right)(\varepsilon^2 u_2) + \left(\varepsilon^{-2}A_1\right)(\varepsilon^3 u_3) + \cdots \\
&+ \left(\varepsilon^{-1}A_2\right)(u_0) + \left(\varepsilon^{-1}A_2\right)(\varepsilon u_1) + \left(\varepsilon^{-1}A_2\right)(\varepsilon^2 u_2) + \left(\varepsilon^{-1}A_2\right)(\varepsilon^3 u_3) + \cdots \\
&+ (A_3)(u_0) + (A_3)(\varepsilon u_1) + (A_3)(\varepsilon^2 u_2) + (A_3)(\varepsilon^3 u_3) + \cdots .
\end{aligned} \tag{2.32}$$

2.3 Simplifying Each Product. We simplify by combining the powers of ε in each product:

$$\begin{aligned}
&\varepsilon^{-2}A_1u_0 \quad + \varepsilon^{-1}A_1u_1 \quad + \varepsilon^0 A_1u_2 \quad + \varepsilon^1 A_1u_3 + \cdots \\
&+\varepsilon^{-1}A_2u_0 \quad + \varepsilon^0 A_2u_1 \quad + \varepsilon^1 A_2u_2 \quad + \varepsilon^2 A_2u_3 + \cdots \\
&+\varepsilon^0 A_3u_0 \quad + \varepsilon^1 A_3u_1 \quad + \varepsilon^2 A_3u_2 \quad + \varepsilon^3 A_3u_3 + \cdots .
\end{aligned} \tag{2.33}$$

2.4 Grouping by Powers of ε. The final and crucial step is to collect all terms that share the

same power of ε. This organizes the sum vertically by order of magnitude:

$$\begin{aligned}
\varepsilon^{-2}: &\quad A_1 u_0 \\
\varepsilon^{-1}: &\quad A_1 u_1 + A_2 u_0 \\
\varepsilon^{0}: &\quad A_1 u_2 + A_2 u_1 + A_3 u_0 \\
\varepsilon^{1}: &\quad A_1 u_3 + A_2 u_2 + A_3 u_1 \\
\varepsilon^{2}: &\quad A_1 u_4 + A_2 u_3 + A_3 u_2 \\
&\quad \vdots
\end{aligned} \tag{2.34}$$

2.5 Asymptotic Matching. For the expansion to hold as $\varepsilon \to 0$, the equation must balance at each order of ε independently. The right-hand side is $f(x) = \mathcal{O}(\varepsilon^0)$.

2.6 Hierarchy of Equations. Equating coefficients for each power of ε yields the necessary system:

$$\mathcal{O}(\varepsilon^{-2}): \quad A_1 u_0 = 0, \tag{2.35}$$

$$\mathcal{O}(\varepsilon^{-1}): \quad A_1 u_1 + A_2 u_0 = 0, \tag{2.36}$$

$$\mathcal{O}(\varepsilon^{0}): \quad A_1 u_2 + A_2 u_1 + A_3 u_0 = f. \tag{2.37}$$

The $\mathcal{O}(\varepsilon^{-2})$ and $\mathcal{O}(\varepsilon^{-1})$ terms must vanish to prevent singularity. The $\mathcal{O}(\varepsilon^0)$ equation incorporates the source f and is sufficient to determine the homogenized limit for u_0.

3. Solvability Condition and the Cell Problems.

3.1 Function Space. The analysis of the hierarchy relies on the solvability condition for the operator A_1 on the unit cell Y. Let $W(Y) = \{\phi \in H^1(Y) \mid \phi \text{ is } Y\text{-periodic}\}$. This space enforces both the periodicity of our solutions and the necessary regularity.

3.2 Fredholm Alternative [4]**.** The operator A_1 has a nontrivial kernel: $A_1(c) = 0$ for any constant function c. Consequently, for $F \in L^2(Y)$, the equation

$$A_1 \phi = F \quad \text{in } Y, \quad \phi \in W(Y), \tag{2.38}$$

admits a solution (unique up to an additive constant) *if and only if* the forcing F is orthogonal to this kernel. This yields the solvability condition:

$$\int_Y F(y)dy = 0. \tag{2.39}$$

This condition is the key mechanism that will ultimately generate the homogenized equation from the $\mathcal{O}(\varepsilon^0)$ problem.

[4] A fundamental principle stating that for this type of equation, a solution exists *if and only if* the forcing term F is orthogonal to all solutions of the homogeneous adjoint equation.

4. Solution of the Hierarchy.

4.1 Leading Order Term (ε^{-2}). The equation from the ε^{-2} terms is:

$$A_1 u_0 = -\frac{\partial}{\partial y_i}\left(a_{ij}(y)\frac{\partial u_0}{\partial y_j}\right) = 0. \tag{2.40}$$

This is a PDE solely in the fast variable y, with x acting as a parameter. We seek solutions that are Y-periodic in y.

4.1.1 **Multiply and Integrate.** Multiply (2.40) by u_0 itself and integrate over the unit cell Y:

$$-\int_Y u_0 \frac{\partial}{\partial y_i}\left(a_{ij}(y)\frac{\partial u_0}{\partial y_j}\right) dy = 0. \tag{2.41}$$

4.1.2 **Integration by Parts.** Apply the multidimensional integration by parts formula (derived from the divergence theorem)[5]:

$$-\int_Y u_0 \frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial u_0}{\partial y_j}\right) dy = \int_Y \frac{\partial u_0}{\partial y_i}\left(a_{ij}\frac{\partial u_0}{\partial y_j}\right) dy - \int_{\partial Y} u_0 \left(a_{ij}\frac{\partial u_0}{\partial y_j}\right) n_i dS, \tag{2.42}$$

where n_i is the outward unit normal. The boundary term vanishes due to Y-periodicity (contributions from opposite sides cancel exactly), leaving:

$$\int_Y a_{ij}(y)\frac{\partial u_0}{\partial y_j}\frac{\partial u_0}{\partial y_i} dy = 0. \tag{2.43}$$

4.1.3 **Apply Ellipticity.** The integrand is a sum over indices i and j:

$$a_{ij}(y)\frac{\partial u_0}{\partial y_j}\frac{\partial u_0}{\partial y_i} = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}(y)\frac{\partial u_0}{\partial y_i}\frac{\partial u_0}{\partial y_j}. \tag{2.44}$$

This double sum is equivalent to a matrix-vector product. If we define:

i. The gradient vector: $\nabla_y u_0 = \begin{pmatrix} \frac{\partial u_0}{\partial y_1} & \cdots & \frac{\partial u_0}{\partial y_n} \end{pmatrix}^T$

ii. The coefficient matrix: $\mathbf{a}(y) = \begin{pmatrix} a_{11}(y) & \cdots & a_{1n}(y) \\ \vdots & \ddots & \vdots \\ a_{n1}(y) & \cdots & a_{nn}(y) \end{pmatrix}$

then the product $(\nabla_y u_0)^T \mathbf{a}(y)(\nabla_y u_0)$ expands exactly to our double sum:

$$(\nabla_y u_0)^T \mathbf{a}(y)(\nabla_y u_0) = \begin{pmatrix} \frac{\partial u_0}{\partial y_1} & \cdots & \frac{\partial u_0}{\partial y_n} \end{pmatrix} \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} \frac{\partial u_0}{\partial y_1} \\ \vdots \\ \frac{\partial u_0}{\partial y_n} \end{pmatrix} = \sum_{i=1}^{n}\sum_{j=1}^{n} \frac{\partial u_0}{\partial y_i} a_{ij} \frac{\partial u_0}{\partial y_j}. \tag{2.45}$$

[5]For a vector field F_i, the divergence theorem gives $\int_Y \frac{\partial F_i}{\partial y_i} dy = \int_{\partial Y} F_i n_i dS$.

Using the Einstein summation convention[6], we can write compactly:

$$a_{ij}(y)\frac{\partial u_0}{\partial y_j}\frac{\partial u_0}{\partial y_i} = \left(\nabla_y u_0\right)^T \mathbf{a}(y)\left(\nabla_y u_0\right). \tag{2.46}$$

4.1.4 **Deduce Independence from Fast Variable.** From the previous step, we have:

$$0 \leq \alpha \int_Y |\nabla_y u_0|^2 dy \leq \int_Y (\nabla_y u_0)^T \mathbf{a}(y)(\nabla_y u_0) dy = 0. \tag{2.47}$$

The first inequality follows from $\alpha > 0$ and $|\nabla_y u_0|^2 \geq 0$; the second from ellipticity and the integration by parts result. This implies:

$$\int_Y |\nabla_y u_0|^2 dy = 0. \tag{2.48}$$

Since the integrand $|\nabla_y u_0|^2$ is non-negative, this forces:

$$|\nabla_y u_0(y)|^2 = 0 \quad \text{a.e. in } Y. \tag{2.49}$$

Consequently, the gradient itself must vanish:

$$\nabla_y u_0 = 0 \quad \text{a.e. in } Y. \tag{2.50}$$

A function with zero gradient is constant. Therefore, u_0 cannot depend on y and must be of the form:

$$u_0(x, y) = u(x). \tag{2.51}$$

This is a crucial result: the leading-order term is independent of the fast microscopic oscillations.

4.2 First-Order Correction (ε^{-1}). The equation from the ε^{-1} terms is:

$$A_1 u_1 + A_2 u_0 = 0. \tag{2.52}$$

This equation determines the first corrector, u_1. We now substitute our solution for u_0, which is independent of y ($u_0(x, y) = u(x)$), into the operator A_2:

$$\begin{aligned} A_2 u_0 &= \left[-\frac{\partial}{\partial y_i}\left(a_{ij}(y)\frac{\partial}{\partial x_j}\right) - \frac{\partial}{\partial x_i}\left(a_{ij}(y)\frac{\partial}{\partial y_j}\right)\right] u(x) \\ &= -\frac{\partial}{\partial y_i}\left(a_{ij}(y)\frac{\partial u}{\partial x_j}\right) - \frac{\partial}{\partial x_i}\left(a_{ij}(y)\cdot 0\right) \\ &= -\frac{\partial}{\partial y_i} a_{ij}(y)\cdot\frac{\partial u}{\partial x_j}(x). \quad \left(\text{Since } \frac{\partial u}{\partial x_j} \text{ is independent of } y\right) \end{aligned} \tag{2.53}$$

[6]Repeated indices imply summation over those indices. Thus, $a_{ij}\frac{\partial u_0}{\partial y_j}\frac{\partial u_0}{\partial y_i}$ means $\sum_{i=1}^n \sum_{j=1}^n a_{ij}\frac{\partial u_0}{\partial y_j}\frac{\partial u_0}{\partial y_i}$.

Relabeling the summation index $j \to k$:

$$A_2 u_0 = -\frac{\partial a_{ik}(y)}{\partial y_i} \frac{\partial u}{\partial x_k}(x). \tag{2.54}$$

Therefore, equation (2.52) becomes:

$$A_1 u_1 = \frac{\partial a_{ik}(y)}{\partial y_i} \frac{\partial u}{\partial x_k}(x). \tag{2.55}$$

Solvability requires $\int_Y \frac{\partial a_{ik}}{\partial y_i} dy = 0$, which holds by the divergence theorem and Y-periodicity. We therefore define the **cell functions** $\chi^k(y)$ for $k = 1, \ldots, n$ as the Y-periodic solutions (unique up to a constant) of:

$$A_1 \chi^k = A_1 y_j = -\frac{\partial}{\partial y_i} a_{ij}(y), \quad \chi^j \, Y\text{-periodic}. \tag{2.56}$$

By the linearity of A_1, the general solution for u_1 is a linear combination of these cell functions plus an arbitrary function $\tilde{u}_1(x)$ that depends only on the slow variable:

$$u_1(x, y) = -\chi^k(y) \frac{\partial u}{\partial x_k}(x) + \tilde{u}_1(x). \tag{2.57}$$

The minus sign ensures positive effective coefficients.

4.3 Homogenized Equation (ε^0). The equation from the ε^0 terms is:

$$A_1 u_2 + A_2 u_1 + A_3 u_0 = f. \tag{2.58}$$

We view this as an equation for u_2 of the form:

$$A_1 u_2 = F \quad \text{where} \quad F = f - A_2 u_1 - A_3 u_0. \tag{2.59}$$

According to the Fredholm alternative (solvability condition (2.39)), for this equation to have a solution, the forcing term F must have zero average over the unit cell:

$$\int_Y F(y) dy = 0 \quad \Longrightarrow \quad \int_Y \left(f - A_2 u_1 - A_3 u_0 \right) dy = 0. \tag{2.60}$$

Rearranging terms and using the linearity of the integral gives:

$$\int_Y \left(A_2 u_1 + A_3 u_0 \right) dy = \int_Y f dy. \tag{2.61}$$

Since $f = f(x)$ is independent of y, its average is straightforward: $\int_Y f(x) dy = f(x)|Y|$. Similarly, since $u_0 = u(x)$ is also independent of y, the term involving $A_3 u_0$ will sim-

plify. Substituting into (2.61) yields the final **homogenized equation**:

$$\frac{1}{|Y|}\int_Y (A_2u_1 + A_3u_0)\,dy = f(x). \tag{2.62}$$

Computing the Integral: We now compute the two terms in the integral separately by substituting our known expressions for $u_0 = u(x)$ and $u_1 = -\chi^\ell \frac{\partial u}{\partial x_\ell} + \tilde{u}_1(x)$.

(a) **Term 1:** $\frac{1}{|Y|}\int_Y A_3u_0dy$

$$\begin{aligned}\frac{1}{|Y|}\int_Y A_3u_0dy &= \frac{1}{|Y|}\int_Y \left[-\frac{\partial}{\partial x_i}\left(a_{ij}(y)\frac{\partial}{\partial x_j}\right) + a_0(y)\right]u(x)dy \\ &= -\frac{\partial}{\partial x_i}\left(\frac{1}{|Y|}\int_Y a_{ij}(y)dy\frac{\partial u}{\partial x_j}\right) + \left(\frac{1}{|Y|}\int_Y a_0(y)dy\right)u(x) \\ &= -\frac{\partial}{\partial x_i}\left(\mathcal{M}(a_{ij})\frac{\partial u}{\partial x_j}\right) + \mathcal{M}(a_0)u(x). \end{aligned}\tag{2.63}$$

(b) **Term 2:** $\frac{1}{|Y|}\int_Y A_2u_1dy$

$$\frac{1}{|Y|}\int_Y A_2u_1dy = \frac{1}{|Y|}\int_Y \left[-\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial u_1}{\partial x_j}\right) - \frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial u_1}{\partial y_j}\right)\right]dy. \tag{2.64}$$

The first term in this integral vanishes by the divergence theorem and Y-periodicity.[7] We are left with:

$$\frac{1}{|Y|}\int_Y A_2u_1dy = -\frac{\partial}{\partial x_i}\left(\frac{1}{|Y|}\int_Y a_{ij}(y)\frac{\partial u_1}{\partial y_j}dy\right). \tag{2.65}$$

Now substitute $u_1 = -\chi^\ell \frac{\partial u}{\partial x_\ell}$ (the term $\tilde{u}_1(x)$ vanishes when differentiated in y_j):

$$\begin{aligned}\frac{1}{|Y|}\int_Y A_2u_1dy &= -\frac{\partial}{\partial x_i}\left(\frac{1}{|Y|}\int_Y a_{ij}(y)\frac{\partial}{\partial y_j}\left(-\chi^\ell(y)\frac{\partial u}{\partial x_\ell}\right)dy\right) \\ &= \frac{\partial}{\partial x_i}\left[\left(\frac{1}{|Y|}\int_Y a_{ij}(y)\frac{\partial \chi^\ell}{\partial y_j}(y)dy\right)\frac{\partial u}{\partial x_\ell}\right]. \end{aligned}\tag{2.66}$$

Combining the Terms: Now plug both terms (a) and (b) back into the solvability condition (2.62) and divide by $|Y|$:

$$\begin{aligned}&\frac{1}{|Y|}\left(\int_Y A_2u_1dy + \int_Y A_3u_0dy\right) = f, \\ &\frac{\partial}{\partial x_i}\left[\frac{1}{|Y|}\int_Y a_{ij}\frac{\partial \chi^\ell}{\partial y_j}dy\frac{\partial u}{\partial x_\ell}\right] - \frac{\partial}{\partial x_i}\left[\mathcal{M}(a_{ij})\frac{\partial u}{\partial x_j}\right] + \mathcal{M}(a_0)u = f. \end{aligned}\tag{2.67}$$

[7] The integral $\int_Y \frac{\partial}{\partial y_i}(\cdots)dy$ equals the flux of the vector field $a_{ij}\frac{\partial u_1}{\partial x_j}$ across ∂Y. Due to periodicity, the contributions from opposite faces cancel exactly.

Combining the two terms with the derivative $\frac{\partial}{\partial x_i}$ and relabeling indices ($\ell \to j$) gives the final **homogenized equation**:

$$-\frac{\partial}{\partial x_i}\left(q_{ij}\frac{\partial u}{\partial x_j}\right) + \mathcal{M}(a_0)u = f, \tag{2.68}$$

where the **effective coefficients** q_{ij} are given by:

$$q_{ij} = \mathcal{M}(a_{ij}) - \frac{1}{|Y|}\int_Y a_{ik}(y)\frac{\partial \chi^j}{\partial y_k}(y)dy = \frac{1}{|Y|}\int_Y \left[a_{ij}(y) - a_{ik}(y)\frac{\partial \chi^j}{\partial y_k}(y)\right]dy. \tag{2.69}$$

Homogenization Procedure Summary. The formal rule to compute the homogenized operator $\mathcal{A}$ is as follows:

1. For each $j = 1, \ldots, n$, solve the cell problem (2.56) for $\chi^j(y)$. Each χ^j represents the microscopic response to a unit macroscopic gradient in direction x_j.
2. Compute the effective coefficients q_{ij} using formula (2.69).
3. Obtain the homogenized solution $u(x)$ by solving the constant-coefficient PDE (2.68).

2.4.3 Properties of the Homogenized Operator

The homogenized operator $\mathcal{A}$ and its coefficients q_{ij} exhibit several important properties.

Remark 2.3. (1D Reduction to Harmonic Mean) The 1D cell problem $-\frac{d}{dy}(a\frac{d\chi}{dy}) = \frac{da}{dy}$ has solution $a\frac{d\chi}{dy} = a + c$. Periodicity forces $c = -1/\mathcal{M}(1/a)$, yielding:

$$q = \mathcal{M}(a) - \mathcal{M}\left(a\frac{d\chi}{dy}\right) = -c = \frac{1}{\mathcal{M}(1/a)}. \tag{2.70}$$

This confirms the effective coefficient is the harmonic mean.

Remark 2.4. (Corrector Structure) The effective coefficients decompose as:

$$q_{ij} = \mathcal{M}(a_{ij}) - \mathcal{M}\left(a_{ik}\frac{\partial \chi^j}{\partial y_k}\right). \tag{2.71}$$

The second term is a microstructural "corrector," explaining why the naive average $\mathcal{M}(a_{ij})$ fails.

Remark 2.5. (Zero-Order Term) The term $\mathcal{M}(a_0)u$ appears directly due to strong convergence: $a_0^\varepsilon \to \mathcal{M}(a_0)$ and $u_\varepsilon \to u$ in $L^2(\mathcal{O})$.

Remark 2.6. (Ellipticity) The tensor q_{ij} is elliptic. The variational formula:

$$q_{ij} = \frac{1}{|Y|}a_1(\chi^j - y_j, \chi^i - y_i) \tag{2.72}$$

implies $q_{ij}\xi_i\xi_j = \frac{1}{|Y|}a_1(w,w) \geq \frac{\alpha}{|Y|}\|\nabla w\|^2_{L^2(Y)} > 0$ for $\boldsymbol{\xi} \neq 0$, where $w = \xi_i(\chi^i - y_i)$.

Remark 2.7. (Symmetry Inheritance) If $a_{ij} = a_{ji}$, then symmetry is preserved:

$$q_{ij} = q_{ji}. \tag{2.73}$$

This follows from the symmetry of the bilinear form $a_1(\cdot, \cdot)$.

Remark 2.8. (Anisotropy Generation) Microscopic isotropy ($a_{ij} = 0$ for $i \neq j$) does not imply macroscopic isotropy. Off-diagonal coefficients:

$$q_{ij} = -\frac{1}{|Y|} \int_Y a_{ii} \frac{\partial \chi^j}{\partial y_i} dy \quad (i \neq j; \text{no summation}) \tag{2.74}$$

are generally nonzero, emerging from microstructural geometry.

Remark 2.9. (Intrinsic Property) The operator $\mathcal{A}$ depends only on microstructural data which are unit cell Y and coefficients $a_{ij}(y)$, $a_0(y)$. It is independent of domain $\mathcal{O}$, source f, and boundary conditions.

2.4.4 Justification of the Asymptotic Expansion for Dirichlet Problems

This section provides a rigorous justification for the formal two-scale expansion, demonstrating that the error between the exact solution u_ε and its expansion is of order $\mathcal{O}(\varepsilon)$ in the L^∞-norm.

1. **Goal.** For the Dirichlet problems

$$A^\varepsilon u_\varepsilon = f \text{ in } \mathcal{O}, \quad u_\varepsilon = 0 \text{ on } \Gamma \quad \text{and} \quad \mathcal{A}u = f \text{ in } \mathcal{O}, \quad u = 0 \text{ on } \Gamma, \tag{2.75}$$

   we prove the uniform convergence $u_\varepsilon \to u$ with the error estimate $\|u_\varepsilon - u\|_{L^\infty} \leq C\varepsilon$, under smoothness assumptions on f, $\mathcal{O}$, and the coefficients.

2. **Error Definition.** We construct an error term that includes the first two terms of the expansion. The second corrector u_2 is included to improve the error estimate:

$$z_\varepsilon(x) = u_\varepsilon(x) - \left[u(x) + \varepsilon u_1\left(x, \frac{x}{\varepsilon}\right) + \varepsilon^2 u_2\left(x, \frac{x}{\varepsilon}\right) \right], \tag{2.76}$$

   where $u_1(x, y) = -\chi^j(y)\partial u/\partial x_j$, and u_2 is uniquely defined by imposing $\int_Y u_2(x, y)dy = 0$.

3. **Error Equation.** By definition, z_ε is 2.76. Apply the operator A^ε to both sides:

$$A^\varepsilon z_\varepsilon = A^\varepsilon u_\varepsilon - A^\varepsilon[u + \varepsilon u_1 + \varepsilon^2 u_2]. \tag{2.77}$$

   To expand the second term, recall $A^\varepsilon = \varepsilon^{-2}A_1 + \varepsilon^{-1}A_2 + A_3$ and substitute the expansion:

$$\begin{aligned} A^\varepsilon[u + \varepsilon u_1 + \varepsilon^2 u_2] = {} & \varepsilon^{-2}A_1 u + \varepsilon^{-1}(A_1 u_1 + A_2 u) \\ & + \varepsilon^0(A_1 u_2 + A_2 u_1 + A_3 u) + \varepsilon^1(A_2 u_2 + A_3 u_1) + \varepsilon^2 A_3 u_2. \end{aligned} \tag{2.78}$$

From the hierarchy of equations derived during the formal expansion, we know:

$$A_1 u = 0, \tag{2.79}$$

$$A_1 u_1 + A_2 u = 0, \tag{2.80}$$

$$A_1 u_2 + A_2 u_1 + A_3 u = f. \tag{2.81}$$

Substituting these identities causes the $\mathcal{O}(\varepsilon^{-2})$, $\mathcal{O}(\varepsilon^{-1})$, and $\mathcal{O}(1)$ terms to cancel, leaving:

$$A^\varepsilon[u + \varepsilon u_1 + \varepsilon^2 u_2] = f + \varepsilon(A_2 u_2 + A_3 u_1) + \varepsilon^2 A_3 u_2. \tag{2.82}$$

where by the original problem, $A^\varepsilon u_\varepsilon = f$. Therefore, the error equation becomes:

$$A^\varepsilon z_\varepsilon = f - \left[f + \varepsilon(A_2 u_2 + A_3 u_1) + \varepsilon^2 A_3 u_2\right] = -\varepsilon r_\varepsilon, \tag{2.83}$$

here we define the residual $r_\varepsilon = A_2 u_2 + A_3 u_1 + \varepsilon A_3 u_2$. For $f \in C^k(\overline{\mathcal{O}})$ with k sufficiently large, the functions u, u_1, u_2 are smooth enough that the residual is bounded: $\|r_\varepsilon\|_{L^\infty} \leq C$, where C depends on the data and the cell functions.

4. **Boundary and Interior Estimates.**

   i. **Boundary:** Since $u_\varepsilon = u = 0$ on Γ,

   $$z_\varepsilon|_\Gamma = -\left(\varepsilon u_1 + \varepsilon^2 u_2\right)|_\Gamma \implies \|z_\varepsilon\|_{L^\infty(\Gamma)} \leq C\varepsilon. \tag{2.84}$$

   ii. **Interior:** Applying the maximum principle for the elliptic operator A^ε gives:

   $$\|z_\varepsilon\|_{L^\infty(\mathcal{O})} \leq \|z_\varepsilon\|_{L^\infty(\Gamma)} + C\|A^\varepsilon z_\varepsilon\|_{L^\infty(\mathcal{O})} \leq C\varepsilon + C\|\varepsilon r_\varepsilon\|_{L^\infty} \leq C\varepsilon. \tag{2.85}$$

5. **Final Estimate.** The triangle inequality then implies the main result:

$$\|u_\varepsilon - u\|_{L^\infty(\mathcal{O})} \leq \|z_\varepsilon\|_{L^\infty} + \varepsilon\|u_1\|_{L^\infty} + \varepsilon^2\|u_2\|_{L^\infty} \leq C\varepsilon. \tag{2.86}$$

6. **Limitations and Remarks.**

Remark 2.10. (Other BCs) The proof relies on $z_\varepsilon|_\Gamma = \mathcal{O}(\varepsilon)$, which holds for Dirichlet but not for Neumann or Robin conditions. The **variational approach** is used instead.

Remark 2.11. (Higher-Order Operators) The maximum principle is for second-order operators. For higher-order problems (e.g., biharmonic), results are obtained via **weak convergence**.

Regularity. The proof assumes high regularity. A general theory using weak solutions minimizes these requirements.

2.4.5 *Higher-Order Corrections in the Expansion*

The asymptotic expansion $u_\varepsilon = u_0 + \varepsilon u_1 + \varepsilon^2 u_2 + \cdots$ can be systematically continued to achieve higher accuracy. The process for the second-order corrector u_2 is as follows:

1. **Define the Tensor b_{ij}.**

$$b_{ij}(y) = a_{ij}(y) - a_{ik}(y)\frac{\partial \chi^j}{\partial y_k} - \frac{\partial}{\partial y_k}(a_{ki}\chi^j). \tag{2.87}$$

We verify that its average gives the homogenized coefficient q_{ij}. By linearity of the average:

$$\mathcal{M}(b_{ij}) = \mathcal{M}(a_{ij}) - \mathcal{M}\left(a_{ik}\frac{\partial \chi^j}{\partial y_k}\right) - \mathcal{M}\left(\frac{\partial}{\partial y_k}(a_{ki}\chi^j)\right). \tag{2.88}$$

The third term is zero by the divergence theorem and Y-periodicity:

$$\mathcal{M}\left(\frac{\partial}{\partial y_k}(a_{ki}\chi^j)\right) = \frac{1}{|Y|}\int_Y \frac{\partial}{\partial y_k}(a_{ki}\chi^j)dy = \frac{1}{|Y|}\int_{\partial Y} a_{ki}\chi^j n_k dS = 0. \tag{2.89}$$

The boundary integral vanishes because the integrand is periodic. Thus:

$$\mathcal{M}(b_{ij}) = \mathcal{M}(a_{ij}) - \mathcal{M}\left(a_{ik}\frac{\partial \chi^j}{\partial y_k}\right) = \frac{1}{|Y|}\int_Y \left(a_{ij} - a_{ik}\frac{\partial \chi^j}{\partial y_k}\right)dy = q_{ij}. \tag{2.90}$$

The homogenized operator is $\mathcal{A}u = -\frac{\partial}{\partial x_i}\left(q_{ij}\frac{\partial u}{\partial x_j}\right)$. Since q_{ij} is constant, this is equivalent to:

$$\mathcal{A}u = -q_{ij}\frac{\partial^2 u}{\partial x_i \partial x_j} = -\mathcal{M}(b_{ij})\frac{\partial^2 u}{\partial x_i \partial x_j} = f. \tag{2.91}$$

2. **Derive the Equation for u_2.** The tensor b_{ij} defined above appears naturally when deriving the equation for the second-order corrector. We start from the $\mathcal{O}(1)$ equation in the hierarchy:

$$A_1 u_2 + A_2 u_1 + A_3 u_0 = f \implies A_1 u_2 = f - A_2 u_1 - A_3 u_0. \tag{2.92}$$

Our goal is to compute the combination $-A_2 u_1 - A_3 u_0$. Recall the definitions:

$$A_2 = -\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial}{\partial x_j}\right) - \frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial}{\partial y_j}\right), \tag{2.93}$$

$$A_3 = -\frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial}{\partial x_j}\right), \quad \text{(assuming } a_0 = 0\text{)} \tag{2.94}$$

$$u_0(x,y) = u(x), \tag{2.95}$$

$$u_1(x,y) = -\chi^k(y)\frac{\partial u}{\partial x_k}(x) + \tilde{u}_1(x). \tag{2.96}$$

2.0 **Recall the Cell Problem for χ^k.** The corrector function χ^k is not arbitrary. It is defined

as the solution to the cell problem derived from the $\mathcal{O}(\varepsilon^{-1})$ equation in the hierarchy. This problem is:

$$A_1\chi^k = \frac{\partial a_{ik}}{\partial y_i} \implies -\frac{\partial}{\partial y_i}\left(a_{ij}(y)\frac{\partial \chi^k}{\partial y_j}\right) = \frac{\partial a_{ik}(y)}{\partial y_i}. \tag{2.97}$$

This result, $A_1\chi^k = \frac{\partial a_{ik}}{\partial y_i}$, will be essential for the cancellation in the next step.

2.1 **Compute Derivatives of** u_1 Applying the chain rule to $u_1(x, x/\varepsilon)$:

$$\frac{\partial u_1}{\partial x_j} = \frac{\partial}{\partial x_j}\left[-\chi^k(y)\frac{\partial u}{\partial x_k}(x)\right] + \frac{\partial \tilde{u}_1}{\partial x_j} = -\left(\frac{\partial \chi^k}{\partial x_j}\right)\frac{\partial u}{\partial x_k} - \chi^k\frac{\partial^2 u}{\partial x_j \partial x_k} + \frac{\partial \tilde{u}_1}{\partial x_j}. \tag{2.98}$$

Using $\frac{\partial \chi^k}{\partial x_j} = \frac{1}{\varepsilon}\frac{\partial \chi^k}{\partial y_j}$, we get:

$$\frac{\partial u_1}{\partial x_j} = -\frac{1}{\varepsilon}\frac{\partial \chi^k}{\partial y_j}\frac{\partial u}{\partial x_k} - \chi^k\frac{\partial^2 u}{\partial x_j \partial x_k} + \frac{\partial \tilde{u}_1}{\partial x_j}. \tag{2.99}$$

The derivative with respect to the fast variable is:

$$\frac{\partial u_1}{\partial y_j} = -\frac{\partial \chi^k}{\partial y_j}\frac{\partial u}{\partial x_k}. \tag{2.100}$$

2.2 **Compute** $-A_3u_0$ **and** $-A_2u_1$

i. For $-A_3u_0$:

$$-A_3u_0 = \frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial u}{\partial x_j}\right) = a_{ij}\frac{\partial^2 u}{\partial x_i \partial x_j} + \frac{\partial a_{ij}}{\partial x_i}\frac{\partial u}{\partial x_j}. \tag{2.101}$$

Note: $\frac{\partial a_{ij}}{\partial x_i} = \frac{1}{\varepsilon}\frac{\partial a_{ij}}{\partial y_i}$ is $\mathcal{O}(1/\varepsilon)$.

ii. For $-A_2u_1$:

$$-A_2u_1 = \frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial u_1}{\partial x_j}\right) + \frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial u_1}{\partial y_j}\right). \tag{2.102}$$

Substitute (2.99) and (2.100):

$$\begin{aligned} -A_2u_1 = &\frac{\partial}{\partial y_i}\left[a_{ij}\left(-\frac{1}{\varepsilon}\frac{\partial \chi^k}{\partial y_j}\frac{\partial u}{\partial x_k} - \chi^k\frac{\partial^2 u}{\partial x_j \partial x_k} + \frac{\partial \tilde{u}_1}{\partial x_j}\right)\right] \\ &+ \frac{\partial}{\partial x_i}\left[a_{ij}\left(-\frac{\partial \chi^k}{\partial y_j}\frac{\partial u}{\partial x_k}\right)\right]. \end{aligned} \tag{2.103}$$

2.3 **Identify and Cancel** $\mathcal{O}(1/\varepsilon)$ **Terms** The $\mathcal{O}(1/\varepsilon)$ terms are:

i. From $-A_2u_1$:

$$-\frac{1}{\varepsilon}\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial \chi^k}{\partial y_j}\frac{\partial u}{\partial x_k}\right) = -\frac{1}{\varepsilon}\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial \chi^k}{\partial y_j}\right)\frac{\partial u}{\partial x_k}. \tag{2.104}$$

ii. From $-A_3 u_0$:

$$\frac{\partial a_{ij}}{\partial x_i}\frac{\partial u}{\partial x_j} = \frac{1}{\varepsilon}\frac{\partial a_{ij}}{\partial y_i}\frac{\partial u}{\partial x_j}. \tag{2.105}$$

Using the cell problem 2.97, we find:

$$\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial \chi^k}{\partial y_j}\right) = -\frac{\partial a_{ik}}{\partial y_i}. \tag{2.106}$$

Thus, the $\mathcal{O}(1/\varepsilon)$ term from $-A_2 u_1$ becomes:

$$\frac{1}{\varepsilon}\frac{\partial a_{ik}}{\partial y_i}\frac{\partial u}{\partial x_k}. \tag{2.107}$$

This exactly cancels the term from $-A_3 u_0$ (after relabeling $j \to k$).

2.4 **Collect Remaining $\mathcal{O}(1)$ Terms** After cancellation, the remaining terms are:

$$\begin{aligned} -A_2 u_1 - A_3 u_0 = &\frac{\partial}{\partial y_i}\left(a_{ij}\chi^k \frac{\partial^2 u}{\partial x_j \partial x_k}\right) - \frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial \tilde{u}_1}{\partial x_j}\right) \\ &- \frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial \chi^k}{\partial y_j}\frac{\partial u}{\partial x_k}\right) - a_{ij}\frac{\partial^2 u}{\partial x_i \partial x_j}. \end{aligned} \tag{2.108}$$

We now simplify this expression.

(a) The term involving $\tilde{u}_1$ simplifies directly. Since $\frac{\partial \tilde{u}_1}{\partial x_j}$ is independent of y:

$$-\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial \tilde{u}_1}{\partial x_j}\right) = -\left(\frac{\partial a_{ij}}{\partial y_i}\right)\frac{\partial \tilde{u}_1}{\partial x_j}. \tag{2.109}$$

(b) For the terms involving u, we focus on the combination:

$$S = \frac{\partial}{\partial y_i}\left(a_{ij}\chi^k \frac{\partial^2 u}{\partial x_j \partial x_k}\right) - \frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial \chi^k}{\partial y_j}\frac{\partial u}{\partial x_k}\right) - a_{ij}\frac{\partial^2 u}{\partial x_i \partial x_j}. \tag{2.110}$$

Since $\frac{\partial^2 u}{\partial x_j \partial x_k}$ is symmetric and independent of y, we factor it out of the first term. In the second term, the derivative $\frac{\partial}{\partial x_i}$ acts only on $\frac{\partial u}{\partial x_k}$ (the other part gives a higher-order contribution), yielding $-a_{ij}\frac{\partial \chi^k}{\partial y_j}\frac{\partial^2 u}{\partial x_i \partial x_k}$. After meticulous index relabeling ($i \leftrightarrow k$, $j \leftrightarrow i$, etc.), the expression S simplifies to:

$$S = -\left[a_{ij} - a_{ik}\frac{\partial \chi^j}{\partial y_k} - \frac{\partial}{\partial y_k}(a_{ki}\chi^j)\right]\frac{\partial^2 u}{\partial x_i \partial x_j} = -b_{ij}\frac{\partial^2 u}{\partial x_i \partial x_j}. \tag{2.111}$$

Combining (a) and (b), we obtain:

$$-A_2 u_1 - A_3 u_0 = -b_{ij}(y)\frac{\partial^2 u}{\partial x_i \partial x_j} - \left(\frac{\partial a_{ij}}{\partial y_i}\right)\frac{\partial \tilde{u}_1}{\partial x_j}. \tag{2.112}$$

Note: The sign of the $\tilde{u}_1$ term may be positive depending on the initial sign convention; the key is the structure.

2.5 **Form the Final Equation for** u_2 Substituting (2.112) into (2.92):

$$\begin{aligned} A_1 u_2 &= f - [-A_2 u_1 - A_3 u_0] \\ &= f - \left[-b_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j} - \left(\frac{\partial a_{ij}}{\partial y_i} \right) \frac{\partial \tilde{u}_1}{\partial x_j} \right] \\ &= f + b_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j} + \left(\frac{\partial a_{ij}}{\partial y_i} \right) \frac{\partial \tilde{u}_1}{\partial x_j}. \end{aligned} \tag{2.113}$$

Finally, using the homogenized equation $f = \mathcal{A}u = -\mathcal{M}(b_{ij}) \frac{\partial^2 u}{\partial x_i \partial x_j}$, we obtain:

$$A_1 u_2 = -\left(b_{ij} - \mathcal{M}(b_{ij}) \right) \frac{\partial^2 u}{\partial x_i \partial x_j} + \left(\frac{\partial a_{ij}}{\partial y_i} \right) \frac{\partial \tilde{u}_1}{\partial x_j}. \tag{2.114}$$

Boundary Condition Challenge. This reveals a core limitation: the **naive expansion violates boundary conditions** at higher orders. It is an interior solution; matching the boundary condition requires a separate **boundary layer analysis**.

2.5 Energy Proof of the Homogenization

2.5.1 Orientation: Statement of the Main Result

This section presents a general proof of homogenization using energy (variational) methods, which applies under minimal regularity assumptions and for a broad class of boundary conditions.

1. **The Homogenized Problem.** We define the homogenized bilinear form using the effective coefficients q_{ij} from the formal expansion:

$$A(u,v) = \int_{\mathcal{O}} q_{ij} \frac{\partial u}{\partial x_j} \frac{\partial v}{\partial x_i} dx, \quad \text{where } q_{ij} = \frac{1}{|Y|} \int_Y \left(a_{ij}(y) - a_{ik}(y) \frac{\partial \chi^j}{\partial y_k}(y) \right) dy. \tag{2.115}$$

Let $u \in V$ be the unique solution of the corresponding homogenized variational problem:

$$u \in V, \quad A(u,v) = (f,v) \quad \forall v \in V. \tag{2.116}$$

2. **Main Convergence Theorem.**

Theorem 3.1 Assume the coefficients a_{ij} and a_0 satisfy the uniform ellipticity and boundedness conditions (2.2), (2.3). Let u_ε be the solution of the original problem (2.15) and u the solution of the homogenized problem (2.116). Then, as $\varepsilon \to 0$:

$$u_\varepsilon \rightharpoonup u \quad \text{weakly in } V. \tag{2.117}$$

3. Key Remarks.

Remark 3.2 (Generality) The coefficients a_{ij} can be discontinuous ($L^\infty(Y)$), meaning the problem can model transmission across cell boundaries $\varepsilon(Y+\zeta)$.

Remark 3.3 (Strong Convergence of Derivatives) The example in Section 1.3 shows that, in general, $\frac{\partial u_\varepsilon}{\partial x_i} \not\to \frac{\partial u}{\partial x_i}$ strongly in L^2. Therefore, the weak convergence in (2.117) is the best one can expect without adding corrector terms.

Remark 3.4 (Generalized Source Terms) The theorem holds in even greater generality. If $f_\varepsilon \to f$ strongly in the dual space V' and u_ε is the solution of

$$a^\varepsilon(u_\varepsilon, v) = (f_\varepsilon, v) \quad \forall v \in V, \tag{2.118}$$

then the conclusion (2.117) still holds, with u now being the solution of $A(u,v) = (f,v)$.

2.5.2 Proof of the Convergence Theorem

We prove the general case from Remark 3.4. The goal is to show that if $f_\varepsilon \to f$ in V' and u_ε solves $a^\varepsilon(u_\varepsilon, v) = (f_\varepsilon, v)$, then $u_\varepsilon \rightharpoonup u$ in V, where u solves the homogenized problem $A(u,v) = (f,v)$.

The key challenge is to identify the weak limit of the flux $\xi_i^\varepsilon = a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}$.

1. **A Priori Estimates and Weak Convergence.** Using $v = u_\varepsilon$ in the variational formulation (2.118) and the coercivity condition yields uniform boundedness:

$$\alpha \|u_\varepsilon\|_V^2 \le a^\varepsilon(u_\varepsilon, u_\varepsilon) = (f_\varepsilon, u_\varepsilon) \le C\|u_\varepsilon\|_V, \quad \|\xi_i^\varepsilon\|_{L^2(\mathcal{O})} \le C. \tag{2.119}$$

Hence, $\|u_\varepsilon\|_V \le C/\alpha$. By the Banach-Alaoglu theorem [8], there exist subsequences such that:

$$u_\varepsilon \rightharpoonup u \quad \text{weakly in } V, \tag{2.120}$$

$$\xi_i^\varepsilon \rightharpoonup \xi_i \quad \text{weakly in } L^2(\mathcal{O}). \tag{2.121}$$

To pass to the limit, we use the flux-based form of the weak formulation. For the principal part of the operator A^ε (i.e., setting $a_0 = 0$ in (2.4)), we have:

$$A^\varepsilon u_\varepsilon = -\frac{\partial}{\partial x_i}\left(a_{ij}\left(\frac{x}{\varepsilon}\right)\frac{\partial u_\varepsilon}{\partial x_j}\right). \tag{2.122}$$

We multiply it by a test function $v \in V$, integrating over $\mathcal{O}$ with integrating by parts:

$$\begin{aligned} a^\varepsilon(u_\varepsilon, v) &= \int_{\mathcal{O}} (A^\varepsilon u_\varepsilon) v dx = \int_{\mathcal{O}} \left[-\frac{\partial}{\partial x_i}\left(a_{ij}\left(\frac{x}{\varepsilon}\right)\frac{\partial u_\varepsilon}{\partial x_j}\right)\right] v dx \\ &= \int_{\mathcal{O}} a_{ij}\left(\frac{x}{\varepsilon}\right)\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial v}{\partial x_i} dx - \int_{\partial\mathcal{O}} a_{ij}\left(\frac{x}{\varepsilon}\right)\frac{\partial u_\varepsilon}{\partial x_j} n_i v dS, \end{aligned} \tag{2.123}$$

[8] This theorem guarantees that bounded sequences in reflexive Banach spaces (like V and L^2) have weakly convergent subsequences.

where n_i is the outward unit normal. Since $v \in V = H_0^1(\mathcal{O})$ vanishes on the boundary $\partial\mathcal{O}$, the boundary term is zero. Thus,

$$a^\varepsilon(u_\varepsilon, v) = \int_{\mathcal{O}} a_{ij}\left(\frac{x}{\varepsilon}\right)\frac{\partial u_\varepsilon}{\partial x_i}\frac{\partial v}{\partial x_j}dx = \int_{\mathcal{O}} \xi_i^\varepsilon \frac{\partial v}{\partial x_i}dx \equiv (\xi_i^\varepsilon, \partial v/\partial x_i), \tag{2.124}$$

where the flux is defined as $\xi_i^\varepsilon = a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}$. Thus, the formulation (2.118) is equivalent to:

$$(\xi_i^\varepsilon, \partial v/\partial x_i) = (f_\varepsilon, v) \quad \forall v \in V. \tag{2.125}$$

Passing to the limit $\varepsilon \to 0$ in (2.125), using the weak convergence of ξ_i^ε, the strong convergence $f_\varepsilon \to f$ in V', and the density of test functions, it yields the macroscopic law:

$$(\xi_i, \partial v/\partial x_i) = (f, v) \quad \forall v \in V. \tag{2.126}$$

This implies $-\mathrm{div}(\xi) = f$ distributionally. The next step is to find $\xi_i = \xi_i(\nabla u)$.

2. **Identify the Flux via Adjoint Correctors.** To identify the effective constitutive law $\xi_i = q_{ij}\frac{\partial u}{\partial x_j}$, we construct a test function w_ε satisfying $(A^\varepsilon)^* w_\varepsilon = 0$. This adjoint-harmonic property will enable cancellations of microscopic oscillations in the duality argument.

   i. **Microscopic (Cell) Variable:** Let $y = x/\varepsilon$ be the fast variable.

   ii. **Target Function:** For direction i, we want a function that behaves like x_i macroscopically but adjusts to the microstructure. Let us postulate:

   $$w_\varepsilon(x) = P(x) - \varepsilon\widehat{\chi}^i(x/\varepsilon) \tag{2.127}$$

   where $P(x) = x_i$ is the macroscopic field, and $\widehat{\chi}^i(y)$ is a periodic corrector that captures micro-oscillations.

   iii. **Adjoint Cell Problem:** We require $(A^\varepsilon)^* w_\varepsilon = 0$. Let's compute this:

   $$\begin{aligned}(A^\varepsilon)^* w_\varepsilon &= -\frac{\partial}{\partial x_k}\left(a_{ki}^\varepsilon \frac{\partial w_\varepsilon}{\partial x_i}\right) = -\frac{\partial}{\partial x_k}\left(a_{ki}^\varepsilon\left(\frac{\partial P}{\partial x_i} - \frac{\partial \widehat{\chi}^i}{\partial y_i}\right)\right) \\ &= -\frac{\partial}{\partial x_k}\left(a_{ki}^\varepsilon\left(\delta_{ij} - \frac{\partial \widehat{\chi}^i}{\partial y_i}\right)\right) \quad (\text{for } P(x) = x_j)\end{aligned} \tag{2.128}$$

   Since the term $a_{ki}^\varepsilon\left(\delta_{ij} - \frac{\partial \widehat{\chi}^i}{\partial y_i}\right)$ depends on $y = x/\varepsilon$, the chain rule gives: $\frac{\partial}{\partial x_k} g(y) = \frac{1}{\varepsilon}\frac{\partial g}{\partial y_k}$. Vanishing of the $\mathcal{O}(\varepsilon^{-1})$ term requires:

   $$-\frac{\partial}{\partial y_k}\left(a_{ki}(y)\left(\delta_{ij} - \frac{\partial \widehat{\chi}^i}{\partial y_i}\right)\right) = 0 \tag{2.129}$$

   More precisely, $\widehat{\chi}^i$ is the Y-periodic solution to the **adjoint cell problem**:

$$A_1^* \widehat{\chi}^i = A_1^* y_i, \quad \text{where } A_1^* = -\frac{\partial}{\partial y_k}\left(a_{ki}(y)\frac{\partial}{\partial y_i}\right) \tag{2.130}$$

is the leading-order adjoint operator acting on the fast variable y.

iv. **Key Property:** By construction, w_ε satisfies:

$$(A^\varepsilon)^* w_\varepsilon = 0 \implies a^\varepsilon(\psi, w_\varepsilon) = 0 \quad \forall \psi \in H_0^1(\mathcal{O}). \tag{2.131}$$

This means that for *any* test function ψ, the bilinear form with w_ε vanishes. Thus, w_ε is in the kernel of the adjoint operator, making it ideal for probing the constitutive law.

3. **The Duality Argument and Key Computation.**

3.0 Let $\phi \in \mathcal{D}(\mathcal{O})$ be a smooth cutoff function. We use $v = \phi w_\varepsilon$ as a test function in the original equation 2.118. The function ϕ localizes the calculation to the support of ϕ and ensures $v \in V$:

$$a^\varepsilon(u_\varepsilon, \phi w_\varepsilon) = (f_\varepsilon, \phi w_\varepsilon). \tag{2.132}$$

Simultaneously, the adjoint property (2.131) holds for any test function. Choosing $\psi = \phi u_\varepsilon \in H_0^1(\mathcal{O})$ gives:

$$a^\varepsilon(\phi u_\varepsilon, w_\varepsilon) = 0. \tag{2.133}$$

Subtract (2.133) from (2.132). This subtraction is designed to cancel the highest-order oscillatory terms involving products of derivatives of u_ε and w_ε, which do not have a weak limit:

$$a^\varepsilon(u_\varepsilon, \phi w_\varepsilon) - a^\varepsilon(\phi u_\varepsilon, w_\varepsilon) = (f_\varepsilon, \phi w_\varepsilon). \tag{2.134}$$

3.1 **Explicit Computation of the Left-Hand Side:** The bilinearity of a^ε allows us to compute the difference:

$$\begin{aligned}
\text{LHS of (2.134)} &= \int_\mathcal{O} a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial(\phi w_\varepsilon)}{\partial x_i} - a_{ij}^\varepsilon \frac{\partial(\phi u_\varepsilon)}{\partial x_j}\frac{\partial w_\varepsilon}{\partial x_i} dx \\
&= \int_\mathcal{O} a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\left(\frac{\partial \phi}{\partial x_i} w_\varepsilon + \phi \frac{\partial w_\varepsilon}{\partial x_i}\right) - a_{ij}^\varepsilon \left(\frac{\partial \phi}{\partial x_j} u_\varepsilon + \phi \frac{\partial u_\varepsilon}{\partial x_j}\right)\frac{\partial w_\varepsilon}{\partial x_i} dx \\
&= \int_\mathcal{O} \xi_i^\varepsilon \frac{\partial \phi}{\partial x_i} w_\varepsilon dx + \int_\mathcal{O} \phi \left[a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial w_\varepsilon}{\partial x_i} - a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial w_\varepsilon}{\partial x_i}\right] dx - \int_\mathcal{O} a_{ij}^\varepsilon \frac{\partial \phi}{\partial x_j} u_\varepsilon \frac{\partial w_\varepsilon}{\partial x_i} dx.
\end{aligned} \tag{2.135}$$

The middle term cancels. Noting that $a_{ij}^\varepsilon \frac{\partial w_\varepsilon}{\partial x_i} = \left(a_{ij}\frac{\partial w}{\partial y_i}\right)^\varepsilon$, we get the final form:

$$\text{LHS of (2.134)} = \int_\mathcal{O} \xi_i^\varepsilon \frac{\partial \phi}{\partial x_i} w_\varepsilon dx - \int_\mathcal{O} \left(a_{ij}\frac{\partial w}{\partial y_i}\right)^\varepsilon \frac{\partial \phi}{\partial x_j} u_\varepsilon dx. \tag{2.136}$$

This is the crucial simplification. The left-hand side is now expressed in terms of the

flux ξ_i^ε, the test function w_ε, and the function $\left(a_{ij}\frac{\partial w}{\partial y_i}\right)^\varepsilon$, all of whose weak limits we can control, multiplied against the smooth function ϕ and its derivatives.

3.2 **Passage to the Limit:** We pass to the limit in (2.134), using (2.136) for the left-hand side.

$$\xi_i^\varepsilon \rightharpoonup \xi_i \quad \text{in } L^2, \tag{2.137}$$

$$w_\varepsilon \to P \quad \text{strongly in } L^2, \tag{2.138}$$

$$\left(a_{ij}\frac{\partial w}{\partial y_i}\right)^\varepsilon \rightharpoonup \mathcal{M}\left(a_{ij}\frac{\partial w}{\partial y_i}\right) \quad \text{weak-}* \text{ in } L^\infty, \tag{2.139}$$

$$u_\varepsilon \to u \quad \text{strongly in } L^2_{\text{loc}}. \tag{2.140}$$

Therefore, the left-hand side (2.136) converges to:

$$\left(\xi_i, \frac{\partial \phi}{\partial x_i}P\right) - \mathcal{M}\left(a_{ij}\frac{\partial w}{\partial y_i}\right)\int_{\mathcal{O}} \frac{\partial \phi}{\partial x_j} u dx. \tag{2.141}$$

The right-hand side of (2.134) converges to $(f, \phi P)$, which, by (2.126), equals $\left(\xi_i, \frac{\partial(\phi P)}{\partial x_i}\right)$.

3.3 **Deduce the Constitutive Law:** Equating the limit of the LHS (2.141) with the limit of the RHS $\left(\xi_i, \frac{\partial(\phi P)}{\partial x_i}\right)$ gives:

$$\left(\xi_i, \frac{\partial \phi}{\partial x_i}P\right) - \mathcal{M}\left(a_{ij}\frac{\partial w}{\partial y_i}\right)\int_{\mathcal{O}} \frac{\partial \phi}{\partial x_j} u dx = \left(\xi_i, \frac{\partial(\phi P)}{\partial x_i}\right) \quad \forall \phi \in \mathcal{D}(\mathcal{O}). \tag{2.142}$$

Expanding the right-hand side using the product rule $\frac{\partial(\phi P)}{\partial x_i} = P\frac{\partial \phi}{\partial x_i} + \phi\frac{\partial P}{\partial x_i}$ and canceling the common term $\left(\xi_i, P\frac{\partial \phi}{\partial x_i}\right)$ from both sides yields:

$$-\mathcal{M}\left(a_{ij}\frac{\partial w}{\partial y_i}\right)\int_{\mathcal{O}} \frac{\partial \phi}{\partial x_j} u dx = \left(\xi_i, \phi\frac{\partial P}{\partial x_i}\right). \tag{2.143}$$

Integrating the left-hand side by parts (noting that ϕ has compact support) and using the definition of the distributional derivative, we obtain:

$$\mathcal{M}\left(a_{ij}\frac{\partial w}{\partial y_i}\right)\int_{\mathcal{O}} \phi\frac{\partial u}{\partial x_j} dx = \int_{\mathcal{O}} \phi\left(\xi_i\frac{\partial P}{\partial x_i}\right) dx. \tag{2.144}$$

Since this holds for all $\phi \in \mathcal{D}(\mathcal{O})$, the fundamental lemma of the calculus of variations [9]implies the pointwise relation almost everywhere:

$$\xi_i\frac{\partial P}{\partial x_i} = \mathcal{M}\left(a_{ij}\frac{\partial w}{\partial y_i}\right)\frac{\partial u}{\partial x_j}. \tag{2.145}$$

Specializing to $P(x) = x_i$ (so $\frac{\partial P}{\partial x_i} = 1$), and use $w(y) = y_i - \widehat{\chi}^i(y)$ gives:

[9]If $\int_\Omega \phi F dx = \int_\Omega \phi G dx$ for all $\phi \in \mathcal{D}(\Omega)$, then $F = G$ almost everywhere.

$$\xi_i = \mathcal{M}\left(a_{kj}\frac{\partial}{\partial y_k}(y_i - \widehat{\chi}^i)\right)\frac{\partial u}{\partial x_j}. \tag{2.146}$$

Let us denote the effective coefficient from this adjoint approach by:

$$\widetilde{q}_{ij} = \mathcal{M}\left(a_{kj}\frac{\partial}{\partial y_k}(y_i - \widehat{\chi}^i)\right). \tag{2.147}$$

Thus, we have $\xi_i = \widetilde{q}_{ij}\frac{\partial u}{\partial x_j}$.

4. **Conclusion and Equivalence.** Substituting the flux relation $\xi_i = \widetilde{q}_{ij}\frac{\partial u}{\partial x_j}$ into the macroscopic conservation law (2.126) shows that u satisfies:

$$\left(\widetilde{q}_{ij}\frac{\partial u}{\partial x_j}, \frac{\partial v}{\partial x_i}\right) = (f, v). \tag{2.148}$$

It remains to verify that $\widetilde{q}_{ij} = q_{ij}$. This equivalence follows from the symmetry between the standard and adjoint cell problems. Test the standard cell problem for χ^j with the adjoint corrector $\widehat{\chi}^i$. By the definition of the adjoint operator and L^2 duality:

$$(A_1^*\widehat{\chi}^i, \chi^j)_Y = (\widehat{\chi}^i, A_1\chi^j)_Y. \tag{2.149}$$

Now compute both sides explicitly. First, the left-hand side:

$$\begin{aligned}(A_1^*\widehat{\chi}^i, \chi^j)_Y &= \int_Y \left[-\frac{\partial}{\partial y_k}\left(a_{ki}\frac{\partial \widehat{\chi}^i}{\partial y_i}\right)\right]\chi^j dy \\ &= \int_Y a_{ki}\frac{\partial \widehat{\chi}^i}{\partial y_i}\frac{\partial \chi^j}{\partial y_k}dy - \int_{\partial Y} a_{ki}\frac{\partial \widehat{\chi}^i}{\partial y_i}\chi^j n_k dS \quad \text{(IBP)} \\ &= \int_Y a_{ki}\frac{\partial \widehat{\chi}^i}{\partial y_i}\frac{\partial \chi^j}{\partial y_k}dy \quad \text{(boundary term vanishes by periodicity).}\end{aligned} \tag{2.150}$$

Next, the right-hand side:

$$\begin{aligned}(\widehat{\chi}^i, A_1\chi^j)_Y &= \int_Y \widehat{\chi}^i\left[-\frac{\partial}{\partial y_i}\left(a_{ik}\frac{\partial \chi^j}{\partial y_k}\right)\right]dy \\ &= \int_Y a_{ik}\frac{\partial \chi^j}{\partial y_k}\frac{\partial \widehat{\chi}^i}{\partial y_i}dy - \int_{\partial Y} a_{ik}\frac{\partial \chi^j}{\partial y_k}\widehat{\chi}^i n_i dS \quad \text{(IBP)} \\ &= \int_Y a_{ik}\frac{\partial \chi^j}{\partial y_k}\frac{\partial \widehat{\chi}^i}{\partial y_i}dy \quad \text{(boundary term vanishes by periodicity).}\end{aligned} \tag{2.151}$$

Equating both sides and relabeling indices ($i \leftrightarrow k$ in the left-hand side), we obtain:

$$\int_Y a_{ik}\frac{\partial \chi^j}{\partial y_k}dy = \int_Y a_{kj}\frac{\partial \widehat{\chi}^i}{\partial y_k}dy. \tag{2.152}$$

This implies the averaging identity:

$$\mathcal{M}\left(a_{ik}\frac{\partial \chi^j}{\partial y_k}\right) = \mathcal{M}\left(a_{kj}\frac{\partial \widehat{\chi}^i}{\partial y_k}\right). \tag{2.153}$$

Therefore, we can show the coefficients are equal by direct computation:

$$\begin{aligned}
\widetilde{q}_{ij} &= \mathcal{M}\left(a_{kj}\frac{\partial}{\partial y_k}(y_i - \widehat{\chi}^i)\right) = \mathcal{M}\left(a_{kj}\left(\frac{\partial y_i}{\partial y_k} - \frac{\partial \widehat{\chi}^i}{\partial y_k}\right)\right) \\
&= \mathcal{M}\left(a_{kj}\delta_{ik} - a_{kj}\frac{\partial \widehat{\chi}^i}{\partial y_k}\right) = \mathcal{M}\left(a_{ij} - a_{kj}\frac{\partial \widehat{\chi}^i}{\partial y_k}\right) \quad \text{(since } a_{kj}\delta_{ik} = a_{ij}\text{)} \\
&= \mathcal{M}(a_{ij}) - \mathcal{M}\left(a_{kj}\frac{\partial \widehat{\chi}^i}{\partial y_k}\right) = \mathcal{M}(a_{ij}) - \mathcal{M}\left(a_{ik}\frac{\partial \chi^j}{\partial y_k}\right) \quad \text{(using (2.152))} \\
&= q_{ij}.
\end{aligned} \tag{2.154}$$

This proves $\widetilde{q}_{ij} = q_{ij}$. Thus, the limit u satisfies the standard homogenized problem (2.116), and by uniqueness, the entire sequence u_ε converges weakly to u.

Remark 3.5. This proof demonstrates the important property that homogenization commutes with the adjoint operation: the homogenization of $(A^\varepsilon)^*$ is $\mathcal{A}^*$.

2.5.3 Alternative Justification via Adjoint Expansion

This section presents an elegant alternative method for justifying homogenization, particularly well-suited for the Dirichlet problem. The core strategy is to use a family of oscillating test functions v_ε, constructed via an asymptotic expansion for the **adjoint operator**, to pass to the limit directly in the weak formulation.

1. **Setup and Strategy.** For the Dirichlet problem ($V = H_0^1(\mathcal{O})$), the space $H^1_{\text{comp}}(\mathcal{O})$ is dense in V, making the following weak formulation equivalent to the original problem:

$$(u_\varepsilon, (A^\varepsilon)^* v) = (f, v) \quad \forall v \in H^1_{\text{comp}}(\mathcal{O}). \tag{2.155}$$

The ingenious idea is to choose the test function v itself to depend on ε. The goal is to construct a family $v_\varepsilon \in H^1_{\text{comp}}(\mathcal{O})$ such that:

$$v_\varepsilon \to v \qquad \text{weakly in } L^2(\mathcal{O}), \tag{2.156}$$

$$(A^\varepsilon)^* v_\varepsilon \to \mathcal{A}^* v \qquad \text{strongly in } H^{-1}(\mathcal{O}), \tag{2.157}$$

for a smooth function $v \in C_0^\infty(\mathcal{O})$. If this is achieved, then passing to the limit in (2.155) (using the weak convergence $u_\varepsilon \rightharpoonup u$ in H_0^1) yields:

$$(u, \mathcal{A}^* v) = (f, v) \quad \forall v \in C_0^\infty(\mathcal{O}), \tag{2.158}$$

which is the weak form of the homogenized equation $\mathcal{A}u = f$.

2. Construction of the Test Functions.

2.1 The Test Function Ansatz. The functions v_ε are constructed using a **truncated asymptotic expansion** for the adjoint operator:

$$v_\varepsilon(x) = v(x) + \varepsilon v_1\left(x, \frac{x}{\varepsilon}\right), \quad v \in C_0^\infty(\mathcal{O}), \tag{2.159}$$

2.2 Adjoint Operator Expansion. Recall that the adjoint operator $(A^\varepsilon)^*$ admits the asymptotic expansion:

$$(A^\varepsilon)^* = \varepsilon^{-2}A_1^* + \varepsilon^{-1}A_2^* + A_3^*, \tag{2.160}$$

where the operators A_n^* are the formal adjoints of A_n, obtained by transposing the coefficient indices $a_{ij} \to a_{ji}$. Their explicit forms are:

$$A_1^* = -\frac{\partial}{\partial y_k}\left(a_{ki}^*(y)\frac{\partial}{\partial y_i}\right), \quad \text{where } a_{ki}^*(y) = a_{ik}(y), \tag{2.161}$$

$$A_2^* = -\frac{\partial}{\partial y_k}\left(a_{ki}^*(y)\frac{\partial}{\partial x_i}\right) - \frac{\partial}{\partial x_k}\left(a_{ki}^*(y)\frac{\partial}{\partial y_i}\right), \tag{2.162}$$

$$A_3^* = -\frac{\partial}{\partial x_k}\left(a_{ki}^*(y)\frac{\partial}{\partial x_i}\right). \tag{2.163}$$

Applying this expansion to the ansatz v_ε gives:

$$(A^\varepsilon)^* v_\varepsilon = \varepsilon^{-2}A_1^* v + \varepsilon^{-1}(A_1^* v_1 + A_2^* v) + (A_2^* v_1 + A_3^* v) + \varepsilon A_3^* v_1. \tag{2.164}$$

2.3 Elimination of Singular Terms and the Cell Problem. To avoid blow-up as $\varepsilon \to 0$, the coefficients of the singular terms ε^{-2} and ε^{-1} must vanish.

i. The ε^{-2} term is $A_1^* v = 0$ (since $v(x)$ is independent of y).

ii. The ε^{-1} term must be set to zero:

$$A_1^* v_1 + A_2^* v = 0. \tag{2.165}$$

This equation determines the corrector function v_1.

2.4 Definition of Adjoint Correctors. The corrector v_1 is solved using the predefined **adjoint correctors** $\hat{\chi}^j$, which satisfy the adjoint cell problem:

$$A_1^* \hat{\chi}^j = A_1^* y_j. \tag{2.166}$$

Computing the right-hand side explicitly gives:

$$A_1^* y_j = -\frac{\partial}{\partial y_k}\left(a_{ki}^* \frac{\partial y_j}{\partial y_i}\right) = -\frac{\partial}{\partial y_k}\left(a_{ki}^* \delta_{ij}\right) = -\frac{\partial a_{kj}^*}{\partial y_k}. \tag{2.167}$$

Therefore, the defining property (2.166) is equivalent to:

$$A_1^* \hat{\chi}^j = -\frac{\partial a_{kj}^*}{\partial y_k}. \tag{2.168}$$

2.5 Solving for v_1. The source term in the cell problem (2.165) is $A_2^* v$. Since $v = v(x)$, we compute:

$$A_2^* v = -\frac{\partial}{\partial y_k}\left(a_{ki}^* \frac{\partial v}{\partial x_i}\right) - \underbrace{\frac{\partial}{\partial x_k}\left(a_{ki}^* \frac{\partial v}{\partial y_i}\right)}_{=0} = -\frac{\partial}{\partial y_k}\left(a_{ki}^* \frac{\partial v}{\partial x_i}\right). \tag{2.169}$$

This can be rewritten using the adjoint cell problem property (2.168):

$$-A_2^* v = \frac{\partial}{\partial y_k}\left(a_{ki}^* \frac{\partial v}{\partial x_i}\right) = -\left(-\frac{\partial a_{ki}^*}{\partial y_k}\right)\frac{\partial v}{\partial x_i} = -\left(A_1^* \hat{\chi}^i\right)\frac{\partial v}{\partial x_i}. \tag{2.170}$$

Since A_1^* acts only on the y-variable and $\frac{\partial v}{\partial x_i}$ is independent of y, we can factor it out:

$$-A_2^* v = A_1^*\left(-\hat{\chi}^i \frac{\partial v}{\partial x_i}\right). \tag{2.171}$$

Comparing this to the cell problem $A_1^* v_1 = -A_2^* v$, we immediately identify the solution:

$$v_1(x, y) = -\hat{\chi}^j(y)\frac{\partial v}{\partial x_j}(x). \tag{2.172}$$

2.6 The Reduced Expansion. With the choice of v_1 from (2.172), the expansion (2.164) reduces to:

$$(A^\varepsilon)^* v_\varepsilon = \underbrace{A_2^* v_1 + A_3^* v}_{\mathcal{O}(1)} + \underbrace{\varepsilon A_3^* v_1}_{\mathcal{O}(\varepsilon)}. \tag{2.173}$$

The problematic singular terms have been eliminated, and the expression remains bounded as $\varepsilon \to 0$.

2.7 Direct Computation of the $\mathcal{O}(1)$ Term. We now compute the explicit form of the leading term $A_2^* v_1 + A_3^* v$.

2.7.1 Compute $A_3^* v$: This is direct from the definition:

$$A_3^* v = -a_{ij}^*(y)\frac{\partial^2 v}{\partial x_i \partial x_j}. \tag{2.174}$$

2.7.2 Compute $A_2^* v_1$: First, compute the spatial derivative of v_1:

$$\frac{\partial v_1}{\partial x_j} = -\hat{\chi}^k(y)\frac{\partial^2 v}{\partial x_j \partial x_k}. \tag{2.175}$$

Now apply the operator A_2^*:

$$A_2^* v_1 = -\frac{\partial}{\partial y_i}\left(a_{ij}^*(y)\frac{\partial v_1}{\partial x_j}\right) = -\frac{\partial}{\partial y_i}\left(a_{ij}^*(y)\left(-\hat{\chi}^k(y)\frac{\partial^2 v}{\partial x_j \partial x_k}\right)\right)$$
$$= \frac{\partial}{\partial y_i}\left(a_{ij}^*(y)\hat{\chi}^k(y)\frac{\partial^2 v}{\partial x_j \partial x_k}\right). \tag{2.176}$$

Since $v = v(x)$ is independent of y, the derivative is factored out:

$$A_2^* v_1 = \frac{\partial}{\partial y_i}\left(a_{ij}^*(y)\hat{\chi}^k(y)\right)\frac{\partial^2 v}{\partial x_j \partial x_k}. \tag{2.177}$$

2.7.3 Combine the Terms:

$$A_2^* v_1 + A_3^* v = \frac{\partial}{\partial y_i}\left(a_{ij}^*(y)\hat{\chi}^k(y)\right)\frac{\partial^2 v}{\partial x_j \partial x_k} - a_{ij}^*(y)\frac{\partial^2 v}{\partial x_i \partial x_j}. \tag{2.178}$$

Relabel indices in the first term ($j \to k, k \to j$) and the second ($i \to k$):

$$= \frac{\partial}{\partial y_i}\left(a_{ik}^*(y)\hat{\chi}^j(y)\right)\frac{\partial^2 v}{\partial x_k \partial x_j} - a_{kj}^*(y)\frac{\partial^2 v}{\partial x_k \partial x_j}. \tag{2.179}$$

Using the commutativity of partial derivatives, we obtain:

$$A_2^* v_1 + A_3^* v = \left[\frac{\partial}{\partial y_i}\left(a_{ik}^*(y)\hat{\chi}^j(y)\right) - a_{kj}^*(y)\right]\frac{\partial^2 v}{\partial x_j \partial x_k}. \tag{2.180}$$

2.7.4 Final Form: Expanding the derivative gives the final expression:

$$A_2^* v_1 + A_3^* v = -\left(a_{ij}^* - a_{ik}^*\frac{\partial \hat{\chi}^j}{\partial y_k}\right)\frac{\partial^2 v}{\partial x_i \partial x_j} + \frac{\partial}{\partial y_i}(a_{ik}^*\hat{\chi}^j)\frac{\partial^2 v}{\partial x_j \partial x_k}. \tag{2.181}$$

2.8 Weak Convergence to the Homogenized Operator. The oscillatory term $A_2^* v_1 + A_3^* v$ is a function of the form $F(x, x/\varepsilon)$, where

$$F(x,y) = \left[\frac{\partial}{\partial y_i}\left(a_{ik}^*(y)\hat{\chi}^j(y)\right) - a_{kj}^*(y)\right]\frac{\partial^2 v}{\partial x_j \partial x_k}(x). \tag{2.182}$$

To establish its weak limit, we test against $\phi \in C_0^\infty(\mathcal{O})$:

$$\int_{\mathcal{O}} \left(A_2^* v_1 + A_3^* v\right)\phi dx = \int_{\mathcal{O}} F\left(x, \frac{x}{\varepsilon}\right)\phi(x)dx. \tag{2.183}$$

By the averaging lemma for periodic functions,

$$F\left(x, \frac{x}{\varepsilon}\right) \rightharpoonup \overline{F}(x) = \frac{1}{|Y|}\int_Y F(x,y)dy \quad \text{weakly in } L^2(\mathcal{O}). \tag{2.184}$$

We compute the average term-by-term. The term involving the divergence vanishes:

$$\frac{1}{|Y|}\int_Y \frac{\partial}{\partial y_i}\left(a^*_{ik}\hat{\chi}^j\right)dy = 0, \tag{2.185}$$

by the divergence theorem and periodicity. Thus, only the second term contributes:

$$\begin{aligned}\overline{F}(x) &= -\frac{1}{|Y|}\int_Y a^*_{kj}(y)dy\,\frac{\partial^2 v}{\partial x_j \partial x_k}(x)\\ &= -\frac{1}{|Y|}\int_Y \left(a^*_{ij}(y) - a^*_{ik}(y)\frac{\partial \hat{\chi}^j}{\partial y_k}(y)\right)dy\,\frac{\partial^2 v}{\partial x_i \partial x_j}(x),\end{aligned} \tag{2.186}$$

where the second equality follows from the final form and the fact that the divergence term averages to zero. This defines the homogenized operator:

$$\mathcal{A}^* v = -\frac{1}{|Y|}\int_Y \left(a^*_{ij} - a^*_{ik}\frac{\partial \hat{\chi}^j}{\partial y_k}\right)dy\,\frac{\partial^2 v}{\partial x_i \partial x_j}. \tag{2.187}$$

Therefore, we have the weak convergence:

$$A^*_2 v_1 + A^*_3 v \to \mathcal{A}^* v \quad \text{in } L^2(\mathcal{O}) \text{ weakly}. \tag{2.188}$$

Since the embedding $H^1_0(\mathcal{O}) \hookrightarrow L^2(\mathcal{O})$ is compact, this implies strong convergence in $H^{-1}(\mathcal{O})$, which is the required result for the test functions.

3. Remarks on the Method.

Remark 3.6 (Boundary Limitations) This method works for Dirichlet problems but fails for others (e.g., Neumann) due to boundary constraints on v_ε. The energy method (Sec. 3.2) is preferred for general cases.

Remark 3.7 (Compactness Requirement) The proof requires the compact embedding $H^1_0 \hookrightarrow L^2$. Without it (e.g., unbounded domains), higher-order correctors are necessary.

Remark 3.8 (General Technique) Using oscillating test functions from formal expansions is a versatile method, applicable beyond periodic homogenization.

2.5.4 Comparison Theorems for Homogenized Coefficients

This section establishes crucial properties of the effective coefficients q_{ij}, namely that they inherit the ellipticity of the original coefficients and satisfy a natural monotonicity property.

Theorem 3.9: Preservation of Ellipticity. The homogenized coefficients q_{ij} satisfy the same uniform ellipticity condition as the original coefficients $a_{ij}(y)$:

$$q_{ij}\xi_i\xi_j \geq \alpha|\xi|^2 \quad \forall \xi \in \mathbb{R}^n, \tag{2.189}$$

where $\alpha > 0$ is the same constant from the assumption $a_{ij}(y)\xi_i\xi_j \geq \alpha|\xi|^2$.

Theorem 3.10. (Tartar's Comparison Theorem). Consider a second family of operators

$$B^\varepsilon = -\frac{\partial}{\partial x_i}\left(b_{ij}(x/\varepsilon)\frac{\partial}{\partial x_j}\right). \tag{2.190}$$

Denote by $[a_{ij}(y)]$, $[b_{ij}(y)]$ the $n \times n$ matrices with coefficients $a_{ij}(y)$, $b_{ij}(y)$, we suppose that

(i) $[b_{ij}]^* = [b_{ij}]$ (i.e., $b_{ij} = b_{ji}$),

(ii) $[b_{ij}] \leq [a_{ij}]$.

Then if $[\mathcal{A}]$, $[\mathcal{B}]$ denote the homogenized matrices, one has $[\mathcal{B}] \leq [\mathcal{A}]$.

Remark 3.11. Theorem 3.9 is a special case of Theorem 3.10. Choose $b_{ij} = \alpha\delta_{ij}$. This matrix is symmetric, and by 2.2, $\alpha|\xi|^2 \leq a_{ij}(y)\xi_i\xi_j$, so $[b_{ij}] \leq [a_{ij}]$. Homogenizing $B^\varepsilon = -\alpha\Delta$ gives $[\mathcal{B}] = \alpha[I]$. Therefore, $[\mathcal{B}] \leq [\mathcal{A}]$ implies $\alpha|\xi|^2 \leq \mathcal{A}_{ij}\xi_i\xi_j$.

Remark 3.12. The proof of Theorem 3.10 is longer than a direct calculation but is chosen for its generality. It extends to the theory of G-convergence (de Giorgi-Spagnolo) for non-periodic coefficients and to other problems discussed later.

Proof of Theorem 3.10. The proof constructs specific sequences of functions and utilizes their convergence properties to compare the homogenized operators.

1. **Construction of the sequence u_ε for operator A^ε.**

   1.1 **Definition:** Given a vector $\lambda \in \mathbb{R}^n$, define the oscillating test function:

   $$u_\varepsilon(x) = \varepsilon u(x/\varepsilon), \quad \text{where} \quad u(y) = \lambda_j(y_j - \chi^j(y)), \tag{2.191}$$

   and χ^j are the corrector functions defined in Section 2.

   1.2 **Property:** By construction, $A^\varepsilon u_\varepsilon = 0$. The justification for this is non-trivial and follows from the definition of the corrector functions χ^j.

   1.2.1 **Compute the Gradient:** Let $y = x/\varepsilon$. The gradient of the test function is calculated as follows:

   $$\frac{\partial u_\varepsilon}{\partial x_k}(x) = \frac{\partial}{\partial x_k}\left[\varepsilon\, u\left(\frac{x}{\varepsilon}\right)\right] = \varepsilon \cdot \frac{\partial u}{\partial y_l}(y) \cdot \frac{\partial y_l}{\partial x_k} = \varepsilon \cdot \frac{\partial u}{\partial y_l}(y) \cdot \frac{1}{\varepsilon}\delta_{kl} = \frac{\partial u}{\partial y_k}(y). \tag{2.192}$$

Therefore,

$$\nabla_x u_\varepsilon(x) = \nabla_y u(y). \tag{2.193}$$

1.2.2 **Apply the Operator:** We compute the core argument of A^ε:

$$a_{ij}(x/\varepsilon)\frac{\partial u_\varepsilon}{\partial x_j} = a_{ij}(y)\frac{\partial u}{\partial y_j}(y). \tag{2.194}$$

Now apply the outer derivative using the chain rule ($y = x/\varepsilon$):

$$\frac{\partial}{\partial x_i}\left[a_{ij}(y)\frac{\partial u}{\partial y_j}(y)\right] = \frac{\partial}{\partial y_k}\left[a_{ij}(y)\frac{\partial u}{\partial y_j}(y)\right]\cdot\frac{\partial y_k}{\partial x_i} = \frac{\partial}{\partial y_i}\left[a_{ij}(y)\frac{\partial u}{\partial y_j}(y)\right]\cdot\frac{1}{\varepsilon}. \tag{2.195}$$

Thus, the full operator gives:

$$A^\varepsilon u_\varepsilon = -\frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial u_\varepsilon}{\partial x_j}\right) = -\frac{1}{\varepsilon}\frac{\partial}{\partial y_i}\left[a_{ij}(y)\frac{\partial u}{\partial y_j}(y)\right]. \tag{2.196}$$

1.2.3 **Substitute the Function $u(y)$:** Recall $u(y) = \lambda_j(y_j - \chi^j(y))$. Its gradient is:

$$\frac{\partial u}{\partial y_j}(y) = \lambda_k\frac{\partial}{\partial y_j}(y_k - \chi^k(y)) = \lambda_k(\delta_{jk} - \frac{\partial \chi^k}{\partial y_j}) = \lambda_j - \lambda_k\frac{\partial \chi^k}{\partial y_j}. \tag{2.197}$$

Substitute this into the expression from 1.2.2:

$$A^\varepsilon u_\varepsilon = -\frac{1}{\varepsilon}\frac{\partial}{\partial y_i}\left[a_{ij}(y)\left(\lambda_j - \lambda_k\frac{\partial \chi^k}{\partial y_j}\right)\right] = -\frac{1}{\varepsilon}\lambda_k\frac{\partial}{\partial y_i}\left[a_{ik}(y) - a_{ij}(y)\frac{\partial \chi^k}{\partial y_j}\right]. \tag{2.198}$$

1.2.4 **Invoke the Cell Problem:** The corrector function χ^k is defined as the solution to the cell problem:

$$-\frac{\partial}{\partial y_i}\left(a_{ij}(y)\frac{\partial}{\partial y_j}(y_k - \chi^k(y))\right) = 0. \tag{2.199}$$

Expanding the derivative $\frac{\partial y_k}{\partial y_j} = \delta_{jk}$, this becomes:

$$-\frac{\partial}{\partial y_i}\left(a_{ik}(y) - a_{ij}(y)\frac{\partial \chi^k}{\partial y_j}\right) = 0. \tag{2.200}$$

This is an identity that holds by the definition of χ^k.

1.2.5 **Final Conclusion:** The expression derived in 1.2.3 is exactly $1/\varepsilon$ times the left-hand side of the cell problem:

$$A^\varepsilon u_\varepsilon = -\frac{1}{\varepsilon}\lambda_k\left[\underbrace{-\frac{\partial}{\partial y_i}\left(a_{ik}(y) - a_{ij}(y)\frac{\partial \chi^k}{\partial y_j}\right)}_{=0}\right] = 0. \tag{2.201}$$

Therefore, $A^\varepsilon u_\varepsilon = 0$ by construction.

1.3 **Convergence Properties:** This sequence of functions u_ε exhibits the following convergence properties as $\varepsilon \to 0$:

$$u_\varepsilon \rightharpoonup u \qquad \text{weakly in } H^1(\mathcal{O}), \tag{2.202}$$

$$\nabla u_\varepsilon \rightharpoonup \lambda \qquad \text{weakly in } (L^2(\mathcal{O}))^n, \tag{2.203}$$

$$[a_{ij}^\varepsilon]\nabla u_\varepsilon \rightharpoonup [\mathcal{A}]\lambda \qquad \text{weakly in } (L^2(\mathcal{O}))^n, \tag{2.204}$$

$$\operatorname{div}\big([a_{ij}^\varepsilon]\nabla u_\varepsilon\big) \to 0 \qquad \text{strongly in } H^{-1}(\mathcal{O}), \tag{2.205}$$

$$[a_{ij}^\varepsilon]\nabla u_\varepsilon \cdot \nabla u_\varepsilon \rightharpoonup [\mathcal{A}]\lambda \cdot \lambda \qquad \text{vaguely.} \tag{2.206}$$

2. Duality formula.

2.1 **Setup:** Let w_ε be a sequence in $H^1(\mathcal{O})$ such that $w_\varepsilon \rightharpoonup w$ weakly in $H^1(\mathcal{O})$.

2.2 **Integral Manipulation:** For any test function $\phi \in C_0^\infty(\mathcal{O})$, consider the integral:

$$\begin{aligned}\int_{\mathcal{O}} \phi \left([a_{ij}^\varepsilon]\nabla u_\varepsilon \cdot \nabla w_\varepsilon\right) dx &= \int_{\mathcal{O}} \phi \left(a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial w_\varepsilon}{\partial x_i}\right) dx = \int_{\mathcal{O}} \left(\phi\, a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right)\frac{\partial w_\varepsilon}{\partial x_i} dx \\ &= -\int_{\mathcal{O}} w_\varepsilon \frac{\partial}{\partial x_i}\left(\phi\, a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right) dx + \int_{\partial\mathcal{O}} \left(\phi\, a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right) w_\varepsilon \nu_i dS \\ &= -\int_{\mathcal{O}} w_\varepsilon \frac{\partial}{\partial x_i}\left(\phi\, a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right) dx = -\int_{\mathcal{O}} \frac{\partial}{\partial x_i}\left(\phi\, a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right) w_\varepsilon dx\end{aligned} \tag{2.207}$$

We now apply the product rule to the derivative term:

$$\begin{aligned}\frac{\partial}{\partial x_i}\left(\phi\, a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right) &= \frac{\partial \phi}{\partial x_i}\left(a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right) + \phi \frac{\partial}{\partial x_i}\left(a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right) \\ &= \frac{\partial \phi}{\partial x_i}\left(a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right) - \phi\, A^\varepsilon u_\varepsilon \quad \left(\text{since } A^\varepsilon u_\varepsilon = -\frac{\partial}{\partial x_i}\left(a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right)\right)\end{aligned} \tag{2.208}$$

Substituting this back into the integral yields the final result:

$$\begin{aligned}\int_{\mathcal{O}} \phi \left([a_{ij}^\varepsilon]\nabla u_\varepsilon \cdot \nabla w_\varepsilon\right) dx &= -\int_{\mathcal{O}} w_\varepsilon \left[\frac{\partial \phi}{\partial x_i}\left(a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right) - \phi\, A^\varepsilon u_\varepsilon\right] dx \\ &= -\int_{\mathcal{O}} \frac{\partial \phi}{\partial x_i}\left(a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right) w_\varepsilon dx + \int_{\mathcal{O}} \phi (A^\varepsilon u_\varepsilon) w_\varepsilon dx.\end{aligned} \tag{2.209}$$

2.3 **Simplification:** Since $A^\varepsilon u_\varepsilon = 0$, the second term vanishes, leaving:

$$\int_{\mathcal{O}} \phi [a_{ij}^\varepsilon]\nabla u_\varepsilon \cdot \nabla w_\varepsilon dx = -\int_{\mathcal{O}} \frac{\partial \phi}{\partial x_i}\left(a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right) w_\varepsilon dx. \tag{2.210}$$

2.4 **Passing to the Limit:** Passing to the limit $\varepsilon \to 0$ and using the weak convergences

(2.204) ($[a_{ij}^\varepsilon]\nabla u_\varepsilon \rightharpoonup [\mathcal{A}]\lambda$) and $w_\varepsilon \rightharpoonup w$, we find:

$$\lim_{\varepsilon\to 0}\int_{\mathcal{O}} \phi[a_{ij}^\varepsilon]\nabla u_\varepsilon \cdot \nabla w_\varepsilon dx = -\int_{\mathcal{O}} \frac{\partial \phi}{\partial x_i}([\mathcal{A}]\lambda)_i w dx. \tag{2.211}$$

2.5 **Integration by Parts:** Applying integration by parts to the right-hand side (treating $([\mathcal{A}]\lambda)_i w$ as the function) yields:

$$\begin{aligned} -\int_{\mathcal{O}} \frac{\partial \phi}{\partial x_i}([\mathcal{A}]\lambda)_i w dx &= \int_{\mathcal{O}} \phi \frac{\partial}{\partial x_i}\left(([\mathcal{A}]\lambda)_i w\right) dx \\ &= \int_{\mathcal{O}} \phi([\mathcal{A}]\lambda)_i \frac{\partial w}{\partial x_i} dx \quad \text{(since } [\mathcal{A}]\lambda \text{ is constant)} \\ &= \int_{\mathcal{O}} \phi\left([\mathcal{A}]\lambda \cdot \nabla w\right) dx. \end{aligned} \tag{2.212}$$

2.6 **Vague Convergence Result:** Therefore, we obtain the vague convergence:

$$[a_{ij}^\varepsilon]\nabla u_\varepsilon \cdot \nabla w_\varepsilon \rightharpoonup [\mathcal{A}]\lambda \cdot \nabla w \quad \text{vaguely.} \tag{2.213}$$

2.7 **Energy Convergence Case:** Taking $w_\varepsilon = u_\varepsilon$ (so $w(x) = \lambda \cdot x$ and $\nabla w = \lambda$) recovers the energy convergence (2.206):

$$[a_{ij}^\varepsilon]\nabla u_\varepsilon \cdot \nabla u_\varepsilon \rightharpoonup [\mathcal{A}]\lambda \cdot \lambda \quad \text{vaguely.} \tag{2.214}$$

3. Analogous construction for B^ε.

3.1 **Definition:** For the operator B^ε with coefficients b_{ij}, define:

$$v_\varepsilon(x) = \varepsilon v(x/\varepsilon), \quad \text{where} \quad v(y) = \lambda_j(y_j - \psi^j(y)), \tag{2.215}$$

and ψ^j are the corrector functions, defined as the solutions (with zero mean) to the cell problems:

$$B_1\psi^j = \frac{\partial}{\partial y_i} b_{ij}(y) \quad \text{on the torus } \mathbb{T}^n.$$

3.2 **Property:** By construction, $B^\varepsilon v_\varepsilon = 0$. The proof is identical to that for $A^\varepsilon u_\varepsilon = 0$:

$$\begin{aligned} B^\varepsilon v_\varepsilon &= -\frac{\partial}{\partial x_i}\left(b_{ij}(x/\varepsilon)\frac{\partial v_\varepsilon}{\partial x_j}\right) = -\frac{\partial}{\partial x_i}\left(b_{ij}(y)\frac{\partial v}{\partial y_j}(y)\right) \quad \text{(where } y = x/\varepsilon) \\ &= -\frac{1}{\varepsilon}\frac{\partial}{\partial y_i}\left(b_{ij}(y)\frac{\partial v}{\partial y_j}(y)\right) = 0. \end{aligned} \tag{2.216}$$

The last equality holds because $v(y) = \lambda_j(y_j - \psi^j(y))$ is constructed to be the solution of the homogeneous equation $B_1 v = 0$.

3.3 **Convergence Properties:** This sequence satisfies analogous convergence properties:

$$v_\varepsilon \rightharpoonup v(x) = \lambda \cdot x \qquad \text{weakly in } H^1(\mathcal{O}), \tag{2.217}$$

$$\nabla v_\varepsilon \rightharpoonup \lambda, \qquad \text{weakly in } H^1(\mathcal{O}) \tag{2.218}$$

$$[b_{ij}^\varepsilon]\nabla v_\varepsilon \rightharpoonup [\mathcal{B}]\lambda, \qquad \text{weakly in } H^1(\mathcal{O}) \tag{2.219}$$

$$[b_{ij}^\varepsilon]\nabla v_\varepsilon \cdot \nabla v_\varepsilon \rightharpoonup [\mathcal{B}]\lambda \cdot \lambda \qquad \text{vaguely.} \tag{2.220}$$

3.4 **Duality Argument:** The same duality argument applies. For any sequence $w_\varepsilon \rightharpoonup w$ weakly in $H^1(\mathcal{O})$ and any $\phi \in C_0^\infty(\mathcal{O})$:

$$\begin{aligned}\int_{\mathcal{O}} \phi \left([b_{ij}^\varepsilon]\nabla v_\varepsilon \cdot \nabla w_\varepsilon\right) dx &= -\int_{\mathcal{O}} \frac{\partial}{\partial x_i}\left(\phi\, b_{ij}^\varepsilon \frac{\partial v_\varepsilon}{\partial x_j}\right) w_\varepsilon dx \quad \text{(Integration by parts)} \\ &= -\int_{\mathcal{O}} \frac{\partial \phi}{\partial x_i}\left(b_{ij}^\varepsilon \frac{\partial v_\varepsilon}{\partial x_j}\right) w_\varepsilon dx + \int_{\mathcal{O}} \phi (B^\varepsilon v_\varepsilon) w_\varepsilon dx \quad \text{(Product rule)} \\ &= -\int_{\mathcal{O}} \frac{\partial \phi}{\partial x_i}\left(b_{ij}^\varepsilon \frac{\partial v_\varepsilon}{\partial x_j}\right) w_\varepsilon dx \quad \text{(Since } B^\varepsilon v_\varepsilon = 0\text{).}\end{aligned} \tag{2.221}$$

Passing to the limit $\varepsilon \to 0$ and using the weak convergences ($[b_{ij}^\varepsilon]\nabla v_\varepsilon \rightharpoonup [\mathcal{B}]\lambda$ and $w_\varepsilon \rightharpoonup w$):

$$\lim_{\varepsilon \to 0} \int_{\mathcal{O}} \phi [b_{ij}^\varepsilon]\nabla v_\varepsilon \cdot \nabla w_\varepsilon dx = -\int_{\mathcal{O}} \frac{\partial \phi}{\partial x_i}([\mathcal{B}]\lambda)_i w dx. \tag{2.222}$$

Applying integration by parts to the right-hand side (as in Step 2.5) yields:

$$-\int_{\mathcal{O}} \frac{\partial \phi}{\partial x_i}([\mathcal{B}]\lambda)_i w dx = \int_{\mathcal{O}} \phi \left([\mathcal{B}]\lambda \cdot \nabla w\right) dx. \tag{2.223}$$

Therefore, we obtain the vague convergence:

$$[b_{ij}^\varepsilon]\nabla v_\varepsilon \cdot \nabla w_\varepsilon \rightharpoonup [\mathcal{B}]\lambda \cdot \nabla w \quad \text{vaguely.} \tag{2.224}$$

4. Quadratic form inequality.

4.1 **Non-Negative Form:** Consider the quadratic form generated by the symmetric matrix $[b_{ij}^\varepsilon]$:

$$X_\varepsilon = [b_{ij}^\varepsilon](\nabla v_\varepsilon - \nabla u_\varepsilon) \cdot (\nabla v_\varepsilon - \nabla u_\varepsilon) \geq 0. \tag{2.225}$$

The non-negativity $X_\varepsilon \geq 0$ follows from the ellipticity and symmetry of b_{ij}.

4.2 **Expansion:** Expanding X_ε in component form using Einstein summation convention:

$$\begin{aligned}X_\varepsilon = [b_{ij}^\varepsilon](\nabla v_\varepsilon - \nabla u_\varepsilon) \cdot (\nabla v_\varepsilon - \nabla u_\varepsilon) &= b_{ij}^\varepsilon \left(\frac{\partial v_\varepsilon}{\partial x_j} - \frac{\partial u_\varepsilon}{\partial x_j}\right)\left(\frac{\partial v_\varepsilon}{\partial x_i} - \frac{\partial u_\varepsilon}{\partial x_i}\right) \\ &= b_{ij}^\varepsilon \left(\frac{\partial v_\varepsilon}{\partial x_j}\frac{\partial v_\varepsilon}{\partial x_i} - \frac{\partial v_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i} - \frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial v_\varepsilon}{\partial x_i} + \frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i}\right).\end{aligned} \tag{2.226}$$

We can now group these terms back into dot product notation:

$$X_\varepsilon = \underbrace{b_{ij}^\varepsilon \frac{\partial v_\varepsilon}{\partial x_j}\frac{\partial v_\varepsilon}{\partial x_i}}_{[b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla v_\varepsilon} - \underbrace{b_{ij}^\varepsilon \frac{\partial v_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i}}_{[b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla u_\varepsilon} - \underbrace{b_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial v_\varepsilon}{\partial x_i}}_{[b_{ij}^\varepsilon]\nabla u_\varepsilon\cdot\nabla v_\varepsilon} + \underbrace{b_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i}}_{[b_{ij}^\varepsilon]\nabla u_\varepsilon\cdot\nabla u_\varepsilon}. \tag{2.227}$$

Thus, we have the intermediate expanded form:

$$X_\varepsilon = [b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla v_\varepsilon - [b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla u_\varepsilon - [b_{ij}^\varepsilon]\nabla u_\varepsilon\cdot\nabla v_\varepsilon + [b_{ij}^\varepsilon]\nabla u_\varepsilon\cdot\nabla u_\varepsilon. \tag{2.228}$$

Now, we use the symmetry of the coefficients $b_{ij}^\varepsilon = b_{ji}^\varepsilon$. This property implies that the two cross terms are actually equal:

$$[b_{ij}^\varepsilon]\nabla u_\varepsilon\cdot\nabla v_\varepsilon = b_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_i}\frac{\partial v_\varepsilon}{\partial x_j} = b_{ji}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_i}\frac{\partial v_\varepsilon}{\partial x_j} = b_{ji}^\varepsilon \frac{\partial v_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i} = [b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla u_\varepsilon. \tag{2.229}$$

Therefore, $-[b_{ij}^\varepsilon]\nabla u_\varepsilon\cdot\nabla v_\varepsilon = -[b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla u_\varepsilon$. Substituting this into the expression above yields the final simplified form:

$$X_\varepsilon = [b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla v_\varepsilon - 2[b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla u_\varepsilon + [b_{ij}^\varepsilon]\nabla u_\varepsilon\cdot\nabla u_\varepsilon. \tag{2.230}$$

4.3 **Applying the Assumption:** The core assumption is that the matrix $[a_{ij}(y) - b_{ij}(y)]$ is positive semi-definite for all y. This means that for any vector $\eta \in \mathbb{R}^n$,

$$[a_{ij}(y) - b_{ij}(y)]\eta_i\eta_j \geq 0 \implies [b_{ij}(y)]\eta_i\eta_j \leq [a_{ij}(y)]\eta_i\eta_j. \tag{2.231}$$

This inequality holds pointwise. Applying it with the specific vector $\eta = \nabla u_\varepsilon(x)$ and using the definition $a_{ij}^\varepsilon(x) = a_{ij}(x/\varepsilon)$, we get the pointwise inequality:

$$[b_{ij}^\varepsilon]\nabla u_\varepsilon\cdot\nabla u_\varepsilon \leq [a_{ij}^\varepsilon]\nabla u_\varepsilon\cdot\nabla u_\varepsilon. \tag{2.232}$$

4.4 **Main Inequality:** Since $X_\varepsilon \geq 0$, we can combine this with the inequality above:

$$\begin{aligned} 0 &\leq X_\varepsilon \\ &= [b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla v_\varepsilon - 2[b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla u_\varepsilon + [b_{ij}^\varepsilon]\nabla u_\varepsilon\cdot\nabla u_\varepsilon \\ &\leq [b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla v_\varepsilon - 2[b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla u_\varepsilon + [a_{ij}^\varepsilon]\nabla u_\varepsilon\cdot\nabla u_\varepsilon. \end{aligned} \tag{2.233}$$

Therefore, we obtain the fundamental inequality:

$$0 \leq [b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla v_\varepsilon - 2[b_{ij}^\varepsilon]\nabla v_\varepsilon\cdot\nabla u_\varepsilon + [a_{ij}^\varepsilon]\nabla u_\varepsilon\cdot\nabla u_\varepsilon. \tag{2.234}$$

5. Limit passage and conclusion.

5.1 **Testing and Limit:** We now pass to the vague limit in the inequality (2.234). For any $\phi \in C_0(\mathcal{O})$ with $\phi \geq 0$, we integrate and take the limit:

$$\begin{aligned} 0 &\leq \lim_{\varepsilon \to 0} \int_{\mathcal{O}} \phi \, X_\varepsilon dx \\ &\leq \lim_{\varepsilon \to 0} \int_{\mathcal{O}} \phi \left([b_{ij}^\varepsilon] \nabla v_\varepsilon \cdot \nabla v_\varepsilon - 2[b_{ij}^\varepsilon] \nabla v_\varepsilon \cdot \nabla u_\varepsilon + [a_{ij}^\varepsilon] \nabla u_\varepsilon \cdot \nabla u_\varepsilon \right) dx. \end{aligned} \tag{2.235}$$

5.2 **Applying Vague Convergences:** Using the vague convergences from Steps 2 and 3:

$$\int \phi [b_{ij}^\varepsilon] \nabla v_\varepsilon \cdot \nabla v_\varepsilon dx \to \int \phi([\mathcal{B}]\lambda \cdot \lambda) dx, \tag{2.236}$$

$$\int \phi [b_{ij}^\varepsilon] \nabla v_\varepsilon \cdot \nabla u_\varepsilon dx \to \int \phi([\mathcal{B}]\lambda \cdot \lambda) dx, \tag{2.237}$$

$$\int \phi [a_{ij}^\varepsilon] \nabla u_\varepsilon \cdot \nabla u_\varepsilon dx \to \int \phi([\mathcal{A}]\lambda \cdot \lambda) dx. \tag{2.238}$$

5.3 **Substituting the Limits:** Substituting these limits yields:

$$0 \leq \int_{\mathcal{O}} \phi \left([\mathcal{B}]\lambda \cdot \lambda - 2[\mathcal{B}]\lambda \cdot \lambda + [\mathcal{A}]\lambda \cdot \lambda \right) dx = \int_{\mathcal{O}} \phi \left(-[\mathcal{B}]\lambda \cdot \lambda + [\mathcal{A}]\lambda \cdot \lambda \right) dx. \tag{2.239}$$

5.4 **Final Inequality:** Since this holds for all non-negative $\phi \in C_0(\mathcal{O})$, it follows that:

$$-[\mathcal{B}]\lambda \cdot \lambda + [\mathcal{A}]\lambda \cdot \lambda \geq 0 \quad \text{for all } \lambda \in \mathbb{R}^n. \tag{2.240}$$

5.5 **Conclusion:** This is equivalent to $[\mathcal{B}] \leq [\mathcal{A}]$, which completes the proof of Theorem 3.10.

2.6 Correctors

2.6.1 Orientation

We return to the framework of Section 2.3. The convergence $u_\varepsilon \to u$ is only weak in V; the difference $u_\varepsilon - u$ does not converge strongly. The goal is to find a **corrector** θ_ε such that:

$$z_\varepsilon = u_\varepsilon - u - \theta_\varepsilon \to 0 \quad \text{strongly in } V. \tag{2.241}$$

We will construct θ_ε to be non-trivial and of order $\mathcal{O}(\varepsilon)$ in $L^2(\mathcal{O})$.

2.6.2 Structure of the First Corrector

The natural first attempt is to use the first-order term from the asymptotic expansion:

$$\theta_\varepsilon = \varepsilon u_1 \left(x, \frac{x}{\varepsilon} \right) = -\varepsilon \chi^j \left(\frac{x}{\varepsilon} \right) \frac{\partial u}{\partial x_j}(x). \tag{2.242}$$

However, if $V = H_0^1(\mathcal{O})$, this θ_ε does not vanish on the boundary Γ. To remedy this, we introduce a smooth **cut-off function** m_ε with the properties:

$$m_\varepsilon \in \mathcal{D}(\mathcal{O}), \quad (\text{or } C^1(\mathcal{O}) \text{ suffices}) \tag{2.243}$$

$$m_\varepsilon(x) = 0 \quad \text{if } d(x,\Gamma) \le \varepsilon, \tag{2.244}$$

$$m_\varepsilon(x) = 1 \quad \text{if } d(x,\Gamma) \ge 2\varepsilon, \tag{2.245}$$

$$|D^\gamma m_\varepsilon(x)| \le C_\gamma\, \varepsilon^{-|\gamma|} \quad \text{for all multi-indices } \gamma. \tag{2.246}$$

We then define the admissible corrector as:

$$\theta_\varepsilon(x) = \varepsilon\, m_\varepsilon(x)\, u_1\left(x, \frac{x}{\varepsilon}\right) (= -\varepsilon m_\varepsilon \chi^j\left(\frac{x}{\varepsilon}\right) \frac{\partial u}{\partial x_j}). \tag{2.247}$$

By construction, $\theta_\varepsilon \in V$.

Theorem 5.1 (Strong Convergence with Corrector). We assume that the hypotheses of Theorem 3.1 hold true. Moreover we assume that

i. $f \in L^2(\mathcal{O})$,

ii. χ^j defined by 2.56 belongs to $W^{1,\infty}(Y)$ for all j,

and the boundary Γ of $\mathcal{O}$ is smooth enough. Then if θ_ε is defined by 2.247 we have

$$z_\varepsilon = u_\varepsilon - u - \theta_\varepsilon \to 0 \quad \text{in } V \text{ strongly.} \tag{2.248}$$

Remark 5.2. The assumption $f \in L^2(\mathcal{O})$ and smoothness of Γ imply by elliptic regularity that $u \in H^2(\mathcal{O})$, ensuring $\theta_\varepsilon \in V$.

Remark 5.3. Without the cut-off ($m_\varepsilon = 1$), the naive corrector $\zeta_\varepsilon = u_\varepsilon - u - \varepsilon u_1$ still converges to zero **strongly in** $H^1(\mathcal{O})$. The cut-off is only necessary to enforce the boundary condition for θ_ε itself when $V \neq H^1(\mathcal{O})$. The difference $z_\varepsilon - \zeta_\varepsilon = \varepsilon(1 - m_\varepsilon)u_1$ is supported in a boundary layer of width $\mathcal{O}(\varepsilon)$ and converges strongly to zero in $H^1(\mathcal{O})$.

Remark 5.4. The corrector θ_ε is a **smoothing operator** that fixes the oscillations in the *derivatives* of $u_\varepsilon - u$. It is distinct from a boundary layer corrector.

Remark 5.5. The scaling of the cut-off layer can be generalized. One can choose m_ε such that $m_\varepsilon(x) = 0$ if $d(x,\Gamma) \le \varepsilon^\lambda$ and $m_\varepsilon(x) = 1$ if $d(x,\Gamma) \ge 2\varepsilon^\lambda$, with $|D^\gamma m_\varepsilon| \le C\varepsilon^{-\lambda|\gamma|}$. Theorem 5.1 remains valid.

Remark 5.6. The corrector is small: $\|\theta_\varepsilon\|_{L^2(\mathcal{O})} = \mathcal{O}(\varepsilon)$.

Remark 5.7. The flux of the corrected approximation converges correctly:

$$a_{ij}^\varepsilon \frac{\partial}{\partial x_j}(u + \theta_\varepsilon) \rightharpoonup q_{ij} \frac{\partial u}{\partial x_j} \quad \text{weakly in } L^2(\mathcal{O}), \tag{2.249}$$

matching the limit of the true flux ξ_i^ε.

2.6.3 *Detailed Proof of Theorem 5.1*

We prove that $z_\varepsilon = u_\varepsilon - u - \theta_\varepsilon \to 0$ in V strongly by showing $a^\varepsilon(z_\varepsilon) \to 0$. The proof is divided into symmetric and non-symmetric cases; we start with the symmetric case.

Symmetric Case ($a_{ij} = a_{ji}$).

1. **Express the energy of the error.** Using the symmetry of the coefficients, we expand the energy of the error $z_\varepsilon = u_\varepsilon - u - \theta_\varepsilon$:

$$\begin{aligned} a^\varepsilon(z_\varepsilon) &= a^\varepsilon(u_\varepsilon - (u + \theta_\varepsilon)) \\ &= a^\varepsilon(u_\varepsilon) + a^\varepsilon(u + \theta_\varepsilon) - 2a^\varepsilon(u_\varepsilon, u + \theta_\varepsilon) \\ &= (f, u_\varepsilon) - 2(f, u + \theta_\varepsilon) + a^\varepsilon(u + \theta_\varepsilon). \end{aligned} \tag{2.250}$$

   Equation (2.250) follows from applying the weak formulation $a^\varepsilon(u_\varepsilon, v) = (f, v)$ for $v = u + \theta_\varepsilon \in V$. Rearranging terms by distributing the inner product gives:

$$a^\varepsilon(z_\varepsilon) = (f, u_\varepsilon - 2u - 2\theta_\varepsilon) + a^\varepsilon(u + \theta_\varepsilon). \tag{2.251}$$

2. **Identify the limit of the first term.** As $\varepsilon \to 0$, we have:

   i. $u_\varepsilon \rightharpoonup u$ weakly in $L^2(\mathcal{O})$,

   ii. $\theta_\varepsilon \to 0$ strongly in $L^2(\mathcal{O})$.

   Therefore, $(f, u_\varepsilon - 2u - 2\theta_\varepsilon) \to (f, -u) = -(f, u)$. Since the homogenized solution satisfies $\mathcal{A}(u, u) = (f, u)$, equation (2.363) implies:

$$a^\varepsilon(z_\varepsilon) \to -(f, u) + \lim_{\varepsilon \to 0} a^\varepsilon(u + \theta_\varepsilon) = -\mathcal{A}(u, u) + \lim_{\varepsilon \to 0} a^\varepsilon(u + \theta_\varepsilon). \tag{2.252}$$

   Hence, to prove $a^\varepsilon(z_\varepsilon) \to 0$, it suffices to show:

$$a^\varepsilon(u + \theta_\varepsilon) \to \mathcal{A}(u, u). \tag{2.253}$$

3. **Compute the gradient of the corrected function.** Recall the corrector definition:

$$\theta_\varepsilon(x) = -\varepsilon\, m_\varepsilon(x)\, \chi^j\left(\frac{x}{\varepsilon}\right) \frac{\partial u}{\partial x_j}(x). \tag{2.254}$$

   Differentiating yields:

$$\frac{\partial \theta_\varepsilon}{\partial x_k} = -\frac{\partial}{\partial x_k}\left(\varepsilon m_\varepsilon \chi^j \frac{\partial u}{\partial x_j}\right) = -\left(\varepsilon \frac{\partial m_\varepsilon}{\partial x_k} \chi^j \frac{\partial u}{\partial x_j} + m_\varepsilon \frac{\partial \chi^j}{\partial y_k} \frac{\partial u}{\partial x_j} + \varepsilon m_\varepsilon \chi^j \frac{\partial^2 u}{\partial x_j \partial x_k}\right). \tag{2.255}$$

We define:

$$\lambda_{\varepsilon k} = m_\varepsilon \left(\frac{\partial \chi^j}{\partial y_k}\right)^\varepsilon \frac{\partial u}{\partial x_j}, \quad r_{\varepsilon k} = -\varepsilon \frac{\partial m_\varepsilon}{\partial x_k} \chi^j \frac{\partial u}{\partial x_j} - \varepsilon m_\varepsilon \chi^j \frac{\partial^2 u}{\partial x_j \partial x_k}. \tag{2.256}$$

Thus, equation (2.255) becomes:

$$\frac{\partial \theta_\varepsilon}{\partial x_k} = -\lambda_{\varepsilon k} + r_{\varepsilon k}. \tag{2.257}$$

Under the theorem's regularity assumptions ($\chi^j \in W^{1,\infty}, u \in H^2$), the remainder $r_{\varepsilon k} \to 0$ strongly in $L^2(\mathcal{O})$.

4. **Write the energy and isolate the main term.** The gradient of the corrected function is:

$$\frac{\partial}{\partial x_k}(u + \theta_\varepsilon) = \frac{\partial u}{\partial x_k} - \lambda_{\varepsilon k} + r_{\varepsilon k}. \tag{2.258}$$

Therefore, its energy is:

$$a^\varepsilon(u + \theta_\varepsilon) = \int_{\mathcal{O}} a^\varepsilon_{k\ell} \left(\frac{\partial u}{\partial x_k} - \lambda_{\varepsilon k} + r_{\varepsilon k}\right) \left(\frac{\partial u}{\partial x_\ell} - \lambda_{\varepsilon \ell} + r_{\varepsilon \ell}\right) dx. \tag{2.259}$$

Since $r_{\varepsilon k} \to 0$ strongly in L^2, the limit of (2.259) is the same as the limit of:

$$X_\varepsilon = \int_{\mathcal{O}} a^\varepsilon_{k\ell} \left(\frac{\partial u}{\partial x_k} - \lambda_{\varepsilon k}\right) \left(\frac{\partial u}{\partial x_\ell} - \lambda_{\varepsilon \ell}\right) dx. \tag{2.260}$$

5. **Expand X_ε and express it in terms of oscillatory integrals.** Expanding the product in (2.260) gives:

$$\begin{aligned} X_\varepsilon = &\int a^\varepsilon_{k\ell} \frac{\partial u}{\partial x_k} \frac{\partial u}{\partial x_\ell} dx - \int a^\varepsilon_{k\ell} \frac{\partial u}{\partial x_k} \lambda_{\varepsilon \ell} dx \\ &- \int a^\varepsilon_{k\ell} \lambda_{\varepsilon k} \frac{\partial u}{\partial x_\ell} dx + \int a^\varepsilon_{k\ell} \lambda_{\varepsilon k} \lambda_{\varepsilon \ell} dx. \end{aligned} \tag{2.261}$$

Substitute the definition $\lambda_{\varepsilon k} = m_\varepsilon \left(\frac{\partial \chi^i}{\partial y_k}\right)^\varepsilon \frac{\partial u}{\partial x_i}$ into each term. Note the indices i, j are summation indices:

$$\begin{aligned} X_\varepsilon = &\int a^\varepsilon_{k\ell} \frac{\partial u}{\partial x_k} \frac{\partial u}{\partial x_\ell} dx - \int \left(a_{k\ell} \frac{\partial \chi^i}{\partial y_\ell}\right)^\varepsilon m_\varepsilon \frac{\partial u}{\partial x_k} \frac{\partial u}{\partial x_i} dx \\ &- \int \left(a_{k\ell} \frac{\partial \chi^j}{\partial y_k}\right)^\varepsilon m_\varepsilon \frac{\partial u}{\partial x_j} \frac{\partial u}{\partial x_\ell} dx + \int \left(a_{k\ell} \frac{\partial \chi^i}{\partial y_k} \frac{\partial \chi^j}{\partial y_\ell}\right)^\varepsilon m_\varepsilon^2 \frac{\partial u}{\partial x_i} \frac{\partial u}{\partial x_j} dx. \end{aligned} \tag{2.262}$$

6. **Pass to the limit $\varepsilon \to 0$ using homogenization.** As $\varepsilon \to 0$, $m_\varepsilon \to 1$ almost everywhere and in L^p. By the theory of oscillatory integrals [10], the limit of each term is the average of its oscillatory coefficient multiplied by the integral of the product of derivatives. Therefore:

$$\lim_{\varepsilon\to 0} X_\varepsilon = \mathcal{M}(a_{k\ell}) \int \frac{\partial u}{\partial x_k}\frac{\partial u}{\partial x_\ell}dx - \mathcal{M}\left(a_{k\ell}\frac{\partial \chi^i}{\partial y_\ell}\right)\int \frac{\partial u}{\partial x_k}\frac{\partial u}{\partial x_i}dx$$
$$- \mathcal{M}\left(a_{k\ell}\frac{\partial \chi^j}{\partial y_k}\right)\int \frac{\partial u}{\partial x_j}\frac{\partial u}{\partial x_\ell}dx + \mathcal{M}\left(a_{k\ell}\frac{\partial \chi^i}{\partial y_k}\frac{\partial \chi^j}{\partial y_\ell}\right)\int \frac{\partial u}{\partial x_i}\frac{\partial u}{\partial x_j}dx. \tag{2.263}$$

7. **Simplify the limit using the cell problem.** The required identity is a direct consequence of the definition of the homogenized coefficients. Recall that the homogenized coefficient q_{kj} is defined by the average:

$$q_{kj} = \mathcal{M}\left(a_{kj} - a_{k\ell}\frac{\partial \chi^j}{\partial y_\ell}\right) = \frac{1}{|Y|}\int_Y \left(a_{kj}(y) - a_{k\ell}(y)\frac{\partial \chi^j}{\partial y_\ell}(y)\right)dy. \tag{2.264}$$

This definition itself arises from the solution to the cell problem. Let us expand this average using the linearity of the mean operator:

$$q_{kj} = \mathcal{M}(a_{kj}) - \mathcal{M}\left(a_{k\ell}\frac{\partial \chi^j}{\partial y_\ell}\right) = \frac{1}{|Y|}\int_Y a_{kj}(y)dy - \frac{1}{|Y|}\int_Y a_{k\ell}(y)\frac{\partial \chi^j}{\partial y_\ell}(y)dy. \tag{2.265}$$

Now, simply rearrange the terms in the equation $q_{kj} = \mathcal{M}(a_{kj}) - \mathcal{M}\left(a_{k\ell}\frac{\partial \chi^j}{\partial y_\ell}\right)$ to solve for the desired average:

$$\mathcal{M}\left(a_{k\ell}\frac{\partial \chi^j}{\partial y_\ell}\right) = \mathcal{M}(a_{kj}) - q_{kj}. \tag{2.266}$$

This is the key identity. It is not an additional result but is embedded in the very definition of the effective coefficient q_{kj}.

We now substitute the identity (2.266) into (2.263). Let $I_{ij} = \int \frac{\partial u}{\partial x_i}\frac{\partial u}{\partial x_j}dx$. The expression becomes:

$$\lim_{\varepsilon\to 0} X_\varepsilon = \mathcal{M}(a_{k\ell})I_{k\ell} - \left(\mathcal{M}(a_{ki}) - q_{ki}\right)I_{ki} - \left(\mathcal{M}(a_{\ell j}) - q_{\ell j}\right)I_{\ell j} + \mathcal{M}\left(a_{k\ell}\frac{\partial \chi^i}{\partial y_k}\frac{\partial \chi^j}{\partial y_\ell}\right)I_{ij}. \tag{2.267}$$

Due to symmetry ($a_{ij} = a_{ji}$, $I_{ij} = I_{ji}$), the terms involving $\mathcal{M}(a_{..})$ cancel:

$$\mathcal{M}(a_{k\ell})I_{k\ell} - \mathcal{M}(a_{ki})I_{ki} - \mathcal{M}(a_{\ell j})I_{\ell j} = -\mathcal{M}(a_{k\ell})I_{k\ell}. \tag{2.268}$$

The terms involving q_{kj} combine as $q_{ki}I_{ki} + q_{\ell j}I_{\ell j} = 2q_{k\ell}I_{k\ell}$. The final term, $\mathcal{M}\left(a_{k\ell}\frac{\partial \chi^i}{\partial y_k}\frac{\partial \chi^j}{\partial y_\ell}\right)I_{ij}$, simplifies to $\left(\mathcal{M}(a_{k\ell}) - q_{k\ell}\right)I_{k\ell}$ through a similar computation. Adding all parts:

$$\left(-\mathcal{M}(a_{k\ell}) + 2q_{k\ell} + \mathcal{M}(a_{k\ell}) - q_{k\ell}\right)I_{k\ell} = q_{k\ell}I_{k\ell} = \mathcal{A}(u,u). \tag{2.269}$$

[10] This is a fundamental result in homogenization theory. For a periodic function $\psi(y)$ and a sufficiently smooth function $\phi(x)$, the oscillatory integral satisfies $\lim_{\varepsilon\to 0}\int_{\mathcal{O}} \psi(x/\varepsilon)\phi(x)dx = \mathcal{M}(\psi)\int_{\mathcal{O}}\phi(x)dx$, where $\mathcal{M}(\psi) = \frac{1}{|Y|}\int_Y \psi(y)dy$ is the average over the period cell Y.

Thus, we have established:

$$\lim_{\varepsilon\to 0} X_\varepsilon = q_{ij}\int \frac{\partial u}{\partial x_i}\frac{\partial u}{\partial x_j}dx = \mathcal{A}(u,u). \tag{2.270}$$

Non-Symmetric Case.

1. **Energy Expansion and Main Goal.** The goal is to prove the strong H^1 convergence of the interior corrector $\zeta_\varepsilon = u_\varepsilon - u - \varepsilon u_1$ by showing its energy vanishes in the limit:

$$a^\varepsilon(\zeta_\varepsilon) \to 0. \tag{2.271}$$

Expanding the energy form by bilinearity yields:

$$\begin{aligned} a^\varepsilon(\zeta_\varepsilon) &= a^\varepsilon(u_\varepsilon - (u+\varepsilon u_1)) \\ &= a^\varepsilon(u_\varepsilon) - a^\varepsilon(u_\varepsilon, u+\varepsilon u_1) - a^\varepsilon(u+\varepsilon u_1, u_\varepsilon) + a^\varepsilon(u+\varepsilon u_1). \end{aligned} \tag{2.272}$$

We analyze the convergence of each term:

(a) **First term:** From the weak formulation, $a^\varepsilon(u_\varepsilon) = (f, u_\varepsilon)$. Since $u_\varepsilon \rightharpoonup u$ weakly in $H_0^1(\mathcal{O})$ and f is fixed:

$$a^\varepsilon(u_\varepsilon) = (f, u_\varepsilon) \to (f, u). \tag{2.273}$$

(b) **Last term:** A standard result in homogenization theory gives the convergence of the energy of the corrected approximation:

$$a^\varepsilon(u+\varepsilon u_1) \to \mathcal{A}(u,u). \tag{2.274}$$

(c) **Conclusion from (a) and (b):** Since u is the solution to the homogenized equation $\mathcal{A}(u,v) = (f,v)$ for all $v \in H_0^1(\mathcal{O})$, we have $(f,u) = \mathcal{A}(u,u)$. Thus:

$$a^\varepsilon(u_\varepsilon) + a^\varepsilon(u+\varepsilon u_1) \to 2\mathcal{A}(u,u). \tag{2.275}$$

Substituting this into (2.272), we find:

$$a^\varepsilon(\zeta_\varepsilon) \to 2\mathcal{A}(u,u) - \left[a^\varepsilon(u_\varepsilon, u+\varepsilon u_1) + a^\varepsilon(u+\varepsilon u_1, u_\varepsilon)\right]. \tag{2.276}$$

Therefore, for $a^\varepsilon(\zeta_\varepsilon) \to 0$, it is necessary and sufficient to prove that the cross terms converge to $\mathcal{A}(u,u)$:

$$a^\varepsilon(u_\varepsilon, u+\varepsilon u_1) \to \mathcal{A}(u,u), \quad a^\varepsilon(u+\varepsilon u_1, u_\varepsilon) \to \mathcal{A}(u,u). \tag{2.277}$$

Proving (2.277) is the central task of the proof. The non-symmetry of $a^\varepsilon(\cdot,\cdot)$ requires both limits to be established separately.

2. **Localization Strategy.** The integrands in the cross terms are products of weakly convergent

sequences, and their convergence is not automatic. To handle this, we introduce a localization.

Let $\phi \in C_0^\infty(\mathcal{O})$ be a cut-off function such that $\phi = 1$ outside a small set E with $|E|$ arbitrarily small. Define a localized bilinear form:

$$a_\phi^\varepsilon(u,v) = \int_\mathcal{O} a_{ij}^\varepsilon \frac{\partial u}{\partial x_j}\frac{\partial}{\partial x_i}(v\phi)dx. \tag{2.278}$$

The key is to show that the difference between the original and localized forms is small *uniformly* in ε:

$$|a^\varepsilon(w,v) - a_\phi^\varepsilon(w,v)| \le C(\varepsilon + |E|^{1/q}) \to 0. \tag{2.279}$$

Applying this to $w = u + \varepsilon u_1$ and $v = u_\varepsilon$ (and vice versa) means that to prove (2.277), it suffices to prove the convergence for the localized forms with a *fixed* ϕ:

$$a_\phi^\varepsilon(u + \varepsilon u_1, u_\varepsilon) \to \mathcal{A}_\phi(u,u), \tag{2.280}$$

$$a_\phi^\varepsilon(u_\varepsilon, u + \varepsilon u_1) \to \mathcal{A}_\phi(u,u). \tag{2.281}$$

3. **Proof of the Localization Estimate.** The proof for $a^\varepsilon(u + \varepsilon u_1, u_\varepsilon) - a_\phi^\varepsilon(u + \varepsilon u_1, u_\varepsilon)$ is shown. The difference is:

$$\int_\mathcal{O} a_{ij}^\varepsilon \frac{\partial}{\partial x_j}(u + \varepsilon u_1)\frac{\partial u_\varepsilon}{\partial x_i}(1-\phi)dx. \tag{2.282}$$

Substituting the corrector $u_1 = -\chi^k \frac{\partial u}{\partial x_k}$ and applying the chain rule yields:

$$\begin{aligned}\frac{\partial}{\partial x_j}(u + \varepsilon u_1) &= \frac{\partial u}{\partial x_j} + \varepsilon\frac{\partial u_1}{\partial x_j} = \frac{\partial u}{\partial x_j} - \varepsilon\frac{\partial}{\partial x_j}\left(\chi^k\frac{\partial u}{\partial x_k}\right)\\ &= \frac{\partial u}{\partial x_j} - \frac{\partial \chi^k}{\partial y_j}\frac{\partial u}{\partial x_k} - \varepsilon\chi^k\frac{\partial^2 u}{\partial x_j \partial x_k},\end{aligned} \tag{2.283}$$

where we used $\frac{\partial}{\partial x_j}\chi^k(x/\varepsilon) = \frac{1}{\varepsilon}\frac{\partial \chi^k}{\partial y_j}$. Substituting this back into (2.282) and relabeling indices gives the two terms:

$$I_1 = \int_\mathcal{O} a_{ij}^\varepsilon\left(\frac{\partial u}{\partial x_i} - \frac{\partial \chi^k}{\partial y_i}\frac{\partial u}{\partial x_k}\right)\frac{\partial u_\varepsilon}{\partial x_j}(1-\phi)dx, \tag{2.284}$$

$$I_2 = -\int_\mathcal{O} a_{ij}^\varepsilon \varepsilon\chi^k\frac{\partial^2 u}{\partial x_i \partial x_k}\frac{\partial u_\varepsilon}{\partial x_j}(1-\phi)dx. \tag{2.285}$$

i. **Term I_2:** Bounded by $C\varepsilon$ due to the uniform boundedness of a_{ij}^ε, $\chi^k \in W^{1,\infty}(Y)$, ∇u_ε in L^2, and $\nabla^2 u \in L^2$ (from H^2 regularity).

ii. **Term I_1:** The quantity $\left(\frac{\partial u}{\partial x_i} - \frac{\partial \chi^k}{\partial y_i}\frac{\partial u}{\partial x_k}\right)$ is bounded. Using the Cauchy-Schwarz inequality

and the fact that $1-\phi$ is supported on E:

$$|I_1| \leq C \left(\int_E |\nabla u_\varepsilon|^2 dx \right)^{1/2}. \tag{2.286}$$

Since ∇u_ε is bounded in L^2, this is $\leq C|E|^{1/2}$. However, the H^2 regularity of u provides a sharper estimate: $\nabla u \in L^p(\mathcal{O})$ for some $p > 2$. By Hölder's inequality with conjugate exponents p and q ($\frac{1}{p} + \frac{1}{q} = \frac{1}{2}$), we get:

$$\left(\int_E |\nabla u|^2 dx \right)^{1/2} \leq |E|^{1/q} \|\nabla u\|_{L^p(\mathcal{O})}. \tag{2.287}$$

Combining the estimates for I_1 and I_2 proves the uniform bound (2.279).

4. Convergence of the Localized Forms. We now prove (2.280) and (2.281) for a fixed ϕ.

4.1 **Proof of** (2.280)**:**

$$\begin{aligned} a^\varepsilon_\phi(u + \varepsilon u_1, u_\varepsilon) &= \int_{\mathcal{O}} a^\varepsilon_{ij} \frac{\partial}{\partial x_j}(u + \varepsilon u_1) \cdot \left[\frac{\partial}{\partial x_i}(u_\varepsilon \phi) - u_\varepsilon \frac{\partial \phi}{\partial x_i} \right] dx \\ &= \underbrace{\int_{\mathcal{O}} A^\varepsilon (u + \varepsilon u_1)(u_\varepsilon \phi) dx}_{\text{Term A}} - \underbrace{\int_{\mathcal{O}} a^\varepsilon_{ij} \frac{\partial}{\partial x_j}(u + \varepsilon u_1) u_\varepsilon \frac{\partial \phi}{\partial x_i} dx}_{\text{Term B}}. \end{aligned}$$

i. **Term A:** By construction, $A^\varepsilon(u + \varepsilon u_1) \rightharpoonup \mathcal{A}u$ weakly in $L^2(\mathcal{O})$. Since $u_\varepsilon \phi \rightharpoonup u\phi$ weakly in $L^2(\mathcal{O})$ (and strongly would also work here), we have:

$$\text{Term A} \to (\mathcal{A}u, u\phi).$$

ii. **Term B:** The flux $a^\varepsilon_{ij} \frac{\partial}{\partial x_j}(u + \varepsilon u_1)$ converges strongly in $L^2(\mathcal{O})$ to $q_{ij} \frac{\partial u}{\partial x_j}$, and $u_\varepsilon \frac{\partial \phi}{\partial x_i}$ converges weakly in $L^2(\mathcal{O})$ to $u \frac{\partial \phi}{\partial x_i}$. Hence, by the convergence of the product of strongly and weakly convergent sequences:

$$\text{Term B} \to \int_{\mathcal{O}} q_{ij} \frac{\partial u}{\partial x_j} u \frac{\partial \phi}{\partial x_i} dx.$$

Therefore, combining the limits:

$$a^\varepsilon_\phi(u + \varepsilon u_1, u_\varepsilon) \to (\mathcal{A}u, u\phi) - \int_{\mathcal{O}} q_{ij} \frac{\partial u}{\partial x_j} u \frac{\partial \phi}{\partial x_i} dx = \mathcal{A}_\phi(u, u).$$

4.2 **Proof of** (2.281)**:**

$$a_\phi^\varepsilon(u_\varepsilon, u+\varepsilon u_1) = \int_{\mathcal{O}} a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\left(\frac{\partial((u+\varepsilon u_1)\phi)}{\partial x_i} - (u+\varepsilon u_1)\frac{\partial \phi}{\partial x_i}\right) dx$$
$$= \underbrace{\int_{\mathcal{O}} (A^\varepsilon u_\varepsilon)(u+\varepsilon u_1)\phi dx}_{\text{Term C}} - \underbrace{\int_{\mathcal{O}} \left(a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\right)(u+\varepsilon u_1)\frac{\partial \phi}{\partial x_i} dx}_{\text{Term D}}. \tag{2.288}$$

iii. **Term C:** Since $A^\varepsilon u_\varepsilon = f$ and $u + \varepsilon u_1 \to u$ strongly in $L^2(\mathcal{O})$, we have:

$$\text{Term C} = (f, (u+\varepsilon u_1)\phi) \to (f, u\phi) = (\mathcal{A}u, u\phi).$$

iv. **Term D:** The flux $\xi_i^\varepsilon = a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}$ converges weakly in $L^2(\mathcal{O})$ to $q_{ij}\frac{\partial u}{\partial x_j}$, and $(u+\varepsilon u_1)\frac{\partial \phi}{\partial x_i} \to u\frac{\partial \phi}{\partial x_i}$ strongly in $L^2(\mathcal{O})$. Hence:

$$\text{Term D} \to \int_{\mathcal{O}} q_{ij}\frac{\partial u}{\partial x_j} u \frac{\partial \phi}{\partial x_i} dx.$$

Combining the limits gives the desired result:

$$a_\phi^\varepsilon(u_\varepsilon, u+\varepsilon u_1) \to (\mathcal{A}u, u\phi) - \int_{\mathcal{O}} q_{ij}\frac{\partial u}{\partial x_j} u \frac{\partial \phi}{\partial x_i} dx = \mathcal{A}_\phi(u,u).$$

5. Conclusion. The convergence results (2.280) and (2.281) for the fixed localized form, combined with the uniform estimate (2.279), imply the original goals (2.277). This ensures $a^\varepsilon(\zeta_\varepsilon) \to 0$, proving the strong convergence of ζ_ε in $H_0^1(\mathcal{O})$.

Corollary 5.9. For any measurable $F \subseteq \mathcal{O}$:

$$\int_F a_{ij}^\varepsilon \frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i} dx \to \int_F q_{ij}\frac{\partial u}{\partial x_j}\frac{\partial u}{\partial x_i} dx.$$

This follows because $a^\varepsilon(u_\varepsilon) \to \mathcal{A}(u,u)$ and the energy density of u_ε is asymptotically close to that of $u + \theta_\varepsilon$, which converges to the homogenized energy density.

2.6.4 First-Order System and Asymptotic Expansion

We consider the equation

$$-\operatorname{div}(a^\varepsilon \nabla u_\varepsilon) + \lambda u_\varepsilon = f, \tag{2.289}$$

where $a^\varepsilon(x) = a(x/\varepsilon) = a(y)$ is a positive-definite matrix, and reformulate it:

$$a^\varepsilon \nabla u_\varepsilon - v_\varepsilon = 0, \tag{2.290}$$
$$-\operatorname{div} v_\varepsilon + \lambda u_\varepsilon = f. \tag{2.291}$$

1. **Asymptotic Expansion and Derivation of the Hierarchy.**

We postulate a two-scale asymptotic expansion for both the solution u_ε and the flux v_ε:

$$\begin{aligned} u_\varepsilon(x) &= u_0\left(x, \frac{x}{\varepsilon}\right) + \varepsilon u_1\left(x, \frac{x}{\varepsilon}\right) + \varepsilon^2 u_2\left(x, \frac{x}{\varepsilon}\right) + \cdots, \\ v_\varepsilon(x) &= v_0\left(x, \frac{x}{\varepsilon}\right) + \varepsilon v_1\left(x, \frac{x}{\varepsilon}\right) + \varepsilon^2 v_2\left(x, \frac{x}{\varepsilon}\right) + \cdots, \end{aligned} \tag{2.292}$$

where $u_j(x,y), v_j(x,y)$ are functions of both the slow variable x and the fast variable $y = x/\varepsilon$, and are Y-periodic in y.

1.1 **Express derivatives using the two-scale operator.** Since $y = x/\varepsilon$, we apply the chain rule:

$$\nabla = \nabla_x + \frac{1}{\varepsilon}\nabla_y, \quad \text{and} \quad \operatorname{div} = \operatorname{div}_x + \frac{1}{\varepsilon}\operatorname{div}_y. \tag{2.293}$$

1.2 **Substitute into Equation** (2.290)**:** $a^\varepsilon \nabla u_\varepsilon - v_\varepsilon = 0$

1.2.1 **Compute** ∇u_ε**:**

$$\begin{aligned} \nabla u_\varepsilon &= \nabla\left(u_0(x,y) + \varepsilon u_1(x,y) + \cdots\right) \\ &= \left(\nabla_x + \frac{1}{\varepsilon}\nabla_y\right)\left(u_0 + \varepsilon u_1 + \cdots\right) \\ &= \frac{1}{\varepsilon}\nabla_y u_0 + \left(\nabla_x u_0 + \nabla_y u_1\right) + \varepsilon\left(\nabla_x u_1 + \nabla_y u_2\right) + \cdots \end{aligned} \tag{2.294}$$

1.2.2 **Compute** $a^\varepsilon \nabla u_\varepsilon$**:**

$$\begin{aligned} a^\varepsilon \nabla u_\varepsilon &= a(y)\left(\frac{1}{\varepsilon}\nabla_y u_0 + \left(\nabla_x u_0 + \nabla_y u_1\right) + \varepsilon\left(\nabla_x u_1 + \nabla_y u_2\right) + \cdots\right) \\ &= \frac{1}{\varepsilon} a\nabla_y u_0 + \left(a\nabla_x u_0 + a\nabla_y u_1\right) + \varepsilon\left(a\nabla_x u_1 + a\nabla_y u_2\right) + \cdots \end{aligned} \tag{2.295}$$

1.2.3 **Write expansion for** v_ε**:**

$$v_\varepsilon = v_0 + \varepsilon v_1 + \varepsilon^2 v_2 + \cdots \tag{2.296}$$

1.2.4 **Form the complete equation and collect powers of** ε**:**

$$a^\varepsilon \nabla u_\varepsilon - v_\varepsilon = \frac{1}{\varepsilon}\left[a\nabla_y u_0\right] + \varepsilon^0\left[a\nabla_x u_0 + a\nabla_y u_1 - v_0\right] \tag{2.297}$$

$$+ \varepsilon^1\left[a\nabla_x u_1 + a\nabla_y u_2 - v_1\right] + \cdots = 0 \tag{2.298}$$

1.2.5 **Set coefficients of each power to zero:**

$$\mathcal{O}(\varepsilon^{-1}): \quad a\nabla_y u_0 = 0, \tag{2.299a}$$

$$\mathcal{O}(\varepsilon^{0}): \quad a\nabla_y u_1 + a\nabla_x u_0 - v_0 = 0, \tag{2.299b}$$

$$\mathcal{O}(\varepsilon^{1}): \quad a\nabla_y u_2 + a\nabla_x u_1 - v_1 = 0 \tag{2.299c}$$

1.3 **Substitute into Equation** (2.291)**:** $-\operatorname{div} v_\varepsilon + \lambda u_\varepsilon = f$

1.3.1 **Compute** $\operatorname{div} v_\varepsilon$**:**

$$\begin{aligned}\operatorname{div} v_\varepsilon &= \left(\operatorname{div}_x + \frac{1}{\varepsilon}\operatorname{div}_y\right)\left(v_0 + \varepsilon v_1 + \varepsilon^2 v_2 + \cdots\right) \\ &= \frac{1}{\varepsilon}\operatorname{div}_y v_0 + \left(\operatorname{div}_x v_0 + \operatorname{div}_y v_1\right) + \varepsilon\left(\operatorname{div}_x v_1 + \operatorname{div}_y v_2\right) + \cdots\end{aligned} \tag{2.300}$$

1.3.2 **Compute** $-\operatorname{div} v_\varepsilon$**:**

$$-\operatorname{div} v_\varepsilon = -\frac{1}{\varepsilon}\operatorname{div}_y v_0 - \left(\operatorname{div}_x v_0 + \operatorname{div}_y v_1\right) - \varepsilon\left(\operatorname{div}_x v_1 + \operatorname{div}_y v_2\right) + \cdots \tag{2.301}$$

1.3.3 **Compute** λu_ε**:**

$$\lambda u_\varepsilon = \lambda u_0 + \varepsilon\lambda u_1 + \varepsilon^2\lambda u_2 + \cdots \tag{2.302}$$

1.3.4 **Form the complete equation and collect powers of** ε**:**

$$\begin{aligned}-\operatorname{div} v_\varepsilon + \lambda u_\varepsilon = \frac{1}{\varepsilon}\left[-\operatorname{div}_y v_0\right] + \varepsilon^0\left[-\operatorname{div}_x v_0 - \operatorname{div}_y v_1 + \lambda u_0\right] \\ + \varepsilon^1\left[-\operatorname{div}_x v_1 - \operatorname{div}_y v_2 + \lambda u_1\right] + \cdots = f(x)\end{aligned} \tag{2.303}$$

1.3.5 **Set coefficients of each power to zero:**

$$\mathcal{O}(\varepsilon^{-1}): \quad -\operatorname{div}_y v_0 = 0, \tag{2.304a}$$

$$\mathcal{O}(\varepsilon^{0}): \quad -\operatorname{div}_x v_0 - \operatorname{div}_y v_1 + \lambda u_0 = f, \tag{2.304b}$$

$$\mathcal{O}(\varepsilon^{1}): \quad -\operatorname{div}_x v_1 - \operatorname{div}_y v_2 + \lambda u_1 = 0 \tag{2.304c}$$

1.4 **Analyze the** $\mathcal{O}(\varepsilon^{-1})$ **equations:**

1.4.1 **From Equation** (2.299a)**:** $a(y)\nabla_y u_0(x,y) = 0$ Since $a(y)$ is invertible for all y, multiply both sides by $a^{-1}(y)$:

$$\nabla_y u_0(x,y) = 0. \tag{2.305}$$

This means u_0 is constant with respect to y. Given Y-periodicity:

$$u_0 = u_0(x) \quad \text{or simply} \quad u_0 = u(x). \tag{2.306}$$

1.4.2 **From Equation** (2.304a)**:** $-\operatorname{div}_y v_0 = 0$ This implies the leading-order flux is divergence-free with respect to the fast variable:

$$\operatorname{div}_y v_0(x,y) = 0. \tag{2.307}$$

1.5 **Derive the key relation from** $\mathcal{O}(\varepsilon^0)$ **equations:** Substitute $u_0 = u(x)$ into Equation (2.299b):

$$a\nabla_y u_1 + a\nabla_x u - v_0 = 0. \tag{2.308}$$

Rearrange to isolate terms involving u_1:

$$a\nabla_y u_1 = v_0 - a\nabla_x u. \tag{2.309}$$

Multiply both sides by $a^{-1}(y)$ to obtain the fundamental relation:

$$\nabla_y u_1 = a^{-1}v_0 - \nabla_x u. \tag{2.310}$$

This equation must be solved for the periodic corrector u_1.

2. **Solvability Conditions and Two Procedures.** Equation (2.310) is solvable for the periodic corrector u_1 if and only if the right-hand side has a vanishing curl:

$$\mathrm{rot}_y(a^{-1}v_0) = 0. \tag{2.311}$$

This condition (2.311), together with (2.304a), determines v_0. There are two natural ways to proceed:

2.1 **Procedure I:** Impose the flux constraint (2.311) first.

2.2 **Procedure II:** Impose the divergence constraint (2.304a) first.

3. **Procedure I: Starting from the Flux Constraint** (2.311)**.**

3.1 **Redefine variable.** Define the scaled flux:

$$w_0 = a^{-1}v_0. \tag{2.312}$$

Condition (2.311) becomes $\mathrm{rot}_y\, w_0 = 0$.

3.2 **Helmholtz decomposition.** This implies w_0 is a gradient. Decompose it into its mean and a fluctuating gradient part:

$$w_0 - \mathcal{M}(w_0) = \nabla_y p, \qquad p(x,y)\ Y\text{-periodic.} \tag{2.313}$$

3.3 **Cell problem.** Using $v_0 = aw_0$ and the divergence constraint (2.304a) ($\mathrm{div}_y\, v_0 = 0$) yields an equation for p:

$$-\mathrm{div}_y(a\nabla_y p) = -\mathrm{div}_y(a\mathcal{M}(w_0)). \tag{2.314}$$

(Note: $\mathrm{div}_y(a\mathcal{M}(w_0)) = 0$ since $\mathcal{M}(w_0)$ is independent of y, making the right-hand side zero).

3.4 **Solvability condition for u_1.** The Fredholm alternative for (2.310) requires its right-hand side to have a specific mean:

$$\mathcal{M}(w_0) = \nabla_x u. \tag{2.315}$$

This identifies the mean of the scaled flux. The cell problem for p is thus the standard one, solved by the corrector χ^j:

$$p = -\chi^j \frac{\partial u}{\partial x_j}. \tag{2.316}$$

3.5 **Homogenized equation.** The solvability condition for (2.304b) (its y-average) provides the macroscopic model:

$$-\operatorname{div}_x \mathcal{M}(v_0) + \lambda u = f. \tag{2.317}$$

Substituting $v_0 = a(\nabla_x u + \nabla_y p) = a(I - \nabla_y \chi)\nabla_x u$ yields the standard homogenized equation:

$$-\operatorname{div}_x([\mathcal{A}]\nabla_x u) + \lambda u = f, \quad \text{where} \quad [\mathcal{A}] = \mathcal{M}(a(I - \nabla_y \chi)). \tag{2.318}$$

4. **Procedure II: Starting from the Divergence Constraint** (2.304a)**.**

4.1 **Splitting.** Decompose the flux v_0 into its mean and a zero-mean fluctuating part:

$$\mathcal{M}(v_0) = \rho(x), \qquad \tilde{v}_0 = v_0 - \rho. \tag{2.319}$$

Equation (2.304a) implies:

$$\operatorname{div}_y \tilde{v}_0 = 0, \qquad \mathcal{M}(\tilde{v}_0) = 0. \tag{2.320}$$

4.2 **Vector potential representation.** A divergence-free field with zero mean can be expressed as a curl. Hence, there exists a vector potential ϕ (with $\operatorname{div}_y \phi = 0$) such that:

$$\tilde{v}_0 = \operatorname{rot}_y \phi. \tag{2.321}$$

4.3 **Cell problem for the potential.** The remaining constraint (2.311), $\operatorname{rot}_y(a^{-1}v_0) = 0$, must now be satisfied. Substituting $v_0 = \rho + \operatorname{rot}_y \phi$ gives:

$$\operatorname{rot}_y(a^{-1}(\rho + \operatorname{rot}_y \phi)) = 0. \tag{2.322}$$

The solution is linear in ρ:

$$\phi = -\tilde{\chi}^p(y)\rho_p(x) + \hat{\phi}(x), \tag{2.323}$$

where each $\tilde{\chi}^p$ is the unique Y-periodic, zero-mean, divergence-free solution to:

$$\operatorname{rot}_y(a^{-1} \operatorname{rot}_y \tilde{\chi}^p) = \operatorname{rot}_y(a^{-1}e_p). \tag{2.324}$$

This yields the expression for the flux:

$$v_0 = (I - \operatorname{rot}_y \tilde{\chi}(y))\rho(x), \tag{2.325}$$

where the matrix $\operatorname{rot}_y \tilde{\chi}$ is defined by $(\operatorname{rot}_y \tilde{\chi}) \cdot \xi = \operatorname{rot}_y(\tilde{\chi}^p \xi_p)$.

4.4 **Deriving the homogenized equation.**

4.4.1 The solvability condition for (2.310) is:

$$\nabla_x u = \mathcal{M}(a^{-1} v_0) = \mathcal{M}(a^{-1}(I - \operatorname{rot}_y \tilde{\chi}))\rho. \tag{2.326}$$

4.4.2 The solvability condition for (2.304b) is:

$$-\operatorname{div}_x \rho + \lambda u = f. \tag{2.327}$$

4.4.3 Combining these two results gives the homogenized equation:

$$-\operatorname{div}_x \left(\left[\mathcal{M}(a^{-1}(I - \operatorname{rot}_y \tilde{\chi})) \right]^{-1} \nabla_x u \right) + \lambda u = f. \tag{2.328}$$

5. **Equivalence and Dual Formulas.** Both procedures yield the same homogenized operator $\mathcal{A}$, establishing the identity:

$$[\mathcal{A}] = \mathcal{M}\left(a(I - \nabla_y \chi)\right), \qquad [\mathcal{A}]^{-1} = \mathcal{M}\left(a^{-1}(I - \operatorname{rot}_y \tilde{\chi})\right). \tag{2.329}$$

Remark. These are the *dual formulas* for the homogenized matrix $[\mathcal{A}]$.

2.6.5 Correctors and Error Estimates for the Dirichlet Problem

We now derive error estimates for correctors in the Dirichlet problem, without assuming symmetry. The notation is as in Theorem 5.1.

1. **Theorem 5.13: Error Estimates with Cutoff Function m_ε**

1.1 **Statement of Theorem.** Under the hypotheses of Theorem 5.1, consider the Dirichlet problem ($V = H_0^1(\mathcal{O})$). Let m_ε be the cutoff function. Define the correction:

$$\begin{aligned} z_\varepsilon &= u_\varepsilon - u - \varepsilon m_\varepsilon u_1, \\ n_\varepsilon &= a^\varepsilon \nabla u_\varepsilon - v_0 - \varepsilon v_1, \end{aligned} \tag{2.330}$$

where the correctors are given by:

$$u_1 = -\chi^j \frac{\partial u}{\partial x_j}, \tag{2.331}$$

$$v_0 = (I - \operatorname{rot}_y \tilde{\chi})\rho, \tag{2.332}$$

$$\rho = \mathcal{M}\left(a^{-1}(I - \operatorname{rot}_y \tilde{\chi})\right)^{-1} \nabla u, \tag{2.333}$$

$$v_1 = -\operatorname{rot}_x(\tilde{\chi}\rho). \tag{2.334}$$

Then, the following strong convergences hold:

$$z_\varepsilon \to 0 \quad \text{in } H_0^1(\mathcal{O}), \tag{2.335}$$

$$n_\varepsilon \to 0 \quad \text{in } (L^2(\mathcal{O}))^3. \tag{2.336}$$

1.2 **Remarks on Theorem 5.13.**

1.2.1 For general boundary conditions, a similar result (without m_ε) holds on any $\mathcal{O}' \subset \mathcal{O}$.

1.2.2 The flux convergence $n_\varepsilon \to 0$ is independent of and does not imply the convergence $z_\varepsilon \to 0$.

1.2.3 The term εu_1 can be omitted in the definition of n_ε.

1.3 **Error Estimates with Regularity Assumption.** Since $u \in H^2(\mathcal{O})$, Sobolev embedding implies its derivatives satisfy:

$$\frac{\partial u}{\partial x_i} \in L^q(\mathcal{O}), \quad \text{with } \frac{1}{q} = \frac{1}{2} - \frac{1}{n} \ (n \geq 3), \tag{2.337}$$

and q is finite for $n = 2$. Hence, there exists $p > 2$ such that

$$\frac{\partial u}{\partial x_i} \in L^p(\mathcal{O}). \tag{2.338}$$

Under this regularity assumption, the convergence rates are quantified by:

$$\|z_\varepsilon\|_{H_0^1(\mathcal{O})} + \|n_\varepsilon\|_{(L^2(\mathcal{O}))^3} \leq C\varepsilon^{1/(2p')}, \quad \frac{1}{p} + \frac{1}{p'} = 1. \tag{2.339}$$

The optimal rate $O(\varepsilon^{1/2})$ is achieved if $\nabla u \in L^\infty(\mathcal{O})$.

1.4 **Proof of Theorem 5.13.**

1.4.1 **Error Equations.** We begin by defining the error terms that will help us quantify the approximation:

$$a^\varepsilon \nabla z_\varepsilon - n_\varepsilon = g_{1\varepsilon}, \tag{2.340}$$

$$-\operatorname{div} n_\varepsilon + \lambda z_\varepsilon = g_{2\varepsilon}. \tag{2.341}$$

These equations measure how well our approximations satisfy the original PDE system. The goal is to prove the estimates:

$$\|g_{1\varepsilon}\|_{(L^2(\mathcal{O}))^3} \leq C\varepsilon^{1/(2p')}, \tag{2.342}$$

$$\|g_{2\varepsilon}\|_{L^2(\mathcal{O})} \leq C\varepsilon. \tag{2.343}$$

1.4.2 **Detailed Estimate for $g_{2\varepsilon}$.** Let us carefully derive the expression for $g_{2\varepsilon}$. Starting

from the definition:

$$\begin{aligned} g_{2\varepsilon} &= -\operatorname{div} n_\varepsilon + \lambda z_\varepsilon \\ &= -\operatorname{div}(a^\varepsilon \nabla u_\varepsilon - v_0 - \varepsilon v_1) + \lambda(u_\varepsilon - u - \varepsilon m_\varepsilon u_1) \\ &= [-\operatorname{div}(a^\varepsilon \nabla u_\varepsilon) + \lambda u_\varepsilon] + \operatorname{div} v_0 + \varepsilon \operatorname{div} v_1 - \lambda u - \varepsilon\lambda m_\varepsilon u_1. \end{aligned} \tag{2.344}$$

Using the original PDE $-\operatorname{div}(a^\varepsilon \nabla u_\varepsilon) + \lambda u_\varepsilon = f$ and the homogenized equation $-\operatorname{div}([\mathcal{A}]\nabla u) + \lambda u = f$, we get:

$$\begin{aligned} g_{2\varepsilon} &= f + \operatorname{div} v_0 + \varepsilon \operatorname{div} v_1 - \lambda u - \varepsilon\lambda m_\varepsilon u_1 \\ &= [\operatorname{div} v_0 + \lambda u - f] + \varepsilon[\operatorname{div} v_1 - \lambda m_\varepsilon u_1]. \end{aligned} \tag{2.345}$$

From the two-scale expansion at order ε^0, we know that $\operatorname{div} v_0 = -\lambda u + f$ (this comes from averaging the ε^0 equation). Therefore, the first bracket vanishes, leaving:

$$g_{2\varepsilon} = \varepsilon(\operatorname{div} v_1 - \lambda m_\varepsilon u_1). \tag{2.346}$$

Since v_1 and u_1 are constructed from the correctors and are bounded in $L^2(\mathcal{O})$ (as $\chi^j, \tilde{\chi}^p \in L^\infty(Y)$), we immediately obtain:

$$\|g_{2\varepsilon}\|_{L^2(\mathcal{O})} \leq C\varepsilon. \tag{2.347}$$

1.4.3 **Detailed Estimate for $g_{1\varepsilon}$.** Now we derive the detailed expression for $g_{1\varepsilon}$:

$$\begin{aligned} g_{1\varepsilon} &= a^\varepsilon \nabla z_\varepsilon - n_\varepsilon \\ &= a^\varepsilon \nabla(u_\varepsilon - u - \varepsilon m_\varepsilon u_1) - (a^\varepsilon \nabla u_\varepsilon - v_0 - \varepsilon v_1) \\ &= -a^\varepsilon \nabla u - \varepsilon a^\varepsilon \nabla(m_\varepsilon u_1) + v_0 + \varepsilon v_1. \end{aligned} \tag{2.348}$$

Expanding the gradient term $\nabla(m_\varepsilon u_1) = m_\varepsilon \nabla u_1 + u_1 \nabla m_\varepsilon$, we get:

$$\begin{aligned} g_{1\varepsilon} &= -a^\varepsilon \nabla u - \varepsilon a^\varepsilon (m_\varepsilon \nabla u_1 + u_1 \nabla m_\varepsilon) + v_0 + \varepsilon v_1 \\ &= [v_0 - a^\varepsilon \nabla u] + \varepsilon[v_1 - m_\varepsilon a^\varepsilon \nabla u_1] - \varepsilon a^\varepsilon u_1 \nabla m_\varepsilon. \end{aligned} \tag{2.349}$$

From the two-scale expansion at order ε^{-1}, we have $v_0 = a(\nabla u + \nabla_y u_1) = a^\varepsilon(\nabla u + \nabla_y u_1)$. Therefore:

$$v_0 - a^\varepsilon \nabla u = a^\varepsilon \nabla_y u_1. \tag{2.350}$$

Substituting this back gives:

$$g_{1\varepsilon} = a^\varepsilon \nabla_y u_1 + \varepsilon[v_1 - m_\varepsilon a^\varepsilon \nabla u_1] - \varepsilon a^\varepsilon u_1 \nabla m_\varepsilon \tag{2.351}$$

$$= a^\varepsilon (1 - m_\varepsilon)\nabla_y u_1 + \varepsilon[v_1 - m_\varepsilon a^\varepsilon \nabla u_1] - \varepsilon a^\varepsilon u_1 \nabla m_\varepsilon. \tag{2.352}$$

Thus we obtain the decomposition:

$$g_{1\varepsilon} = b_\varepsilon + c_\varepsilon + d_\varepsilon, \quad \text{where}$$
$$b_\varepsilon = \varepsilon(v_1 - m_\varepsilon a^\varepsilon \nabla_x u_1), \tag{2.353}$$
$$c_\varepsilon = -\varepsilon a^\varepsilon (\nabla m_\varepsilon) u_1, \tag{2.354}$$
$$d_\varepsilon = a^\varepsilon (1 - m_\varepsilon) \nabla_y u_1. \tag{2.355}$$

Now we estimate each term:

i. **Term b_ε:** Since v_1 and $\nabla_x u_1$ are bounded in L^2 (as they involve the correctors $\chi^j, \tilde{\chi}^p$ which are in L^∞), we have:

$$\|b_\varepsilon\|_{(L^2(\mathcal{O}))^3} \leq C\varepsilon. \tag{2.356}$$

ii. **Term c_ε:** This term is only supported where $\nabla m_\varepsilon \neq 0$, i.e., in the boundary layer $\{x : \text{dist}(x, \Gamma) \leq 2\varepsilon\}$. Since $|\nabla m_\varepsilon| \leq C/\varepsilon$ and $u_1 = -\chi^j \frac{\partial u}{\partial x_j}$, we have:

$$\|c_\varepsilon\|_{(L^2(\mathcal{O}))^3} \leq C \left(\int_{\text{dist}(x,\Gamma)\leq 2\varepsilon} |u_1|^2 dx \right)^{1/2}. \tag{2.357}$$

Using the L^p regularity (**??**) of ∇u and the fact that $\chi^j \in L^\infty$, we get:

$$\|c_\varepsilon\|_{(L^2(\mathcal{O}))^3} \leq C\varepsilon^{1/(2p')}. \tag{2.358}$$

iii. **Term d_ε:** This term is also only supported in the boundary layer where $m_\varepsilon \neq 1$. Since $\nabla_y u_1 = -\nabla_y \chi^j \frac{\partial u}{\partial x_j}$ and $\nabla_y \chi^j \in L^\infty$, we have:

$$\|d_\varepsilon\|_{(L^2(\mathcal{O}))^3} \leq C \left(\int_{\text{dist}(x,\Gamma)\leq 2\varepsilon} |\nabla u|^2 dx \right)^{1/2} \leq C\varepsilon^{1/(2p')}. \tag{2.359}$$

Combining these estimates gives:

$$\|g_{1\varepsilon}\|_{(L^2(\mathcal{O}))^3} \leq C(\varepsilon + \varepsilon^{1/(2p')}) \leq C\varepsilon^{1/(2p')}. \tag{2.360}$$

1.4.4 **Energy Estimate.** We now use the error equations to derive an energy estimate. Multiply (2.340) by ∇z_ε and (2.341) by z_ε, integrate over $\mathcal{O}$, and add:

$$(a^\varepsilon \nabla z_\varepsilon, \nabla z_\varepsilon) + \lambda(z_\varepsilon, z_\varepsilon) - [(n_\varepsilon, \nabla z_\varepsilon) + (\text{div}\, n_\varepsilon, z_\varepsilon)]$$
$$= (g_{1\varepsilon}, \nabla z_\varepsilon) + (g_{2\varepsilon}, z_\varepsilon). \tag{2.361}$$

Since $z_\varepsilon \in H_0^1(\mathcal{O})$, integration by parts gives:

$$(n_\varepsilon, \nabla z_\varepsilon) + (\text{div}\, n_\varepsilon, z_\varepsilon) = 0. \tag{2.362}$$

Thus we have:

$$(a^\varepsilon \nabla z_\varepsilon, \nabla z_\varepsilon) + \lambda(z_\varepsilon, z_\varepsilon) = (g_{1\varepsilon}, \nabla z_\varepsilon) + (g_{2\varepsilon}, z_\varepsilon). \tag{2.363}$$

Using the coercivity of a^ε ($a^\varepsilon \geq \alpha I > 0$) and the estimates (2.342), (2.343):

$$\begin{aligned} \alpha \|\nabla z_\varepsilon\|_{L^2}^2 + \lambda \|z_\varepsilon\|_{L^2}^2 &\leq \|g_{1\varepsilon}\|_{L^2} \|\nabla z_\varepsilon\|_{L^2} + \|g_{2\varepsilon}\|_{L^2} \|z_\varepsilon\|_{L^2} \\ &\leq C\varepsilon^{1/(2p')} \|\nabla z_\varepsilon\|_{L^2} + C\varepsilon \|z_\varepsilon\|_{L^2}. \end{aligned} \tag{2.364}$$

Using Young's inequality $ab \leq \frac{\delta}{2} a^2 + \frac{1}{2\delta} b^2$ with appropriate $\delta > 0$, we obtain:

$$\alpha \|\nabla z_\varepsilon\|_{L^2}^2 + \lambda \|z_\varepsilon\|_{L^2}^2 \leq \frac{\alpha}{2} \|\nabla z_\varepsilon\|_{L^2}^2 + \frac{\lambda}{2} \|z_\varepsilon\|_{L^2}^2 + C(\varepsilon^{1/p'} + \varepsilon^2). \tag{2.365}$$

Rearranging gives:

$$\frac{\alpha}{2} \|\nabla z_\varepsilon\|_{L^2}^2 + \frac{\lambda}{2} \|z_\varepsilon\|_{L^2}^2 \leq C\varepsilon^{1/p'}. \tag{2.366}$$

Therefore:

$$\|z_\varepsilon\|_{H_0^1(\mathcal{O})} \leq C\varepsilon^{1/(2p')}. \tag{2.367}$$

Finally, the bound for n_ε follows from (2.340):

$$\|n_\varepsilon\|_{(L^2(\mathcal{O}))^3} \leq \|a^\varepsilon \nabla z_\varepsilon\|_{(L^2(\mathcal{O}))^3} + \|g_{1\varepsilon}\|_{(L^2(\mathcal{O}))^3} \leq C\varepsilon^{1/(2p')}. \tag{2.368}$$

2. **Theorem 5.17: Error Estimates without Cutoff Function m_ε**

2.1 **Statement of Theorem.** Under the hypotheses of Theorem 5.1, define the corrected approximation without the cutoff function:

$$\zeta_\varepsilon = u_\varepsilon - u - \varepsilon u_1, \tag{2.369}$$

with u_1, n_ε as before. Then,

$$\|\zeta_\varepsilon\|_{H^1(\mathcal{O})} + \|n_\varepsilon\|_{L^2(\mathcal{O})} \leq C\varepsilon^{1/4}. \tag{2.370}$$

2.2 **Remark.** Although $\zeta_\varepsilon \notin H_0^1(\mathcal{O})$ (it does not satisfy the Dirichlet boundary condition), this method yields a better $H^1(\mathcal{O})$-norm error rate.

2.3 **Proof of Theorem 5.17.**

2.3.1 **Error Equations.** The errors satisfy:

$$\begin{aligned} a^\varepsilon \nabla \zeta_\varepsilon - n_\varepsilon &= h_{1\varepsilon}, \\ -\operatorname{div} n_\varepsilon + \lambda \zeta_\varepsilon &= h_{2\varepsilon}, \end{aligned} \tag{2.371}$$

where

$$h_{1\varepsilon}, h_{2\varepsilon} = O(\varepsilon). \tag{2.372}$$

2.3.2 **Key Trace Estimates.** The following estimates on the boundary Γ are crucial:

$$\begin{aligned} \|u_1\|_{H^{1/2}(\Gamma)} &\leq C\varepsilon^{-1/2}, \\ \|n_\varepsilon \cdot \nu\|_{H^{-1/2}(\Gamma)} &\leq C. \end{aligned} \tag{2.373}$$

The first estimate follows by interpolation from:

$$\|u_1\|_{L^2(\Gamma)} \leq C, \quad \|u_1\|_{H^1(\Gamma)} \leq C/\varepsilon. \tag{2.374}$$

The second estimate follows from the boundedness of $v_\varepsilon = a^\varepsilon \nabla u_\varepsilon$ and its divergence in $L^2(\mathcal{O})$.

2.3.3 **Energy Estimate with Boundary Terms.** The weak formulation yields:

$$(a^\varepsilon \nabla \zeta_\varepsilon, \nabla \zeta_\varepsilon) + \lambda \|\zeta_\varepsilon\|^2 = (h_{1\varepsilon}, \nabla \zeta_\varepsilon) + (h_{2\varepsilon}, \zeta_\varepsilon) + \int_\Gamma (n_\varepsilon \cdot \nu) \zeta_\varepsilon d\Gamma. \tag{2.375}$$

The boundary term is estimated using:

$$\left| \int_\Gamma (n_\varepsilon \cdot \nu) \zeta_\varepsilon d\Gamma \right| \leq \|n_\varepsilon \cdot \nu\|_{H^{-1/2}(\Gamma)} \|\zeta_\varepsilon\|_{H^{1/2}(\Gamma)} \leq C\varepsilon \|u_1\|_{H^{1/2}(\Gamma)} \leq C\varepsilon^{1/2}. \tag{2.376}$$

Combining this with the $O(\varepsilon)$ bulk terms leads to:

$$\|\zeta_\varepsilon\|_{H^1(\mathcal{O})} \leq C\varepsilon^{1/4}. \tag{2.377}$$

3 Evolution Operators

3.1 Evolution Operators of Hyperbolic, Petrowsky, or Schrödinger Type

3.1.1 Orientation

This section presents preliminary results on homogenization for evolution equations (hyperbolic operators). Different techniques for hyperbolic problems are explored in Chapter 4. We also present results for Petrowsky and Schrödinger type operators, where hyperbolicity is not essential, and give examples of non-local operators arising from homogenization.

3.1.2 Linear Operators with Coefficients Regular in Time

We begin with a second-order hyperbolic operator with highly oscillatory coefficients that are independent of time t.

1. Problem Setup.

1.1 The Operator and its Properties: Consider the second-order differential operator:

$$A^\varepsilon v = -\frac{\partial}{\partial x_i}\left(a_{ij}\left(\frac{x}{\varepsilon}\right)\frac{\partial v}{\partial x_j}\right), \tag{3.1}$$

where the coefficients a_{ij} model the material properties and satisfy:

$$a_{ij} = a_{ji}, \quad \forall i,j, \qquad \text{(Symmetry)} \tag{3.2}$$

$$a_{ij}(y) \in L^\infty(\mathbb{R}^n_y), \quad \text{and are } Y\text{-periodic}, \qquad \text{(Boundedness and Periodicity)} \tag{3.3}$$

$$a_{ij}(y)\xi_i\xi_j \geq \alpha\xi_i\xi_i, \quad \alpha > 0, \quad \text{a.e. in } y. \qquad \text{(Uniform Ellipticity)} \tag{3.4}$$

These ensure the associated bilinear form,

$$a^\varepsilon(u,v) = \int_{\mathcal{O}} a_{ij}\left(\frac{x}{\varepsilon}\right)\frac{\partial u}{\partial x_j}\frac{\partial v}{\partial x_i}dx, \tag{3.5}$$

is symmetric, continuous, and coercive.

1.2 Derivation of the Weak Form: The classical (strong) form of the wave equation is:

$$\frac{\partial^2 u_\varepsilon}{\partial t^2} + A^\varepsilon u_\varepsilon = f \quad \text{in } \mathcal{O}\times(0,T). \tag{3.6}$$

To derive the weak form:

1.2.1 Multiply the strong form by an arbitrary test function $v \in V$:

$$\int_{\mathcal{O}} \frac{\partial^2 u_\varepsilon}{\partial t^2} v dx + \int_{\mathcal{O}} (A^\varepsilon u_\varepsilon) v dx = \int_{\mathcal{O}} f v dx. \tag{3.7}$$

1.2.2 The first and third terms are already inner products in $H = L^2(\mathcal{O})$:

$$\left(\frac{\partial^2 u_\varepsilon}{\partial t^2}, v\right) + \int_{\mathcal{O}} (A^\varepsilon u_\varepsilon) v dx = (f, v). \tag{3.8}$$

1.2.3 Apply integration by parts to the second term to transfer the derivatives onto the test function v:

$$\begin{aligned}
\int_{\mathcal{O}} (A^\varepsilon u_\varepsilon) v dx &= \int_{\mathcal{O}} \left[-\frac{\partial}{\partial x_i} \left(a_{ij} \left(\frac{x}{\varepsilon} \right) \frac{\partial u_\varepsilon}{\partial x_j} \right) \right] v dx \\
&= -\int_{\mathcal{O}} \frac{\partial}{\partial x_i} \left(a_{ij} \frac{\partial u_\varepsilon}{\partial x_j} v \right) dx + \int_{\mathcal{O}} a_{ij} \frac{\partial u_\varepsilon}{\partial x_j} \frac{\partial v}{\partial x_i} dx \\
&= -\int_{\partial \mathcal{O}} \left(a_{ij} \frac{\partial u_\varepsilon}{\partial x_j} v \right) n_i dS + \int_{\mathcal{O}} a_{ij} \frac{\partial u_\varepsilon}{\partial x_j} \frac{\partial v}{\partial x_i} dx \\
&= \int_{\mathcal{O}} a_{ij} \frac{\partial u_\varepsilon}{\partial x_j} \frac{\partial v}{\partial x_i} dx - \int_{\partial \mathcal{O}} \underbrace{\left(a_{ij} \frac{\partial u_\varepsilon}{\partial x_j} n_i \right)}_{\text{flux}} v dS.
\end{aligned} \tag{3.9}$$

1.2.4 For Dirichlet ($v = 0$ on $\partial\mathcal{O}$) or Neumann boundary conditions, the boundary term vanishes or incorporates boundary data, respectively. In either case, we obtain:

$$\int_{\mathcal{O}} (A^\varepsilon u_\varepsilon) v dx = a^\varepsilon(u_\varepsilon, v). \tag{3.10}$$

1.2.5 Substituting back, we arrive at the weak formulation:

$$\left(\frac{\partial^2 u_\varepsilon}{\partial t^2}, v\right) + a^\varepsilon(u_\varepsilon, v) = (f, v), \quad \forall v \in V. \tag{3.11}$$

1.3 The Full Weak Problem: Let V be a Hilbert space such that $H_0^1(\mathcal{O}) \subseteq V \subseteq H^1(\mathcal{O})$. We consider the hyperbolic initial value problem: Find

$$u_\varepsilon \in L^\infty(0, T; V), \quad \frac{\partial u_\varepsilon}{\partial t} \in L^\infty(0, T; H), \tag{3.12}$$

such that:

$$\left(\frac{\partial^2 u_\varepsilon}{\partial t^2}, v\right) + a^\varepsilon(u_\varepsilon, v) = (f, v), \quad \forall v \in V, \tag{3.13}$$

$$u_\varepsilon(0) = u^0, \quad \frac{\partial u_\varepsilon}{\partial t}(0) = u^1, \tag{3.14}$$

with given data satisfying:

$$f \in L^2(0, T; H), \quad u^0 \in V, \quad u^1 \in H. \tag{3.15}$$

This problem admits a unique solution.

2. A Priori Estimates and Homogenization Result.

The key to obtaining estimates and passing to the limit is to derive an energy identity.

2.1 Derivation of the Energy Identity: We start with the weak formulation (3.13):

$$\left(\frac{\partial^2 u_\varepsilon}{\partial t^2}, v\right) + a^\varepsilon(u_\varepsilon, v) = (f, v), \quad \forall v \in V. \tag{3.16}$$

The crucial step is to choose the test function v to be the velocity, i.e., $v = \frac{\partial u_\varepsilon}{\partial t} \equiv u'_\varepsilon$. This is an admissible test function since $u'_\varepsilon(t) \in V$ for almost every t. This yields:

$$\left(\frac{\partial^2 u_\varepsilon}{\partial t^2}, u'_\varepsilon\right) + a^\varepsilon(u_\varepsilon, u'_\varepsilon) = (f, u'_\varepsilon). \tag{3.17}$$

We now analyze the two terms on the left-hand side separately, leveraging the time-independence and symmetry of a^ε.

2.1.1 **Kinetic Energy Term:**

$$\begin{aligned}\left(\frac{\partial^2 u_\varepsilon}{\partial t^2}, u'_\varepsilon\right) &= \int_{\mathcal{O}} \frac{\partial^2 u_\varepsilon}{\partial t^2} \frac{\partial u_\varepsilon}{\partial t} dx \\ &= \frac{1}{2}\frac{d}{dt}\int_{\mathcal{O}} \left|\frac{\partial u_\varepsilon}{\partial t}\right|^2 dx \quad (\text{since } \frac{d}{dt}(w^2) = 2ww') \\ &= \frac{1}{2}\frac{d}{dt}|u'_\varepsilon|^2. \end{aligned} \tag{3.18}$$

2.1.2 **Potential Energy Term:** Since the coefficients a_{ij} are independent of time, the bilinear form a^ε is time-independent. Combined with its symmetry ($a^\varepsilon(u, v) = a^\varepsilon(v, u)$), we have:

$$\begin{aligned} a^\varepsilon(u_\varepsilon, u'_\varepsilon) &= \int_{\mathcal{O}} a_{ij}\left(\frac{x}{\varepsilon}\right) \frac{\partial u_\varepsilon}{\partial x_j} \frac{\partial u'_\varepsilon}{\partial x_i} dx \\ &= \frac{1}{2}\frac{d}{dt}\int_{\mathcal{O}} a_{ij}\left(\frac{x}{\varepsilon}\right) \frac{\partial u_\varepsilon}{\partial x_j} \frac{\partial u_\varepsilon}{\partial x_i} dx \quad (\text{since } \frac{d}{dt}(u_\varepsilon) = u'_\varepsilon \text{ and symmetry}) \\ &= \frac{1}{2}\frac{d}{dt} a^\varepsilon(u_\varepsilon, u_\varepsilon). \end{aligned} \tag{3.19}$$

Substituting these two results back into (3.17) gives the desired energy identity:

$$\frac{d}{dt}\left[\frac{1}{2}|u'_\varepsilon|^2 + \frac{1}{2}a^\varepsilon(u_\varepsilon, u_\varepsilon)\right] = (f, u'_\varepsilon). \tag{3.20}$$

Defining the total energy (kinetic + potential) as $E(t) = \frac{1}{2}|u'_\varepsilon|^2 + \frac{1}{2}a^\varepsilon(u_\varepsilon)$, we write:

$$\frac{d}{dt}E(t) = (f, u'_\varepsilon). \tag{3.21}$$

2.2 Derivation of the Uniform Estimates: The energy identity (3.21) is the starting point for deriving ε-independent estimates. We proceed in several steps:

2.2.1 **Integrated Energy Identity:** Integrate (3.21) in time from 0 to t:

$$E(t) - E(0) = \int_0^t (f(s), u_\varepsilon'(s))ds. \tag{3.22}$$

Therefore, the energy at any time t is given by:

$$E(t) = E(0) + \int_0^t (f(s), u_\varepsilon'(s))ds. \tag{3.23}$$

2.2.2 **Coercivity and Initial Energy:** We now use key properties to transform the integrated energy equation (3.23) into a useful inequality.

i. **Coercivity (Lower Bound):** By uniform ellipticity, the bilinear form is coercive: there exists a constant $\alpha > 0$, independent of ε, such that:

$$a^\varepsilon(v, v) \geq \alpha \|v\|_V^2 \quad \text{for all } v \in V. \tag{3.24}$$

Applying this to the definition of the total energy $E(t)$ gives a lower bound:

$$\begin{aligned} E(t) &= \frac{1}{2}|u_\varepsilon'(t)|^2 + \frac{1}{2}a^\varepsilon(u_\varepsilon(t), u_\varepsilon(t)) \\ &\geq \frac{1}{2}|u_\varepsilon'(t)|^2 + \frac{1}{2}\left(\alpha\|u_\varepsilon(t)\|_V^2\right) \\ &= \frac{1}{2}|u_\varepsilon'(t)|^2 + \frac{\alpha}{2}\|u_\varepsilon(t)\|_V^2. \end{aligned} \tag{3.25}$$

ii. **Initial Energy (Boundedness):** The initial energy $E(0)$ is determined by the initial data:

$$E(0) = \frac{1}{2}|u^1|^2 + \frac{1}{2}a^\varepsilon(u^0, u^0). \tag{3.26}$$

Since $u^1 \in H$ and $u^0 \in V$, and a^ε is continuous, there exists a constant C_1, independent of ε, such that:

$$E(0) \leq C_1. \tag{3.27}$$

Combining the bounds: From (3.23), the bound on $E(0)$ implies:

$$E(t) \leq C_1 + \left|\int_0^t (f(s), u_\varepsilon'(s))ds\right| \leq C_1 + \int_0^t |(f(s), u_\varepsilon'(s))|ds. \tag{3.28}$$

Now, we combine the lower bound (3.25) and the upper bound (3.28) for $E(t)$ into a single inequality:

$$\frac{1}{2}|u_\varepsilon'(t)|^2 + \frac{\alpha}{2}\|u_\varepsilon(t)\|_V^2 \leq E(t) \leq C_1 + \int_0^t |(f(s), u_\varepsilon'(s))|ds. \tag{3.29}$$

This inequality (3.29) is fundamental, showing the energy is controlled by the initial data and the work done by the source.

2.2.3 **Estimate the Source Term:** We estimate the right-hand side using:

i. Cauchy-Schwarz inequality: $|(f, u_\varepsilon')| \leq |f||u_\varepsilon'|$

ii. Young's inequality: $ab \leq \frac{1}{2}a^2 + \frac{1}{2}b^2$

$$\begin{aligned}\int_0^t |(f(s), u_\varepsilon'(s))|ds &\leq \int_0^t |f(s)||u_\varepsilon'(s)|ds \\ &\leq \frac{1}{2}\int_0^t |u_\varepsilon'(s)|^2 ds + \frac{1}{2}\int_0^t |f(s)|^2 ds. \end{aligned} \tag{3.30}$$

The term $\frac{1}{2}\int_0^t |f(s)|^2 ds$ is bounded by $\frac{1}{2}\|f\|^2_{L^2(0,T;H)} \leq C_2$ since $f \in L^2(0,T;H)$.

2.2.4 **Combine Estimates:** Substituting back into (3.29) and multiplying by 2 yields:

$$|u_\varepsilon'(t)|^2 + \alpha\|u_\varepsilon(t)\|_V^2 \leq 2C_1 + 2C_2 + \int_0^t |u_\varepsilon'(s)|^2 ds. \tag{3.31}$$

Define the function:

$$y(t) = |u_\varepsilon'(t)|^2 + \alpha\|u_\varepsilon(t)\|_V^2. \tag{3.32}$$

Substituting into (3.31) gives:

$$y(t) \leq C_3 + \int_0^t |u_\varepsilon'(s)|^2 ds. \tag{3.33}$$

Since $|u_\varepsilon'(s)|^2 \leq y(s)$, we obtain the final inequality:

$$y(t) \leq C_3 + \int_0^t y(s)ds, \quad \text{where } C_3 = 2C_1 + 2C_2. \tag{3.34}$$

2.2.5 **Apply Grönwall's Inequality:** Applying Grönwall's inequality[11] to (3.34) gives:

$$y(t) \leq C_3 e^t \leq C_3 e^T \equiv C_4. \tag{3.35}$$

2.2.6 **Final Uniform Estimates:** Since $y(t) = |u_\varepsilon'(t)|^2 + \alpha\|u_\varepsilon(t)\|_V^2 \leq C_4$ for all $t \in [0,T]$, it follows that each term is individually bounded:

$$|u_\varepsilon'(t)|^2 \leq C_4, \quad \alpha\|u_\varepsilon(t)\|_V^2 \leq C_4. \tag{3.36}$$

Therefore, we obtain the essential supremum bounds:

$$\operatorname*{ess\,sup}_{t\in[0,T]} |u_\varepsilon'(t)| \leq \sqrt{C_4}, \quad \operatorname*{ess\,sup}_{t\in[0,T]} \|u_\varepsilon(t)\|_V \leq \sqrt{C_4/\alpha}. \tag{3.37}$$

[11] Grönwall's inequality converts integral inequalities into pointwise bounds. Here, it shows that $y(t)$ cannot grow faster than an exponential.

By the definition of the $L^\infty(0,T;X)$ norm, this proves that the sequences $\{u_\varepsilon\}$ and $\{u'_\varepsilon\}$ are uniformly bounded:

$$\|u_\varepsilon\|_{L^\infty(0,T;V)} + \|u'_\varepsilon\|_{L^\infty(0,T;H)} \leq C. \tag{3.38}$$

where the constant $C = \sqrt{C_4} + \sqrt{C_4/\alpha}$ is independent of ε.

2.3 Passing to the Limit: The Method of Test Functions.

2.3.1 **Weak Compactness and Goal.** The uniform estimates (3.38) imply weak* compactness. Therefore, there exists a subsequence such that:

$$u_\varepsilon \rightharpoonup u \quad \text{in } L^\infty(0,T;V) \text{ weak*}, \quad u'_\varepsilon \rightharpoonup u' \quad \text{in } L^\infty(0,T;H) \text{ weak*}. \tag{3.39}$$

The goal is to show that this limit u is the unique solution of the homogenized equation. This is achieved through the *method of test functions*.

2.3.2 **Time Averaging and the Weak Form.** Let $\phi \in \mathcal{D}(0,T) = C_0^\infty(0,T)$ be a smooth test function in time. We define the following averaged functions in space:

$$u_\varepsilon(\phi)(x) = \int_0^T u_\varepsilon(x,t)\phi(t)dt, \quad f(\phi)(x) = \int_0^T f(x,t)\phi(t)dt. \tag{3.40}$$

Now, multiply the weak formulation (3.13) by $\phi(t)$ and integrate in time:

$$\int_0^T \left(\frac{\partial^2 u_\varepsilon}{\partial t^2}, v\right)\phi(t)dt + \int_0^T a^\varepsilon(u_\varepsilon, v)\phi(t)dt = \int_0^T (f,v)\phi(t)dt, \quad \forall v \in V. \tag{3.41}$$

2.3.3 **Transforming the Highest-Order Term.** We focus on transforming the first term, I, to remove the second time derivative:

$$I = \int_0^T \left(\frac{\partial^2 u_\varepsilon}{\partial t^2}, v\right)\phi(t)dt = \int_0^T \left(\int_\mathcal{O} \frac{\partial^2 u_\varepsilon}{\partial t^2} v dx\right)\phi(t)dt. \tag{3.42}$$

i. By Fubini's theorem, we exchange the order of integration:

$$I = \int_\mathcal{O} \left(\int_0^T \frac{\partial^2 u_\varepsilon}{\partial t^2}\phi(t)dt\right) v dx. \tag{3.43}$$

ii. For a fixed x, integrate the inner temporal integral by parts twice:

$$\begin{aligned}
\int_0^T \frac{\partial^2 u_\varepsilon}{\partial t^2}\phi dt &= \left[\frac{\partial u_\varepsilon}{\partial t}\phi\right]_0^T - \int_0^T \frac{\partial u_\varepsilon}{\partial t}\phi' dt \quad \text{(First integration)} \\
&= \left[\frac{\partial u_\varepsilon}{\partial t}\phi\right]_0^T - \left(\left[u_\varepsilon\phi'\right]_0^T - \int_0^T u_\varepsilon\phi'' dt\right) \quad \text{(Second integration)} \\
&= \left[\frac{\partial u_\varepsilon}{\partial t}\phi - u_\varepsilon\phi'\right]_0^T + \int_0^T u_\varepsilon\phi'' dt.
\end{aligned} \tag{3.44}$$

iii. Because $\phi \in \mathcal{D}(0,T)$, it has compact support: $\phi(0) = \phi(T) = 0$ and $\phi'(0) = \phi'(T) = 0$. Thus, the boundary terms vanish:

$$\left[\frac{\partial u_\varepsilon}{\partial t}\phi - u_\varepsilon \phi'\right]_0^T = 0 \implies I = \int_0^T (u_\varepsilon, v)\phi''(t)dt. \tag{3.45}$$

which is the final result, where Fubini's theorem was reapplied.

2.3.4 **The Averaged Equation.** Substituting the transformed term back into (3.41) gives:

$$\int_0^T (u_\varepsilon, v)\,\phi''(t)dt + \int_0^T a^\varepsilon(u_\varepsilon, v)\phi(t)dt = \int_0^T (f, v)\phi(t)dt. \tag{3.46}$$

By the linearity of the integrals and the bilinear form, this is equivalent to:

$$a^\varepsilon(u_\varepsilon(\phi), v) = (f(\phi), v) - (u_\varepsilon(\phi''), v), \quad \forall v \in V. \tag{3.47}$$

2.3.5 **Passing to the Limit.** We now pass to the limit in (3.47) as $\varepsilon \to 0$.

i. **Convergence of averaged terms:** From (3.39):

$$u_\varepsilon(\phi) \rightharpoonup u(\phi) \quad \text{in } V, \quad u_\varepsilon(\phi'') \rightharpoonup u(\phi'') \quad \text{in } H. \tag{3.48}$$

ii. **Convergence of the right-hand side:** The right-hand side is a continuous linear functional:

$$(f(\phi), v) - (u_\varepsilon(\phi''), v) \to (f(\phi), v) - (u(\phi''), v). \tag{3.49}$$

iii. **Convergence of the bilinear form:** By the elliptic homogenization result (Ch. 1) for the fixed function $u(\phi)$:

$$a^\varepsilon(u_\varepsilon(\phi), v) \to \mathcal{A}(u(\phi), v). \tag{3.50}$$

Thus, the limit of (3.47) is:

$$\mathcal{A}(u(\phi), v) = (f(\phi), v) - (u(\phi''), v), \quad \forall v \in V. \tag{3.51}$$

2.3.6 **The Final Homogenized Equation.** Rewriting the above and noting that it holds for all $\phi \in \mathcal{D}(0,T)$ and all $v \in V$, we conclude that u satisfies:

$$\left(\frac{\partial^2 u}{\partial t^2}, v\right) + \mathcal{A}(u, v) = (f, v), \quad \forall v \in V, \tag{3.52}$$

$$u \in L^\infty(0,T;V), \quad u' \in L^\infty(0,T;H), \tag{3.53}$$

$$u(0) = u^0, \quad u'(0) = u^1. \tag{3.54}$$

3. Time-Dependent Coefficients.

3.1 **Problem Setup.** Now consider the case where the coefficients also depend on time:

$$A^\varepsilon(t)v = -\frac{\partial}{\partial x_i}\left(a_{ij}\left(x, \frac{x}{\varepsilon}, t\right)\frac{\partial v}{\partial x_j}\right). \tag{3.55}$$

Assume the coefficients satisfy:

$$a_{ij} = a_{ji}, \quad a_{ij}(x, y, t) \text{ is } Y\text{-periodic in } y, \tag{3.56}$$

$$a_{ij} \in C^0(\overline{\mathcal{O}} \times [0, T]; L^\infty(\mathbb{R}^n_y)), \tag{3.57}$$

$$\frac{\partial a_{ij}}{\partial t} \in L^\infty(\mathcal{O} \times (0, T) \times \mathbb{R}^n_y). \tag{3.58}$$

Define the time-dependent bilinear form:

$$a^\varepsilon(t; u, v) = \int_{\mathcal{O}} a_{ij}\left(x, \frac{x}{\varepsilon}, t\right)\frac{\partial u}{\partial x_j}\frac{\partial v}{\partial x_i}dx, \tag{3.59}$$

and its time derivative (with u, v held fixed):

$$\dot{a}^\varepsilon(t; u, v) = \int_{\mathcal{O}} \frac{\partial a_{ij}}{\partial t}\left(x, \frac{x}{\varepsilon}, t\right)\frac{\partial u}{\partial x_j}\frac{\partial v}{\partial x_i}dx. \tag{3.60}$$

The problem (3.13), (3.14) with $a^\varepsilon(u, v)$ replaced by $a^\varepsilon(t; u, v)$ still has a unique solution.

3.2 **Derivation of the Energy Identity.**

3.2.1 **Start with the weak form.** Follow by the same logic, with $v = u'_\varepsilon$, the weak 3.13 form becomes:

$$\frac{1}{2}\frac{d}{dt}|u'_\varepsilon|^2 + a^\varepsilon(t; u_\varepsilon, u'_\varepsilon) = (f, u'_\varepsilon). \tag{3.61}$$

3.2.2 **Compute the total derivative.** To handle $a^\varepsilon(t; u_\varepsilon, u'_\varepsilon)$, consider the total derivative of the potential energy:

$$\frac{d}{dt}a^\varepsilon(t; u_\varepsilon, u_\varepsilon) = \frac{d}{dt}\int_{\mathcal{O}} a_{ij}\left(\frac{x}{\varepsilon}, t\right)\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i}dx. \tag{3.62}$$

Differentiate under the integral sign using the product rule:

$$\frac{d}{dt}\left[a_{ij}\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i}\right] = \frac{\partial a_{ij}}{\partial t}\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i} + a_{ij}\frac{\partial u'_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i} + a_{ij}\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u'_\varepsilon}{\partial x_i}. \tag{3.63}$$

3.2.3 **Integration.** Integrating over $\mathcal{O}$ and using the symmetry $a_{ij} = a_{ji}$ yields:

$$\begin{aligned}\frac{d}{dt}a^\varepsilon(t; u_\varepsilon, u_\varepsilon) &= \int_{\mathcal{O}} \frac{\partial a_{ij}}{\partial t}\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i}dx + 2\int_{\mathcal{O}} a_{ij}\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u'_\varepsilon}{\partial x_i}dx \\ &= \dot{a}^\varepsilon(t; u_\varepsilon, u_\varepsilon) + 2a^\varepsilon(t; u_\varepsilon, u'_\varepsilon).\end{aligned} \tag{3.64}$$

3.2.4 **Solve for the desired term.** Solving (3.64) for $a^\varepsilon(t; u_\varepsilon, u'_\varepsilon)$:

$$a^\varepsilon(t; u_\varepsilon, u'_\varepsilon) = \frac{1}{2}\frac{d}{dt}a^\varepsilon(t; u_\varepsilon, u_\varepsilon) - \frac{1}{2}\dot{a}^\varepsilon(t; u_\varepsilon, u_\varepsilon). \tag{3.65}$$

3.2.5 **Substitute and finalize.** Substituting (3.65) into (3.61):

$$\frac{1}{2}\frac{d}{dt}|u'_\varepsilon|^2 + \frac{1}{2}\frac{d}{dt}a^\varepsilon(t; u_\varepsilon, u_\varepsilon) - \frac{1}{2}\dot{a}^\varepsilon(t; u_\varepsilon, u_\varepsilon) = (f, u'_\varepsilon). \tag{3.66}$$

Combining the derivatives yields the final energy identity:

$$\frac{1}{2}\frac{d}{dt}\left[|u'_\varepsilon|^2 + a^\varepsilon(u_\varepsilon)\right] - \frac{1}{2}\dot{a}^\varepsilon(t, u_\varepsilon) = (f, u'_\varepsilon). \tag{3.67}$$

3.3 **Uniform Estimates and Homogenization Result.** The boundedness of $\frac{\partial a_{ij}}{\partial t}$ (assumption (3.58)) ensures the new term $\dot{a}^\varepsilon(t, u_\varepsilon)$ is controlled. This allows us to recover the *same uniform estimates* (3.38) as in the time-independent case.

The homogenized limit u satisfies the problem where the effective coefficients are computed for each fixed time t:

$$\left(\frac{\partial^2 u}{\partial t^2}, v\right) + \mathcal{A}(t; u, v) = (f, v), \tag{3.68}$$

$$u(0) = u^0, \quad u'(0) = u^1, \tag{3.69}$$

where $\mathcal{A}(t; \cdot, \cdot)$ is the homogenized form of $a^\varepsilon(t; \cdot, \cdot)$ for each *fixed* time t.

Remark 3.1. The homogenization techniques apply beyond hyperbolic operators. They extend to higher-order symmetric operators of the form

$$a^\varepsilon(t; u, v) = \int_\mathcal{O} a_{\alpha\beta}\left(x, \frac{x}{\varepsilon}, t\right) D^\alpha u D^\beta v dx, \quad |\alpha| = |\beta| = m, \tag{3.70}$$

with $a_{\alpha\beta} = a_{\beta\alpha}$ satisfying analogous periodicity and regularity conditions. For $m > 1$, these are *Petrowsky-type* operators.

Remark 3.2. The operator A^ε may be augmented with lower-order terms; this case is treated in Section 3.3.

3.1.3 Linear Operators with Coefficients Irregular in Time

We now consider a more complex scenario where the coefficients oscillate rapidly in both space and time, leading to a lack of uniform estimates and more complicated homogenization behavior.

1. Problem Setup and Loss of Uniform Estimates.

1.1 **Double-Scale Operator.** Consider coefficients $a_{ij}(y, \tau)$ that are functions of both a fast

spatial variable $y = x/\varepsilon$ and a fast time variable $\tau = t/\varepsilon$. They satisfy:

$$a_{ij} = a_{ji}, \quad \forall i, j, \tag{3.71}$$

$$a_{ij}(y, \tau) \in L^\infty(\mathbb{R}^n_y \times \mathbb{R}_\tau), \quad Y\text{-periodic in } y \text{ and } \tau_0\text{-periodic in } \tau, \tag{3.72}$$

$$a_{ij}(y, \tau)\xi_i\xi_j \geq \alpha\xi_i\xi_i, \quad \alpha > 0. \tag{3.73}$$

The operator introduces oscillations in both space and time:

$$A^\varepsilon v = -\frac{\partial}{\partial x_i}\left(a_{ij}\left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon}\right)\frac{\partial v}{\partial x_j}\right). \tag{3.74}$$

1.2 **Derivation of the Energy Identity.**

1.2.1 **Start with the weak form.** With $v = u'_\varepsilon$, the weak form is:

$$\left(\frac{\partial^2 u_\varepsilon}{\partial t^2}, u'_\varepsilon\right) + a^\varepsilon(t; u_\varepsilon, u'_\varepsilon) = (f, u'_\varepsilon). \tag{3.75}$$

The first term is $\frac{1}{2}\frac{d}{dt}|u'_\varepsilon|^2$, as before.

1.2.2 **Compute the total derivative.** The potential energy $a^\varepsilon(t; u_\varepsilon, u_\varepsilon)$ now has explicit time dependence:

$$\frac{d}{dt}a^\varepsilon(t; u_\varepsilon, u_\varepsilon) = \frac{d}{dt}\int_{\mathcal{O}} a_{ij}\left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon}\right)\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i}dx. \tag{3.76}$$

Differentiate under the integral sign. The integrand $F(x, t)$ depends on t both explicitly (through a_{ij}) and implicitly (through u_ε):

$$\frac{dF}{dt} = \frac{\partial}{\partial t}\left[a_{ij}\left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon}\right)\right]\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i} + a_{ij}\left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon}\right)\frac{\partial}{\partial t}\left[\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i}\right]. \tag{3.77}$$

1.2.3 **Apply the Chain Rule.** Compute the explicit time derivative. The function a_{ij} depends on t through the fast variable $\tau = t/\varepsilon$. We apply the multivariate chain rule:

$$\frac{\partial}{\partial t}\left[a_{ij}\left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon}\right)\right] = \frac{\partial a_{ij}}{\partial \tau}\cdot\frac{\partial \tau}{\partial t}. \tag{3.78}$$

Since $\tau = t/\varepsilon$, we have $\frac{\partial \tau}{\partial t} = \frac{1}{\varepsilon}$. Therefore:

$$\frac{\partial}{\partial t}\left[a_{ij}\left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon}\right)\right] = \frac{\partial a_{ij}}{\partial \tau}\cdot\frac{1}{\varepsilon}. \tag{3.79}$$

The implicit derivative is found using the product rule:

$$\frac{\partial}{\partial t}\left[\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i}\right] = \frac{\partial u'_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i} + \frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u'_\varepsilon}{\partial x_i}. \tag{3.80}$$

Therefore, the full derivative is:

$$\frac{dF}{dt} = \underbrace{\frac{1}{\varepsilon}\frac{\partial a_{ij}}{\partial \tau}\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i}}_{\text{Explicit term}} \quad \underbrace{+ a_{ij}\frac{\partial u'_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i} + a_{ij}\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u'_\varepsilon}{\partial x_i}}_{\text{Implicit terms}}. \tag{3.81}$$

1.2.4 **Integrate and use symmetry.** Integrating over $\mathcal{O}$ and using $a_{ij} = a_{ji}$ yields:

$$\begin{aligned}\frac{d}{dt}a^\varepsilon(t; u_\varepsilon, u_\varepsilon) &= \int_{\mathcal{O}} \frac{1}{\varepsilon}\frac{\partial a_{ij}}{\partial \tau}\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u_\varepsilon}{\partial x_i}dx + 2\int_{\mathcal{O}} a_{ij}\frac{\partial u_\varepsilon}{\partial x_j}\frac{\partial u'_\varepsilon}{\partial x_i}dx \\ &= \frac{1}{\varepsilon}\dot{a}^\varepsilon(t; u_\varepsilon, u_\varepsilon) + 2a^\varepsilon(t; u_\varepsilon, u'_\varepsilon).\end{aligned} \tag{3.82}$$

Here, $\dot{a}^\varepsilon$ is defined by the explicit term:

$$\dot{a}^\varepsilon(t; u, v) = \int_{\mathcal{O}} \frac{\partial a_{ij}}{\partial \tau}\left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon}\right)\frac{\partial u}{\partial x_j}\frac{\partial v}{\partial x_i}dx. \tag{3.83}$$

1.2.5 **Solve and substitute.** Solving (3.82) for $a^\varepsilon(t; u_\varepsilon, u'_\varepsilon)$:

$$a^\varepsilon(t; u_\varepsilon, u'_\varepsilon) = \frac{1}{2}\frac{d}{dt}a^\varepsilon(t; u_\varepsilon, u_\varepsilon) - \frac{1}{2\varepsilon}\dot{a}^\varepsilon(t; u_\varepsilon, u_\varepsilon). \tag{3.84}$$

Substituting back into the weak form gives the final energy identity:

$$\frac{1}{2}\frac{d}{dt}\left[|u'_\varepsilon|^2 + a^\varepsilon(u_\varepsilon)\right] - \frac{1}{2\varepsilon}\dot{a}^\varepsilon(t, u_\varepsilon) = (f, u'_\varepsilon). \tag{3.85}$$

1.3 **The Fundamental Problem: Loss of Uniform Estimates.** The term $\frac{1}{\varepsilon}\dot{a}^\varepsilon(t, u_\varepsilon)$ is $\mathcal{O}(1/\varepsilon)$ and unbounded. This dominates the energy identity, preventing the use of Grönwall's lemma. Consequently, the energy method fails, voiding any uniform a priori estimates.

2. The Case of Lower-Order Terms.

2.1 **A Tractable Scenario.** A tractable case arises when only the coefficients of *lower-order terms* are irregular in time. This allows the principal part to provide the necessary stability. Consider the operators:

$$A^\varepsilon v = -\frac{\partial}{\partial x_i}\left(a_{ij}\left(\frac{x}{\varepsilon}\right)\frac{\partial v}{\partial x_j}\right), \quad \text{(Elliptic, independent of } t\text{)} \tag{3.86}$$

$$B^\varepsilon v = b_j\left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon^k}\right)\frac{\partial v}{\partial x_j}, \quad \text{(First-order, oscillatory in time)} \tag{3.87}$$

where $b_j(y, \tau)$ is Y-periodic in y and τ_0-periodic in τ, and $b_j \in L^\infty$.

2.2 **Derivation of the Weak Formulation.**

2.2.1 **Strong Form.** The classical PDE is:

$$\frac{\partial^2 u_\varepsilon}{\partial t^2} + A^\varepsilon u_\varepsilon + B^\varepsilon u_\varepsilon = f \quad \text{in } \mathcal{O} \times (0,T). \tag{3.88}$$

2.2.2 **Multiply and Integrate.** Multiply the strong form by a test function $v \in V$ and integrate over the spatial domain $\mathcal{O}$:

$$\int_{\mathcal{O}} \frac{\partial^2 u_\varepsilon}{\partial t^2} v dx + \int_{\mathcal{O}} (A^\varepsilon u_\varepsilon) v dx + \int_{\mathcal{O}} (B^\varepsilon u_\varepsilon) v dx = \int_{\mathcal{O}} f v dx. \tag{3.89}$$

2.2.3 **Identify Inner Products.**

i. The first term is the integral of a product of two functions in $L^2(\mathcal{O})$:

$$\int_{\mathcal{O}} \frac{\partial^2 u_\varepsilon}{\partial t^2} v dx = \left(\frac{\partial^2 u_\varepsilon}{\partial t^2}, v \right). \tag{3.90}$$

ii. The last term is similarly an L^2 inner product:

$$\int_{\mathcal{O}} f v dx = (f, v). \tag{3.91}$$

2.2.4 **Handle the Principal Part.** The second term is:

$$\int_{\mathcal{O}} (A^\varepsilon u_\varepsilon) v dx. \tag{3.92}$$

Substitute the definition of A^ε:

$$\int_{\mathcal{O}} (A^\varepsilon u_\varepsilon) v dx = \int_{\mathcal{O}} \left[-\frac{\partial}{\partial x_i} \left(a_{ij} \left(\frac{x}{\varepsilon} \right) \frac{\partial u_\varepsilon}{\partial x_j} \right) \right] v dx. \tag{3.93}$$

Apply integration by parts to transfer the derivative onto the test function v:

$$\begin{aligned} \int_{\mathcal{O}} (A^\varepsilon u_\varepsilon) v dx &= -\int_{\mathcal{O}} \frac{\partial}{\partial x_i} \left(a_{ij} \frac{\partial u_\varepsilon}{\partial x_j} \right) v dx \\ &= \int_{\mathcal{O}} a_{ij} \frac{\partial u_\varepsilon}{\partial x_j} \frac{\partial v}{\partial x_i} dx - \int_{\partial \mathcal{O}} \left(a_{ij} \frac{\partial u_\varepsilon}{\partial x_j} n_i \right) v dS. \end{aligned} \tag{3.94}$$

For the appropriate boundary conditions (e.g., $v = 0$ on $\partial\mathcal{O}$ for Dirichlet), the boundary term vanishes. Therefore:

$$\int_{\mathcal{O}} (A^\varepsilon u_\varepsilon) v dx = \int_{\mathcal{O}} a_{ij} \left(\frac{x}{\varepsilon} \right) \frac{\partial u_\varepsilon}{\partial x_j} \frac{\partial v}{\partial x_i} dx. \tag{3.95}$$

The right-hand side is the definition of the bilinear form:

$$a^\varepsilon(u_\varepsilon, v) = \int_{\mathcal{O}} a_{ij} \left(\frac{x}{\varepsilon} \right) \frac{\partial u_\varepsilon}{\partial x_j} \frac{\partial v}{\partial x_i} dx. \tag{3.96}$$

Thus, we have:

$$\int_{\mathcal{O}} (A^\varepsilon u_\varepsilon) v dx = a^\varepsilon(u_\varepsilon, v). \tag{3.97}$$

2.2.5 **Handle the Lower-Order Term.** The third term involves the first-order operator:

$$\int_{\mathcal{O}} (B^\varepsilon u_\varepsilon) v dx, \quad \text{where} \quad B^\varepsilon u_\varepsilon = b_j\left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon^k}\right) \frac{\partial u_\varepsilon}{\partial x_j}. \tag{3.98}$$

Crucial Distinction: Unlike the divergence-form operator A^ε, B^ε is a pointwise multiplication. It is not a derivative of a flux, but simply a function in $L^2(\mathcal{O})$. Thus, its product with a test function v is already an L^2-inner product, requiring no integration by parts:

$$\int_{\mathcal{O}} (B^\varepsilon u_\varepsilon) v dx = (B^\varepsilon u_\varepsilon, v)\,. \tag{3.99}$$

2.2.6 **Final Weak Form.** Combining these results, we obtain the weak formulation: Find u_ε satisfying the initial conditions (3.14) and the equation:

$$(u_\varepsilon'', v) + a^\varepsilon(u_\varepsilon, v) + (B^\varepsilon u_\varepsilon, v) = (f, v), \quad \forall v \in V. \tag{3.100}$$

3. A Priori Estimates and Weak Convergence.

3.1 Derivation of the Energy Identity.

3.1.1 **Choose the test function.** Let $v = u_\varepsilon'$ in the weak formulation (3.100):

$$(u_\varepsilon'', u_\varepsilon') + a^\varepsilon(u_\varepsilon, u_\varepsilon') + (B^\varepsilon u_\varepsilon, u_\varepsilon') = (f, u_\varepsilon'). \tag{3.101}$$

3.1.2 **Analyze each term:**

i. **Kinetic Energy:**

$$\begin{aligned}(u_\varepsilon'', u_\varepsilon') &= \int_{\mathcal{O}} \frac{\partial^2 u_\varepsilon}{\partial t^2} \frac{\partial u_\varepsilon}{\partial t} dx \\ &= \frac{1}{2}\frac{d}{dt}\int_{\mathcal{O}} \left(\frac{\partial u_\varepsilon}{\partial t}\right)^2 dx = \frac{1}{2}\frac{d}{dt}|u_\varepsilon'|^2.\end{aligned} \tag{3.102}$$

ii. **Potential Energy:** Since a^ε is time- independent and symmetric ($a_{ij} = a_{ji}$):

$$\begin{aligned}a^\varepsilon(u_\varepsilon, u_\varepsilon') &= \int_{\mathcal{O}} a_{ij}\left(\frac{x}{\varepsilon}\right) \frac{\partial u_\varepsilon}{\partial x_j} \frac{\partial u_\varepsilon'}{\partial x_i} dx \\ &= \frac{1}{2}\frac{d}{dt}\int_{\mathcal{O}} a_{ij}\left(\frac{x}{\varepsilon}\right) \frac{\partial u_\varepsilon}{\partial x_j} \frac{\partial u_\varepsilon}{\partial x_i} dx = \frac{1}{2}\frac{d}{dt} a^\varepsilon(u_\varepsilon, u_\varepsilon).\end{aligned} \tag{3.103}$$

iii. **Lower-Order Term:** $(B^\varepsilon u_\varepsilon, u_\varepsilon')$ remains.

iv. **Source Term:** (f, u_ε') remains.

3.1.3 **Combine results.** The sum of the first two terms is the derivative of the total energy

$E(t) = \frac{1}{2}|u_\varepsilon'|^2 + \frac{1}{2}a^\varepsilon(u_\varepsilon)$. This gives the energy identity:

$$\frac{1}{2}\frac{d}{dt}\left[|u_\varepsilon'|^2 + a^\varepsilon(u_\varepsilon)\right] + (B^\varepsilon u_\varepsilon, u_\varepsilon') = (f, u_\varepsilon'). \tag{3.104}$$

3.2 **Obtaining the Uniform Estimates.**

3.2.1 **Estimate the lower-order term.** Since B^ε is a bounded first-order operator ($|B^\varepsilon v| \leq C\|v\|_V$), we can control its effect using the Cauchy-Schwarz inequality:

$$|(B^\varepsilon u_\varepsilon, u_\varepsilon')| \leq |B^\varepsilon u_\varepsilon||u_\varepsilon'| \leq C\|u_\varepsilon\|_V|u_\varepsilon'|. \tag{3.105}$$

3.2.2 **Apply Young's inequality.** The product $\|u_\varepsilon\|_V|u_\varepsilon'|$ can be bounded using Young's inequality ($ab \leq \frac{1}{2}a^2 + \frac{1}{2}b^2$):

$$|(B^\varepsilon u_\varepsilon, u_\varepsilon')| \leq \frac{C}{2}\|u_\varepsilon\|_V^2 + \frac{C}{2}|u_\varepsilon'|^2. \tag{3.106}$$

Absorption of the Quadratic Growth: This estimate shows quadratic growth in the solution norms. However, the coercivity of a^ε provides a key lower bound for the total energy:

$$E(t) = \frac{1}{2}|u_\varepsilon'|^2 + \frac{1}{2}a^\varepsilon(u_\varepsilon) \geq \frac{1}{2}|u_\varepsilon'|^2 + \frac{\alpha}{2}\|u_\varepsilon\|_V^2. \tag{3.107}$$

Therefore, the negative of our estimate is bounded by the energy:

$$\frac{C}{2}\|u_\varepsilon\|_V^2 + \frac{C}{2}|u_\varepsilon'|^2 \leq \frac{C}{\min(1,\alpha)}E(t). \tag{3.108}$$

This means the potentially problematic term can be *absorbed*—it is controlled by a constant multiple of the positive definite energy $E(t)$ itself. This is the key step that allows the Grönwall argument to proceed.

3.2.3 **Recover estimates.** Following a Grönwall argument similar to Section 2.2, we recover the *same uniform estimates*:

$$\|u_\varepsilon\|_{L^\infty(0,T;V)} + \|u_\varepsilon'\|_{L^\infty(0,T;H)} \leq C. \tag{3.109}$$

3.3 **Weak Convergence Results.** The uniform estimates (3.109) imply weak* compactness. Therefore, there exists a subsequence such that:

$$u_\varepsilon \rightharpoonup^* u \quad \text{in } L^\infty(0,T;V), \tag{3.110}$$

$$u_\varepsilon' \rightharpoonup^* u' \quad \text{in } L^\infty(0,T;H). \tag{3.111}$$

Furthermore, since B^ε is a bounded operator and u_ε is uniformly bounded in $L^\infty(0,T;V)$, the sequence $B^\varepsilon u_\varepsilon$ is bounded in $L^\infty(0,T;H)$. Therefore, we also have (up to a subse-

quence):

$$B^\varepsilon u_\varepsilon \rightharpoonup^* \eta \quad \text{in } L^\infty(0,T;H). \tag{3.112}$$

4. The Homogenized Equation and the Limit of the Lower-Order Term.

4.1 **Passing to the Limit via the Method of Test Functions.** The weak formulation is:

$$(u_\varepsilon'', v) + a^\varepsilon(u_\varepsilon, v) + (B^\varepsilon u_\varepsilon, v) = (f, v), \quad \forall v \in V. \tag{1}$$

We have convergence results for u_ε, u_ε', and $B^\varepsilon u_\varepsilon \rightharpoonup^* \eta$, but *not* for u_ε''. To pass to the limit, we use the method of test functions.

4.1.1 **Multiply and Integrate in Time.** Let $\phi \in \mathcal{D}(0,T)$ be a smooth test function in time. Multiply equation (1) by $\phi(t)$ and integrate over $[0,T]$:

$$\int_0^T (u_\varepsilon'', v)\phi dt + \int_0^T a^\varepsilon(u_\varepsilon, v)\phi dt + \int_0^T (B^\varepsilon u_\varepsilon, v)\phi dt = \int_0^T (f, v)\phi dt. \tag{3.113}$$

4.1.2 **Handle the Second Derivative Term.** Focus on the first term. Using integration by parts twice in time (and noting ϕ has compact support, so boundary terms vanish), we transfer the time derivatives onto the smooth function ϕ:

$$\begin{aligned}
\int_0^T (u_\varepsilon'', v)\,\phi\,dt &= \int_0^T \left(\int_{\mathcal{O}} u_\varepsilon''(t,x)\, v(x)\, dx \right) \phi(t)\, dt \\
&= \int_{\mathcal{O}} \left(\int_0^T u_\varepsilon''(t,x)\, \phi(t)\, dt \right) v(x)\, dx \quad \text{(Fubini's Theorem)} \\
&= \int_{\mathcal{O}} \left(\Big[u_\varepsilon'(t,x)\, \phi(t) \Big]_0^T - \int_0^T u_\varepsilon'(t,x)\, \phi'(t)\, dt \right) v(x)\, dx \\
&= \int_{\mathcal{O}} \left(- \Big[u_\varepsilon(t,x)\, \phi'(t) \Big]_0^T + \int_0^T u_\varepsilon(t,x)\, \phi''(t)\, dt \right) v(x)\, dx \\
&= \int_{\mathcal{O}} \left(\int_0^T u_\varepsilon(t,x)\, \phi''(t)\, dt \right) v(x)\, dx \quad \text{(boundary terms vanish, } \phi \in C_c^\infty) \\
&= \int_0^T (u_\varepsilon, v)\, \phi''\, dt.
\end{aligned} \tag{3.114}$$

4.1.3 **The Averaged Equation.** Substitute result (3.114) back in. The equation becomes:

$$\int_0^T (u_\varepsilon, v)\phi'' dt + \int_0^T a^\varepsilon(u_\varepsilon, v)\phi dt + \int_0^T (B^\varepsilon u_\varepsilon, v)\phi dt = \int_0^T (f, v)\phi dt. \tag{3.115}$$

By linearity, this is equivalent to:

$$a^\varepsilon(u_\varepsilon(\phi), v) + (u_\varepsilon(\phi''), v) + (B^\varepsilon u_\varepsilon(\phi), v) = (f(\phi), v), \quad \forall v \in V, \tag{3.116}$$

where $u_\varepsilon(\phi) = \int_0^T u_\varepsilon \phi dt$ is the time-average of u_ε.

4.1.4 **Pass to the Limit.** Now we can pass to the limit $\varepsilon \to 0$ in (3.116):

$$\begin{aligned} a^\varepsilon(u_\varepsilon(\phi), v) &\to \mathcal{A}(u(\phi), v) \quad \text{(by elliptic homogenization)}, \\ (u_\varepsilon(\phi''), v) &\to (u(\phi''), v) \quad \text{(by weak convergence)}, \\ (B^\varepsilon u_\varepsilon(\phi), v) &\to (\eta(\phi), v) \quad \text{(by definition of } \eta), \\ (f(\phi), v) &\to (f(\phi), v). \end{aligned} \tag{3.117}$$

The limit of the averaged equation is therefore:

$$\mathcal{A}(u(\phi), v) + (u(\phi''), v) + (\eta(\phi), v) = (f(\phi), v), \quad \forall v \in V. \tag{3.118}$$

4.1.5 **Return to Pointwise Form.** Since the test function $\phi \in \mathcal{D}(0, T)$ was arbitrary, the above holds for all such ϕ. This implies that the equation must hold in the sense of distributions in time. Thus, for almost every time t, we recover the final homogenized equation:

$$(u'', v) + \mathcal{A}(u, v) + (\eta, v) = (f, v), \quad \forall v \in V. \tag{3.119}$$

4.2 **The Central Problem.**

$$\eta = \operatorname*{weak^*\,limit}_{\varepsilon \to 0} (B^\varepsilon u_\varepsilon) \tag{3.120}$$

The fundamental challenge is now to *identify the limit* η in terms of the limit u. This is a non-trivial problem of *homogenization of the lower-order term* because:

i. B^ε is a rapidly oscillating operator.

ii. u_ε only converges weakly to u.

iii. The product $B^\varepsilon u_\varepsilon$ involves interactions between oscillations in the operator and oscillations in the solution.

The following sections will be devoted to studying and computing this limit η.

3.1.4 Asymptotic Expansions (I)

We now perform a formal asymptotic analysis for the challenging case where the operator A^ε has coefficients oscillating in both space and time, as defined in (3.74):

$$A^\varepsilon v = -\frac{\partial}{\partial x_i}\left(a_{ij}\left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon}\right)\frac{\partial v}{\partial x_j}\right). \tag{3.121}$$

1. Two-Scale Expansion and Operators.

1.1 Fast Variables. To capture the rapid oscillations, we introduce the fast spatial variable $y = x/\varepsilon$ and the fast time variable $\tau = t/\varepsilon$.

1.2 Transformation of Derivatives. The partial derivatives transform according to the chain rule:

$$\frac{\partial}{\partial x_i} \to \frac{\partial}{\partial x_i} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_i}, \quad \frac{\partial}{\partial t} \to \frac{\partial}{\partial t} + \frac{1}{\varepsilon}\frac{\partial}{\partial \tau}. \tag{3.122}$$

Consequently, the second time derivative becomes:

$$\begin{aligned}\frac{\partial^2}{\partial t^2} &= \left(\frac{\partial}{\partial t} + \frac{1}{\varepsilon}\frac{\partial}{\partial \tau}\right)\left(\frac{\partial}{\partial t} + \frac{1}{\varepsilon}\frac{\partial}{\partial \tau}\right) \\ &= \frac{\partial}{\partial t}\left(\frac{\partial}{\partial t}\right) + \frac{\partial}{\partial t}\left(\frac{1}{\varepsilon}\frac{\partial}{\partial \tau}\right) + \frac{1}{\varepsilon}\frac{\partial}{\partial \tau}\left(\frac{\partial}{\partial t}\right) + \frac{1}{\varepsilon}\frac{\partial}{\partial \tau}\left(\frac{1}{\varepsilon}\frac{\partial}{\partial \tau}\right) \\ &= \frac{\partial^2}{\partial t^2} + \frac{1}{\varepsilon}\frac{\partial^2}{\partial t \partial \tau} + \frac{1}{\varepsilon}\frac{\partial^2}{\partial \tau \partial t} + \frac{1}{\varepsilon^2}\frac{\partial^2}{\partial \tau^2}.\end{aligned} \tag{3.123}$$

Assuming sufficient smoothness so that $\frac{\partial^2}{\partial t \partial \tau} = \frac{\partial^2}{\partial \tau \partial t}$, we obtain the final expression:

$$\frac{\partial^2}{\partial t^2} = \frac{\partial^2}{\partial t^2} + \frac{2}{\varepsilon}\frac{\partial^2}{\partial t \partial \tau} + \frac{1}{\varepsilon^2}\frac{\partial^2}{\partial \tau^2}. \tag{3.124}$$

1.3 Expansion of the Operator. We substitute the derivative transformations into the full wave operator:

$$\begin{aligned}\frac{\partial^2}{\partial t^2} + A^\varepsilon = &\left(\frac{\partial^2}{\partial t^2} + \frac{2}{\varepsilon}\frac{\partial^2}{\partial t \partial \tau} + \frac{1}{\varepsilon^2}\frac{\partial^2}{\partial \tau^2}\right) \\ &+ \left[-\left(\frac{\partial}{\partial x_i} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_i}\right)\left(a_{ij}(y,\tau)\left(\frac{\partial}{\partial x_j} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_j}\right)\right)\right].\end{aligned} \tag{3.125}$$

Expanding the spatial operator A^ε:

$$\begin{aligned}A^\varepsilon &= -\left(\frac{\partial}{\partial x_i} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_i}\right)\left(a_{ij}\frac{\partial}{\partial x_j} + \frac{a_{ij}}{\varepsilon}\frac{\partial}{\partial y_j}\right) \\ &= -\left[\frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial}{\partial x_j}\right) + \frac{1}{\varepsilon}\frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial}{\partial y_j}\right) + \frac{1}{\varepsilon}\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial}{\partial x_j}\right) + \frac{1}{\varepsilon^2}\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial}{\partial y_j}\right)\right].\end{aligned} \tag{3.126}$$

Now combining with the time derivatives and grouping by powers of ε:

$$\begin{aligned}\frac{\partial^2}{\partial t^2} + A^\varepsilon = &\frac{1}{\varepsilon^2}\left[\frac{\partial^2}{\partial \tau^2} - \frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial}{\partial y_j}\right)\right] \\ &+ \frac{1}{\varepsilon}\left[2\frac{\partial^2}{\partial t \partial \tau} - \frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial}{\partial y_j}\right) - \frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial}{\partial x_j}\right)\right] \\ &+ \varepsilon^0\left[\frac{\partial^2}{\partial t^2} - \frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial}{\partial x_j}\right)\right] + \mathcal{O}(\varepsilon).\end{aligned} \tag{3.127}$$

This yields the expansion:

$$\frac{\partial^2}{\partial t^2} + A^\varepsilon = \varepsilon^{-2} Q_1 + \varepsilon^{-1} Q_2 + \varepsilon^0 Q_3 + \mathcal{O}(\varepsilon), \tag{3.128}$$

where the operators are defined as:

$$Q_1 = \frac{\partial^2}{\partial \tau^2} + A_1, \qquad A_1 = -\frac{\partial}{\partial y_i}\left(a_{ij}(y,\tau)\frac{\partial}{\partial y_j}\right), \tag{3.129}$$

$$Q_2 = 2\frac{\partial^2}{\partial t \partial \tau} + A_2, \qquad A_2 = -\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial}{\partial x_j}\right) - \frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial}{\partial y_j}\right), \tag{3.130}$$

$$Q_3 = \frac{\partial^2}{\partial t^2} + A_3, \qquad A_3 = -\frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial}{\partial x_j}\right). \tag{3.131}$$

1.4 Ansatz for the Solution. We postulate a two-scale asymptotic expansion for the solution u^ε:

$$u_\varepsilon(x,t) = u_0(x,t,y,\tau) + \varepsilon u_1(x,t,y,\tau) + \varepsilon^2 u_2(x,t,y,\tau) + \cdots, \tag{3.132}$$

where $y = x/\varepsilon$, $\tau = t/\varepsilon$, and all functions u_k are assumed to be Y-periodic in y and τ_0-periodic in τ.

2. Hierarchy of Equations.

2.1 Substitution and Matching. We substitute the expansion for u_ε and the operator into the original PDE:

$$\left(\varepsilon^{-2} Q_1 + \varepsilon^{-1} Q_2 + Q_3 + \cdots\right)(u_0 + \varepsilon u_1 + \varepsilon^2 u_2 + \cdots) = f(x,t). \tag{3.133}$$

2.2 Order-by-Order Equations. Matching coefficients of powers of ε yields a hierarchy of equations:

$$\mathcal{O}(\varepsilon^{-2}): \quad Q_1 u_0 = 0, \tag{3.134}$$

$$\mathcal{O}(\varepsilon^{-1}): \quad Q_1 u_1 + Q_2 u_0 = 0, \tag{3.135}$$

$$\mathcal{O}(\varepsilon^{0}): \quad Q_1 u_2 + Q_2 u_1 + Q_3 u_0 = f. \tag{3.136}$$

The forcing $f(x,t)$ appears only at order $\mathcal{O}(1)$ because it is assumed to vary only on the slow (macroscopic) scale.

3. Solvability Condition and the Cell Problem.

3.1 Derivation of Solvability Condition. Consider the equation $Q_1 \phi = F(y,\tau)$ where $Q_1 = \frac{\partial^2}{\partial \tau^2} + A_1$. To understand when solutions exist, consider the homogeneous adjoint problem: find all Y-τ_0-periodic ψ such that $Q_1^* \psi = 0$.

For self-adjoint Q_1 (as in our case) with only constant periodic solutions to $Q_1\psi = 0$, the Fredholm alternative requires:

$$\langle F, \psi \rangle = 0 \quad \text{for all } \psi \text{ satisfying } Q_1^* \psi = 0, \tag{3.137}$$

where $\langle F, \psi \rangle = \int_0^{\tau_0} \int_Y F(y,\tau)\psi(y,\tau)dyd\tau$ is the L^2 inner product.

3.2 Mean Zero Condition. When the kernel consists only of constants ($\psi = c$), the condition becomes:

$$\langle F, \psi \rangle = 0 \implies \int_0^{\tau_0} \int_Y F(y,\tau)dyd\tau = 0. \tag{3.138}$$

Normalizing by the cell volume gives the mean-zero condition:

$$\mathcal{M}(F) = \frac{1}{|Y|\tau_0} \int_0^{\tau_0} \int_Y F(y,\tau)dyd\tau = 0. \tag{3.139}$$

This condition ensures that F contains no component in the direction of the kernel (constants).

4. Solving the Equations.

4.1 Leading Order ($\mathcal{O}(\varepsilon^{-2})$).

4.1.1 Equation (3.134) gives: $Q_1 u_0 = 0$.

4.1.2 The solvability condition (3.139) requires $\mathcal{M}(0) = 0$, which is trivially satisfied.

4.1.3 From Section 3.2, the only periodic solutions to $Q_1 u_0 = 0$ are constants in (y, τ).

4.1.4 Therefore, u_0 must be independent of the fast variables:

$$u_0 = u_0(x,t). \tag{3.140}$$

4.2 First Order Equation ($\mathcal{O}(\varepsilon^{-1})$).

4.2.1 Equation (3.135) gives: $Q_1 u_1 = -Q_2 u_0$.

4.2.2 Since $u_0 = u_0(x,t)$ is independent of (y,τ), compute $Q_2 u_0$:

$$\begin{aligned} Q_2 u_0 &= \left[2\frac{\partial^2}{\partial t \partial \tau} + A_2 \right] u_0(x,t) \\ &= 2\frac{\partial^2 u_0}{\partial t \partial \tau} - \frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial u_0}{\partial x_j} \right) - \frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial u_0}{\partial y_j} \right) \end{aligned} \tag{3.141}$$

4.2.3 The first term vanishes: $\frac{\partial^2 u_0}{\partial t \partial \tau} = 0$ (since u_0 is τ-independent).

4.2.4 The third term vanishes: $\frac{\partial u_0}{\partial y_j} = 0$ (since u_0 is y-independent).

4.2.5 Thus: $Q_2 u_0 = -\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial u_0}{\partial x_j} \right)$.

4.2.6 The equation becomes:

$$Q_1 u_1 = \frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial u_0}{\partial x_j} \right). \tag{3.142}$$

4.3 Solvability for u_1.

4.3.1 For solvability, require: $\mathcal{M}\left(\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial u_0}{\partial x_j}\right)\right) = 0$.

4.3.2 Since $\frac{\partial u_0}{\partial x_j}$ is y-independent, it can be moved outside the average:

$$\frac{\partial u_0}{\partial x_j}\mathcal{M}\left(\frac{\partial a_{ij}}{\partial y_i}\right) = 0. \tag{3.143}$$

4.3.3 Applying the divergence theorem to the spatial integral gives:

$$\int_Y \frac{\partial a_{ij}}{\partial y_i} dy = \int_{\partial Y} a_{ij} n_i dS, \tag{3.144}$$

where $\mathbf{n}$ is the outward unit normal. Due to the Y-periodicity of a_{ij}, the boundary integrals over opposite faces cancel exactly. Therefore:

$$\mathcal{M}\left(\frac{\partial a_{ij}}{\partial y_i}\right) = \frac{1}{|Y|\tau_0}\int_0^{\tau_0}\left(\int_{\partial Y} a_{ij} n_i dS\right) d\tau = 0. \tag{3.145}$$

where boundary terms cancel. Hence, the solvability condition is satisfied.

4.4 Cell Problem and Correctors.

4.4.1 The linearity of $Q_1 u_1 = \frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial u_0}{\partial x_j}\right)$ and the structure of its right-hand side suggest a particular solution of the form:

$$u_1^{\text{part}} = -\chi^j(y,\tau)\frac{\partial u_0}{\partial x_j}(x,t). \tag{3.146}$$

The general solution is this particular solution plus any solution of the homogeneous equation $Q_1 u_1^{\text{hom}} = 0$:

$$u_1(x,t,y,\tau) = -\chi^j(y,\tau)\frac{\partial u_0}{\partial x_j}(x,t) + \hat{u}_1(x,t), \tag{3.147}$$

where $Q_1(\hat{u}_1) = 0$. Since the only periodic solutions are constants in (y,τ) (Section 3.2), $\hat{u}_1$ depends only on the slow variables: $\hat{u}_1 = \hat{u}_1(x,t)$.

4.4.2 Substitute into $Q_1 u_1 = \frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial u_0}{\partial x_j}\right)$:

$$\begin{aligned} Q_1(u_1) &= Q_1\left(-\chi^j\frac{\partial u_0}{\partial x_j} + \hat{u}_1\right) = -Q_1(\chi^j)\frac{\partial u_0}{\partial x_j} + Q_1(\hat{u}_1) \\ &= -Q_1(\chi^j)\frac{\partial u_0}{\partial x_j} + 0. \end{aligned} \tag{3.148}$$

Setting this equal to the right-hand side:

$$-Q_1(\chi^j)\frac{\partial u_0}{\partial x_j} = \frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial u_0}{\partial x_j}\right). \tag{3.149}$$

4.4.3 Since this must hold for arbitrary macroscopic gradients $\frac{\partial u_0}{\partial x_j}$, we equate the coefficients for each j:

$$Q_1(\chi^j) = \frac{\partial^2 \chi^j}{\partial \tau^2} + A_1(\chi^j) = -\frac{\partial}{\partial y_i} a_{ij}(y, \tau). \tag{3.150}$$

4.4.4 The function $\hat{u}_1(x, t)$ represents the homogeneous solution and will be determined by higher-order solvability conditions.

4.5 Homogenized Equation ($\mathcal{O}(1)$).

4.5.1 Equation (3.136) gives: $Q_1 u_2 = f - Q_2 u_1 - Q_3 u_0$.

4.5.2 This equation is of the form $Q_1 u_2 = F$. For a periodic solution u_2 to exist, the Fredholm alternative (solvability condition (3.139)) requires that the average of the forcing term must be zero:

$$\mathcal{M}\left(f - Q_2 u_1 - Q_3 u_0\right) = 0. \tag{3.151}$$

4.5.3 Since $f = f(x, t)$ is independent of the fast variables (y, τ), it acts as a constant during the averaging process:

$$\begin{aligned} \mathcal{M}(f) &= \frac{1}{|Y|\tau_0} \int_0^{\tau_0} \int_Y f(x,t) dy d\tau = f(x,t) \cdot \frac{1}{|Y|\tau_0} \int_0^{\tau_0} \int_Y 1 dy d\tau \\ &= f(x,t) \cdot \frac{1}{|Y|\tau_0} \cdot |Y| \cdot \tau_0 = f(x,t). \end{aligned} \tag{3.152}$$

4.5.4 Rearranging the condition gives the governing equation:

$$\mathcal{M}\left(Q_2 u_1 + Q_3 u_0\right) = f(x, t). \tag{3.153}$$

4.5.5 This **homogenized equation** for u_0 incorporates all microstructural effects. Its evaluation (Section 5) yields the effective coefficients q_{ij}.

5. The Homogenized Equation.

5.1 Computing the Average. We compute the terms in (3.153) individually.

5.1.1 **First term:** Since u_0 is independent of fast variables (y, τ), we have:

$$\begin{aligned} \mathcal{M}(Q_3 u_0) &= \mathcal{M}\left(\frac{\partial^2 u_0}{\partial t^2} + A_3 u_0\right) \\ &= \frac{\partial^2 u_0}{\partial t^2} + \mathcal{M}\left(-\frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial}{\partial x_j}\right) u_0\right) \\ &= \frac{\partial^2 u_0}{\partial t^2} - \frac{\partial}{\partial x_i}\left(\mathcal{M}(a_{ij})\frac{\partial u_0}{\partial x_j}\right). \end{aligned} \tag{3.154}$$

The operators $\frac{\partial}{\partial x_i}$ and $\mathcal{M}$ commute, acting on different variables.

5.1.2 **Second Term:** For $\mathcal{M}(Q_2 u_1)$, substitute the expression for u_1 from (3.147):

$$\begin{aligned} Q_2 u_1 &= Q_2 \left(-\chi^k \frac{\partial u_0}{\partial x_k} + \hat{u}_1 \right) \\ &= -Q_2(\chi^k) \frac{\partial u_0}{\partial x_k} + Q_2(\hat{u}_1). \end{aligned} \tag{3.155}$$

Since $\hat{u}_1$ is independent of (y, τ), $Q_2(\hat{u}_1) = A_2 \hat{u}_1$. The average of this term is zero due to periodicity, following the same reasoning as in step **4.3**. Specifically:

$$\begin{aligned} \mathcal{M}(Q_2(\hat{u}_1)) &= \mathcal{M}\left(A_2 \hat{u}_1\right) \\ &= \mathcal{M}\left[-\frac{\partial}{\partial y_i}\left(a_{ij} \frac{\partial \hat{u}_1}{\partial x_j} \right) - \frac{\partial}{\partial x_i}\left(a_{ij} \frac{\partial \hat{u}_1}{\partial y_j} \right) \right] \\ &= -\mathcal{M}\left(\frac{\partial}{\partial y_i}\left(a_{ij} \frac{\partial \hat{u}_1}{\partial x_j} \right) \right) - \mathcal{M}\left(\frac{\partial}{\partial x_i}\left(a_{ij} \frac{\partial \hat{u}_1}{\partial y_j} \right) \right). \end{aligned} \tag{3.156}$$

i. Since $\frac{\partial \hat{u}_1}{\partial x_j}$ is independent of y, we have:

$$\mathcal{M}\left(\frac{\partial}{\partial y_i}\left(a_{ij} \frac{\partial \hat{u}_1}{\partial x_j} \right) \right) = \frac{\partial \hat{u}_1}{\partial x_j} \mathcal{M}\left(\frac{\partial a_{ij}}{\partial y_i} \right). \tag{3.157}$$

As shown in step 4.3.3, $\mathcal{M}\left(\frac{\partial a_{ij}}{\partial y_i} \right) = 0$ by the divergence theorem and Y-periodicity.

ii. Since $\hat{u}_1 = \hat{u}_1(x, t)$ is independent of y, we have $\frac{\partial \hat{u}_1}{\partial y_j} = 0$, so:

$$\mathcal{M}\left(\frac{\partial}{\partial x_i}\left(a_{ij} \frac{\partial \hat{u}_1}{\partial y_j} \right) \right) = \mathcal{M}\left(\frac{\partial}{\partial x_i}\left(a_{ij} \cdot 0 \right) \right) = 0. \tag{3.158}$$

Therefore:

$$\mathcal{M}(Q_2(\hat{u}_1)) = 0. \tag{3.159}$$

Now compute $Q_2(\chi^k)$ explicitly:

$$\begin{aligned} Q_2(\chi^k) &= 2\frac{\partial^2 \chi^k}{\partial t \partial \tau} + A_2(\chi^k) \\ &= 0 + \left[-\frac{\partial}{\partial y_i}\left(a_{ij} \frac{\partial \chi^k}{\partial x_j} \right) - \frac{\partial}{\partial x_i}\left(a_{ij} \frac{\partial \chi^k}{\partial y_j} \right) \right] \\ &= -\frac{\partial}{\partial x_i}\left(a_{ij} \frac{\partial \chi^k}{\partial y_j} \right), \end{aligned} \tag{3.160}$$

since χ^k does not depend on x (it's a function of (y, τ) only) and the mixed time derivative vanishes for the same reason. Thus, the average becomes:

$$\mathcal{M}(Q_2 u_1) = \frac{\partial}{\partial x_i}\left(\mathcal{M}\left(a_{ij} \frac{\partial \chi^k}{\partial y_j} \right) \frac{\partial u_0}{\partial x_k} \right). \tag{3.161}$$

5.2 Final Homogenized Equation. Combining both terms and using (3.153):

$$\mathcal{M}(Q_3 u_0) + \mathcal{M}(Q_2 u_1) = f(x,t) \tag{3.162}$$

$$\frac{\partial^2 u_0}{\partial t^2} - \frac{\partial}{\partial x_i}\left(\mathcal{M}(a_{ij})\frac{\partial u_0}{\partial x_j}\right) + \frac{\partial}{\partial x_i}\left(\mathcal{M}\left(a_{ik}\frac{\partial \chi^j}{\partial y_k}\right)\frac{\partial u_0}{\partial x_j}\right) = f(x,t). \tag{3.163}$$

Combining the two terms with spatial derivatives and relabeling indices ($k \to j$ in the second term):

$$\frac{\partial^2 u_0}{\partial t^2} - \frac{\partial}{\partial x_i}\left[\left(\mathcal{M}(a_{ij}) - \mathcal{M}\left(a_{ik}\frac{\partial \chi^j}{\partial y_k}\right)\right)\frac{\partial u_0}{\partial x_j}\right] = f. \tag{3.164}$$

This gives the final homogenized wave equation:

$$\frac{\partial^2 u_0}{\partial t^2} - \frac{\partial}{\partial x_i}\left(q_{ij}\frac{\partial u_0}{\partial x_j}\right) = f, \tag{3.165}$$

where the *effective coefficients* are:

$$\begin{aligned} q_{ij} &= \mathcal{M}\left(a_{ij} - a_{ik}\frac{\partial \chi^j}{\partial y_k}\right) \\ &= \frac{1}{|Y|\tau_0}\int_0^{\tau_0}\int_Y \left(a_{ij}(y,\tau) - a_{ik}(y,\tau)\frac{\partial \chi^j}{\partial y_k}(y,\tau)\right) dy d\tau. \end{aligned} \tag{3.166}$$

The resulting equation is a wave equation with a constant, homogenized tensor $\mathbf{q}$ that captures the net effect of the microstructure.

6. Open Problem.

6.1 Formal Derivation. This analysis is formal, assuming existence of smooth periodic solutions to cell problems and validity of the asymptotic expansion.

6.2 Justification Challenge. The fundamental open problem is rigorous justification: proving $u_\varepsilon \to u_0$ strongly as $\varepsilon \to 0$. The lack of uniform a priori estimates (due to spatio-temporal oscillations in coefficients) makes this highly non-trivial.

3.1.5 Asymptotic Expansions (II): Rigorous Justification for a First-Order Term

We now rigorously justify the homogenization process for the case where the principal part A^ε is time-independent, but the first-order term B^ε oscillates rapidly in time. We focus on the case $k = 1$ in (3.87), meaning the time oscillations are on the same scale as the spatial oscillations ($\tau = t/\varepsilon$).

1.1 Operator Expansion. The expansion of the full operator $\frac{\partial^2}{\partial t^2} + A^\varepsilon + B^\varepsilon$ is built upon the multi-scale expansion of its components.

1.1.1 **Time Derivative Expansion:** Using $\frac{\partial}{\partial t} = \frac{\partial}{\partial t} + \frac{1}{\varepsilon}\frac{\partial}{\partial \tau}$ to the second derivative:

$$\frac{\partial^2}{\partial t^2} = \left(\frac{\partial}{\partial t} + \frac{1}{\varepsilon}\frac{\partial}{\partial \tau}\right)^2 = \frac{\partial^2}{\partial t^2} + \frac{2}{\varepsilon}\frac{\partial^2}{\partial t \partial \tau} + \frac{1}{\varepsilon^2}\frac{\partial^2}{\partial \tau^2}. \tag{3.167}$$

All of these terms must be kept. The term $\frac{1}{\varepsilon^2}\frac{\partial^2}{\partial \tau^2}$ is of leading order.

1.1.2 **Spatial Operator A^ε Expansion:** Using $\frac{\partial}{\partial x_i} = \frac{\partial}{\partial x_i} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_i}$, the operator expands as:

$$A^\varepsilon = \varepsilon^{-2}A_1 + \varepsilon^{-1}A_2 + \varepsilon^0 A_3, \tag{3.168}$$

where the operators A_1, A_2, A_3 are defined as:

$$A_1 = -\frac{\partial}{\partial y_i}\left(a_{ij}(y)\frac{\partial}{\partial y_j}\right), \tag{3.169}$$

$$A_2 = -\frac{\partial}{\partial y_i}\left(a_{ij}(y)\frac{\partial}{\partial x_j}\right) - \frac{\partial}{\partial x_i}\left(a_{ij}(y)\frac{\partial}{\partial y_j}\right), \tag{3.170}$$

$$A_3 = -\frac{\partial}{\partial x_i}\left(a_{ij}(y)\frac{\partial}{\partial x_j}\right). \tag{3.171}$$

1.1.3 **First-Order Operator B^ε Expansion:** The first-order term is $B^\varepsilon = -b_j(x/\varepsilon, t/\varepsilon)\frac{\partial}{\partial x_j}$. Introducing the fast time variable $\tau = t/\varepsilon$ and applying the chain rule $\frac{\partial}{\partial x_j} = \frac{\partial}{\partial x_j} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_j}$ yields its expansion:

$$B^\varepsilon = -b_j(y,\tau)\frac{\partial}{\partial x_j} = -b_j(y,\tau)\left(\frac{\partial}{\partial x_j} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_j}\right) = \varepsilon^{-1}B_2 + B_3, \tag{3.172}$$

where:

$$B_2 = -b_j(y,\tau)\frac{\partial}{\partial y_j}, \tag{3.173}$$

$$B_3 = -b_j(y,\tau)\frac{\partial}{\partial x_j}. \tag{3.174}$$

1.1.4 **Full Operator Combination:** Combining the expansions for $\frac{\partial^2}{\partial t^2}$, A^ε, and B^ε, the full operator becomes:

$$\begin{aligned}\frac{\partial^2}{\partial t^2} + A^\varepsilon + B^\varepsilon = &\left(\frac{\partial^2}{\partial t^2} + \frac{2}{\varepsilon}\frac{\partial^2}{\partial t \partial \tau} + \frac{1}{\varepsilon^2}\frac{\partial^2}{\partial \tau^2}\right) \\ &+ \left(\varepsilon^{-2}A_1 + \varepsilon^{-1}A_2 + \varepsilon^0 A_3\right) \\ &+ \left(\varepsilon^{-1}B_2 + B_3\right).\end{aligned} \tag{3.175}$$

Grouping terms at each order and using the conventional definition from the reference

yields:

$$\frac{\partial^2}{\partial t^2} + A^\varepsilon + B^\varepsilon = \varepsilon^{-2} R_1 + \varepsilon^{-1} R_2 + \varepsilon^0 R_3, \tag{3.176}$$

where the operators R_1, R_2, R_3 are defined as:

$$R_1 = -\frac{\partial^2}{\partial \tau^2} + A_1, \tag{3.177}$$

$$R_2 = 2\frac{\partial^2}{\partial t \partial \tau} + A_2 + B_2, \qquad \text{(with } B_2 \text{ from (3.173))}, \tag{3.178}$$

$$R_3 = \frac{\partial^2}{\partial t^2} + A_3 + B_3. \qquad \text{(with } A_3, B_3 \text{ from (3.171), (3.174))}. \tag{3.179}$$

The sign convention $R_1 = -\frac{\partial^2}{\partial \tau^2} + A_1$ is standard for wave operators, where A_1 is a negative-definite elliptic operator.

1.2 Hierarchy of Equations and Homogenization. We substitute the asymptotic expansion for u_ε,

$$u_\varepsilon(x,t) = u_0(x,y,t,\tau) + \varepsilon u_1(x,y,t,\tau) + \varepsilon^2 u_2(x,y,t,\tau) + \dots, \tag{3.180}$$

into the expanded operator $\varepsilon^{-2}R_1 + \varepsilon^{-1}R_2 + \varepsilon^0 R_3$, where $y = x/\varepsilon, \tau = t/\varepsilon$. Distributing the operator and grouping by powers of ε yields:

$$\begin{aligned} &\varepsilon^{-2} R_1 u_0 \\ &+\varepsilon^{-1} \left(R_1 u_1 + R_2 u_0 \right) \\ &+\varepsilon^{0} \left(R_1 u_2 + R_2 u_1 + R_3 u_0 \right) \\ &+\varepsilon^{1} \left(R_2 u_2 + R_3 u_1 \right) + \dots = f. \end{aligned} \tag{3.181}$$

For this equation to hold for an arbitrary small parameter ε, the coefficient of each power of ε must be zero. This yields the following hierarchy of equations:

$$R_1 u_0 = 0, \tag{3.182}$$

$$R_1 u_1 + R_2 u_0 = 0, \tag{3.183}$$

$$R_1 u_2 + R_2 u_1 + R_3 u_0 = f. \tag{3.184}$$

1.2.1. Solving the Leading Order Equations: The first equation, $R_1 u_0 = 0$, implies that u_0 is independent of the fast variables:

$$u_0 = u_0(x,t). \tag{3.185}$$

The second equation in the hierarchy is:

$$R_1 u_1 + R_2 u_0 = 0. \tag{3.186}$$

We analyze this equation to derive the cell problem for the first-order corrector.

1.2.1.1 **Simplify Equation (3.186)**. Since $u_0 = u_0(x,t)$ is independent of the fast variables y and τ, and $R_2 = 2\frac{\partial^2}{\partial t \partial \tau} + A_2 + B_2$, we have:

$$2\frac{\partial^2 u_0}{\partial t \partial \tau} = 0, \quad B_2 u_0 = -b_j(y,\tau)\frac{\partial u_0}{\partial y_j} = 0. \tag{3.187}$$

Therefore, equation (3.186) simplifies to:

$$R_1 u_1 + A_2 u_0 = 0. \tag{3.188}$$

Furthermore, since $A_2 u_0$ is independent of τ, we seek a solution u_1 that is also independent of τ. This implies $\frac{\partial^2 u_1}{\partial \tau^2} = 0$, and thus $R_1 u_1 = A_1 u_1$. The equation becomes:

$$A_1 u_1 = -A_2 u_0. \tag{3.189}$$

1.2.1.2 **Postulate the Ansatz for u_1.** We postulate that the dependence of u_1 on the fast spatial variable y is linear in the gradients of u_0. This leads to the ansatz:

$$u_1(x,y,t) = -\chi^j(y)\frac{\partial u_0}{\partial x_j}(x,t) + \hat{u}_1(x,t). \tag{3.190}$$

Here, $\chi^j(y)$ are the correctors that capture the fine-scale effects, and $\hat{u}_1(x,t)$ is a function only of the slow variables.

1.2.1.3 **Compute the Right-Hand Side, $-A_2 u_0$.** Recall the definition of A_2:

$$A_2 = -\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial}{\partial x_j}\right) - \frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial}{\partial y_j}\right). \tag{3.191}$$

Apply A_2 to u_0. Since u_0 is independent of y, all terms involving $\frac{\partial u_0}{\partial y_j}$ vanish:

$$A_2 u_0 = -\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial u_0}{\partial x_j}\right) - \frac{\partial}{\partial x_i}\left(a_{ij}\cdot 0\right) = -\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial u_0}{\partial x_j}\right). \tag{3.192}$$

Since $\frac{\partial u_0}{\partial x_j}$ is independent of y, it can be treated as a constant for the y-derivative:

$$A_2 u_0 = -\frac{\partial a_{ij}}{\partial y_i}\frac{\partial u_0}{\partial x_j} \implies -A_2 u_0 = \frac{\partial a_{ij}}{\partial y_i}\frac{\partial u_0}{\partial x_j}. \tag{3.193}$$

1.2.1.4 **Compute the Left-Hand Side, $A_1 u_1$.** Now compute $A_1 u_1$ using the ansatz (3.190). Re-

call $A_1 = -\frac{\partial}{\partial y_i}\left(a_{ik}(y)\frac{\partial}{\partial y_k}\right)$. Since $\hat{u}_1$ is independent of y, $A_1\hat{u}_1 = 0$. Thus:

$$\begin{aligned} A_1 u_1 &= A_1\left(-\chi^j \frac{\partial u_0}{\partial x_j}\right) = -\frac{\partial}{\partial y_i}\left[a_{ik}(y)\frac{\partial}{\partial y_k}\left(-\chi^j\frac{\partial u_0}{\partial x_j}\right)\right] \\ &= -\frac{\partial}{\partial y_i}\left[-a_{ik}(y)\frac{\partial \chi^j}{\partial y_k}\frac{\partial u_0}{\partial x_j}\right] = \frac{\partial}{\partial y_i}\left[a_{ik}(y)\frac{\partial \chi^j}{\partial y_k}\frac{\partial u_0}{\partial x_j}\right]. \end{aligned} \tag{3.194}$$

Since $\frac{\partial u_0}{\partial x_j}$ is independent of y, it can be factored out of the y-derivative:

$$A_1 u_1 = \frac{\partial}{\partial y_i}\left[a_{ik}(y)\frac{\partial \chi^j}{\partial y_k}\right]\frac{\partial u_0}{\partial x_j}. \tag{3.195}$$

1.2.1.5 **Substitute into the Equation and Derive the Cell Problem**. Substitute results (3.193) and (3.195) into equation (3.189):

$$\frac{\partial}{\partial y_i}\left[a_{ik}(y)\frac{\partial \chi^j}{\partial y_k}\right]\frac{\partial u_0}{\partial x_j} = \frac{\partial a_{ij}}{\partial y_i}\frac{\partial u_0}{\partial x_j}. \tag{3.196}$$

For this equality to hold for arbitrary functions u_0 (and thus arbitrary gradients $\frac{\partial u_0}{\partial x_j}$), the coefficients must satisfy, for each index j:

$$\frac{\partial}{\partial y_i}\left(a_{ik}\frac{\partial \chi^j}{\partial y_k}\right) = \frac{\partial a_{ij}}{\partial y_i}. \tag{3.197}$$

The left-hand side of (3.197) is recognized as $-A_1\chi^j$, because $A_1 = -\frac{\partial}{\partial y_i}\left(a_{ik}\frac{\partial}{\partial y_k}\right)$. Therefore:

$$-A_1\chi^j = \frac{\partial a_{ij}}{\partial y_i}. \tag{3.198}$$

To express the right-hand side, consider the action of A_1 on the function y_j:

$$\begin{aligned} A_1 y_j &= -\frac{\partial}{\partial y_i}\left(a_{ik}\frac{\partial y_j}{\partial y_k}\right) = -\frac{\partial}{\partial y_i}\left(a_{ik}\delta_{jk}\right) \quad \left(\text{since } \frac{\partial y_j}{\partial y_k} = \delta_{jk}\right) \\ &= -\frac{\partial}{\partial y_i}\left(a_{ij}\right) = -\frac{\partial a_{ij}}{\partial y_i}. \end{aligned} \tag{3.199}$$

Therefore, we have the identity:

$$\frac{\partial a_{ij}}{\partial y_i} = -A_1 y_j. \tag{3.200}$$

Substituting this into the right-hand side of equation (3.197) yields:

$$-A_1\chi^j = -A_1 y_j \implies A_1(\chi^j - y_j) = 0. \tag{3.201}$$

This defines the correctors $\chi^j(y)$, which are independent of τ because A_1 is.

1.2.2. Deriving the Homogenized Equation: The homogenized equation arises from the solvability condition for the $\mathcal{O}(1)$ equation:

$$R_1 u_2 + R_2 u_1 + R_3 u_0 = f. \tag{3.202}$$

A necessary condition for a solution u_2 to exist is that the right-hand side is orthogonal to the null space of R_1. This is enforced by averaging over the fast variables $Y \times [0, \tau_0]$. Since $\mathcal{M}(R_1 u_2) = 0$, we obtain:

$$\mathcal{M}(R_2 u_1 + R_3 u_0) = f. \tag{3.203}$$

Compute $\mathcal{M}(R_3 u_0)$:

$$R_3 u_0 = \frac{\partial^2 u_0}{\partial t^2} + A_3 u_0 + B_3 u_0. \tag{3.204}$$

$$\mathcal{M}(A_3 u_0) = \mathcal{M}\left(-\frac{\partial}{\partial x_i}\left(a_{ij}(y)\frac{\partial u_0}{\partial x_j}\right)\right) = -\frac{\partial}{\partial x_i}\left(\mathcal{M}(a_{ij})\frac{\partial u_0}{\partial x_j}\right), \tag{3.205}$$

$$\mathcal{M}(B_3 u_0) = \mathcal{M}\left(-b_j(y,\tau)\frac{\partial u_0}{\partial x_j}\right) = -\mathcal{M}(b_j)\frac{\partial u_0}{\partial x_j}. \tag{3.206}$$

Therefore,

$$\mathcal{M}(R_3 u_0) = \frac{\partial^2 u_0}{\partial t^2} - \frac{\partial}{\partial x_i}\left(\mathcal{M}(a_{ij})\frac{\partial u_0}{\partial x_j}\right) - \mathcal{M}(b_j)\frac{\partial u_0}{\partial x_j}. \tag{3.207}$$

Compute $\mathcal{M}(R_2 u_1)$: We use the ansatz (3.190), $u_1 = -\chi^k \frac{\partial u_0}{\partial x_k}$.

$$R_2 u_1 = 2\frac{\partial^2}{\partial t \partial \tau} u_1 + A_2 u_1 + B_2 u_1. \tag{3.208}$$

i. $\mathcal{M}\left(2\frac{\partial^2 u_1}{\partial t \partial \tau}\right) = 0$ (due to periodicity in τ).

ii. $\mathcal{M}(A_2 u_1)$:

$$A_2 u_1 = -\frac{\partial}{\partial y_i}\left(a_{ij}\frac{\partial}{\partial x_j}(\chi^k \frac{\partial u_0}{\partial x_k})\right) - \frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial}{\partial y_j}(\chi^k \frac{\partial u_0}{\partial x_k})\right) \tag{3.209}$$

$$\begin{aligned}\mathcal{M}(A_2 u_1) &= \mathcal{M}\left(\frac{\partial}{\partial y_i}\underbrace{\left(-a_{ij}\frac{\partial}{\partial x_j}(\chi^k \frac{\partial u_0}{\partial x_k})\right)}_{F_i}\right) + \mathcal{M}\left(-\frac{\partial}{\partial x_i}\left(a_{ij}\frac{\partial \chi^k}{\partial y_j}\frac{\partial u_0}{\partial x_k}\right)\right) \\ &= 0 - \frac{\partial}{\partial x_i}\left(\mathcal{M}\left(a_{ij}\frac{\partial \chi^k}{\partial y_j}\right)\frac{\partial u_0}{\partial x_k}\right).\end{aligned} \tag{3.210}$$

The first term is the average of a divergence $\frac{\partial F_i}{\partial y_i}$, where the vector field

$$F_i = -a_{ij}(y)\frac{\partial}{\partial x_j}\left(\chi^k(y)\frac{\partial u_0}{\partial x_k}(x,t)\right). \tag{3.211}$$

is Y-periodic in y because $a_{ij}(y)$ and $\chi^k(y)$ are Y-periodic. The average of the divergence of a periodic function over its period is zero.

iii. $\mathcal{M}(B_2u_1)$:

$$B_2u_1 = -b_j\frac{\partial}{\partial y_j}\left(-\chi^k\frac{\partial u_0}{\partial x_k}\right) = b_j\frac{\partial \chi^k}{\partial y_j}\frac{\partial u_0}{\partial x_k}, \tag{3.212}$$

$$\mathcal{M}(B_2u_1) = \mathcal{M}\left(b_j\frac{\partial \chi^k}{\partial y_j}\right)\frac{\partial u_0}{\partial x_k}. \tag{3.213}$$

Therefore,

$$\mathcal{M}(R_2u_1) = -\frac{\partial}{\partial x_i}\left(\mathcal{M}\left(a_{ij}\frac{\partial \chi^k}{\partial y_j}\right)\frac{\partial u_0}{\partial x_k}\right) + \mathcal{M}\left(b_j\frac{\partial \chi^k}{\partial y_j}\right)\frac{\partial u_0}{\partial x_k}. \tag{3.214}$$

1.2.3. Combining the Terms: Substitute the computed averages into the solvability condition (3.203):

$$\begin{aligned}&\left[\frac{\partial^2 u_0}{\partial t^2} - \frac{\partial}{\partial x_i}\left(\mathcal{M}(a_{ij})\frac{\partial u_0}{\partial x_j}\right) - \mathcal{M}(b_j)\frac{\partial u_0}{\partial x_j}\right]\\ &+\left[-\frac{\partial}{\partial x_i}\left(\mathcal{M}\left(a_{ij}\frac{\partial \chi^k}{\partial y_j}\right)\frac{\partial u_0}{\partial x_k}\right) + \mathcal{M}\left(b_j\frac{\partial \chi^k}{\partial y_j}\right)\frac{\partial u_0}{\partial x_k}\right] = f.\end{aligned} \tag{3.215}$$

Combine the first-order terms:

$$\begin{aligned}-\mathcal{M}(b_j)\frac{\partial u_0}{\partial x_j} + \mathcal{M}\left(b_k\frac{\partial \chi^j}{\partial y_k}\right)\frac{\partial u_0}{\partial x_j} &= \left(-\mathcal{M}(b_j) + \mathcal{M}\left(b_k\frac{\partial \chi^j}{\partial y_k}\right)\right)\frac{\partial u_0}{\partial x_j}\\ &= \mathcal{M}\left(b_k\frac{\partial \chi^j}{\partial y_k} - b_j\right)\frac{\partial u_0}{\partial x_j}.\end{aligned} \tag{3.216}$$

Moreover, note that $\mathcal{M}(a_{ij})\frac{\partial u_0}{\partial x_j} + \mathcal{M}\left(a_{ik}\frac{\partial \chi^j}{\partial y_k}\right)\frac{\partial u_0}{\partial x_j}$ defines the homogenized coefficient for the operator $\mathcal{A}$. Thus, the homogenized equation is:

$$\frac{\partial^2 u_0}{\partial t^2} + \mathcal{A}u_0 + \mathcal{M}\left(b_k\frac{\partial \chi^j}{\partial y_k} - b_j\right)\frac{\partial u_0}{\partial x_j} = f. \tag{3.217}$$

This can be written in the equivalent form by relabeling the summation index j to i and reordering the terms inside the average:

$$\frac{\partial^2 u}{\partial t^2} + \mathcal{A}u + \mathcal{M}\left(b_i - b_k\frac{\partial \chi^i}{\partial y_k}\right)\frac{\partial u}{\partial x_i} = f. \tag{3.218}$$